CHARACTERIZATION OF *IN VITRO* FITNESS AND ANTIGENIC DRIFT DIFFERENCES BETWEEN CIRCULATING H1N1 INFLUENZA CLADES

by
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Abstract

Influenza viruses are a significant contributor to morbidity and mortality, with multiple subtypes circulating seasonally and causing annual epidemics. New viral clades emerge regularly through the acquisition of mutations, which have the potential to cause antigenic drift leading to reduced recognition by host immunity. New mutations may also alter virological fitness, and due to the overlap between antigenic and functional regions a mutation may alter both phenotypes simultaneously. Studying how these separate components are balanced in circulating viruses is crucial to understanding the processes driving viral evolution.

The 2019-20 Northern Hemisphere Influenza Season was characterized by a predominance of the H1N1 influenza subtype, and was one of the most severe seasons of the past decade. During this season, the predominant 5a clade branched into two cocirculating clades, 5a.1 and 5a.2, which were differentiated by several clade-defining mutations on their hemagglutinin protein. While the 5a.1 clade quickly predominated, 5a.2 performed poorly and was almost nonexistent by the end of 2020. The differing success of these two clades indicate phenotypic differences, and this dissertation identifies a reduction in *in vitro* fitness of the 5a.2 clade that likely contributed to its failure to predominate.

At the start of 2021 the 5a.2 clade rapidly acquired a series of hemagglutinin mutations, and by 2022 the 5a.2a subclade quickly overtook 5a.1 to become the predominant H1N1 clade. 5a.2a then further branched into 5a.2a.1 which has continued to cocirculate with 5a.2a. The rapid success of this clade indicates phenotypic advantages over its 5a.2 predecessor, and this dissertation identifies improvements in

in vitro fitness and antigenic drift of 5a.2a and 5a.2a.1 compared with 5a.2. An infectious clone system of H1N1 was employed to study the phenotypic effects of individual clade-defining mutations, and it was found that the mutations that contributed to antigenic drift reduced *in vitro* fitness, while the mutation that improved viral fitness actually increased virus neutralization by human serum antibodies. These findings highlight the dual evolutionary pressures on influenza to evade host immunity while maintaining viral fitness and point to epistatic interactions that optimize both components while mitigating negative pleiotropic effects of any single mutation.

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Preface

“ ‘Cela est bien dit’, répondit Candide, ‘mais il faut cultiver notre jardin.’ ”

-Voltaire, *Candide*

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List of Abbreviations

Abbreviation	Definition
AFM	Atomic Force Microscopy
ALI	Air-Liquid Interface
AUC	Area Under the Curve
BLI	Biolayer Interferometry
CDC	United States Centers for Disease Control and Prevention
CFG	Consortium for Functional Glycomics
CM	Complete Medium
cRNA	Complimentary Viral RNA
DC-SIGN	Dendritic Cell-Specific Intercellular Adhesion Molecule-3 Grabbing Nonintegrin
DMEM	Dulbecco's Modified Eagle Medium
ER	Endoplasmic Reticulum
GISRS	Global Influenza Surveillance and Response System
GLAD	Glycan Array Dashboard
HA	Hemagglutinin
HAI	Hemagglutinin Inhibition
HAT	Human Airway Trypsin-Like Protease
HCW	Healthcare Workers
HEF	Hemagglutinin-Esterase Fusion Protein
hNEC	Human Nasal Epithelial Cell
HPAI	Highly Pathogenic Avian Influenza
HPLC	High Performance Liquid Chromatography
IAV	Influenza A Virus
IBV	Influenza B Virus
IIV	Inactivated Influenza Vaccine
JH-CEIRR	Johns Hopkins Center for Excellence in Influenza Research and Response
JH-CEIRS	Johns Hopkins Center for Excellence in Influenza Research and Surveillance
JHH	Johns Hopkins Hospital
JHU	Johns Hopkins University
LAIV	Live Attenuated Influenza Vaccine
MDCK	Madin-Darby Canine Kidney
MDCK-SIAT	Madin-Darby Canine Kidney Sialtransferase 1
MHC	Major Histocompatibility Complex
MMR	Macrophage mannose receptor
MOI	Multiplicity of Infection
mRNA	Messenger RNA
MS	Mass Spectrometry
NA	Neuraminidase
NCFG	National Center for Functional Glycomics

NEP	Nuclear Export Protein
Neu5Gc	N-Glycolylneuraminic Acid
NH	Northern Hemisphere
NIH	National Institutes of Health
NLS	Nuclear Localization Signal
NP	Nucleoprotein
NS1	Nonstructural Protein 1
NS2	Nonstructural Protein 2
PA	Polymerase Acidic Protein
PB1	Polymerase Basic 1 Protein
PB2	Polymerase Basic 2 Protein
PBS	Phosphate Buffered Saline
Pdm2009	2009 Pandemic-Origin Human H1N1 Subtype
RBS	Receptor Binding Site
RDE	Receptor Destroying Enzyme
RFU	Relative Fluorescence Units
RNA	Ribonucleic Acid
SA	Sialic Acid
TCID50	Tissue Culture Infectious Dose 50
TMPRSS2	Transmembrane Protease Serine 2
vRNA	Viral RNA
vRNP	Viral Ribonucleoprotein
WHO	World Health Organization

CHAPTER 1: Introduction

Influenza Epidemiology

Influenza viruses are among the largest contributors to morbidity and mortality from infectious respiratory diseases. While disease burden varies from year to year (Figure 1.1), on average influenza causes over one billion infections and hundreds of thousands of deaths annually (1, 2). In the temperate climates of the Northern and Southern Hemispheres influenza peaks seasonally during cold months, and in tropical climates circulation can occur consistently throughout the year (3, 4). Disease severity depends on a variety of both viral and host factors; H1N1 tends to cause a more severe infection than H3N2 (1, 5, 6), and there is a higher risk of severe outcomes in children infected with influenza B viruses (1, 7). On the host side, age and immune status are often indicators of disease severity, with older and immunocompromised individuals typically exhibiting more severe symptoms (1, 5, 6, 8). Interestingly, this trend was reversed during the 1918 H1N1 pandemic, where there was a striking increase in mortality among healthy, young adults (9). Biological sex also plays a role in both disease severity and response to vaccination; women tend to exhibit more severe disease symptoms, but also receive more protection after vaccination (10, 11).

Influenza disease pathology mirrors that of many other respiratory viruses. While transmission dynamics are still not completely understood, viruses are thought to generally spread through respiratory droplets, aerosols, or direct contact (12-14). Fomites, or surfaces, may also contribute to viral transmission (15-17). Exposure to the virus generally leads to symptom presentation between one and three days (18, 19), and symptoms in self-limiting infections usually resolve after one to two weeks (18, 19). Symptoms include fever, cough, chills, congestion, and muscle aches, and infected

individuals are usually infectious for one to two days preceding symptom onset and continue to shed viruses for up to a week (18-21). Asymptomatic infections are also possible, with carriers shedding virus despite not exhibiting any symptoms of influenza (20, 22, 23). Complications and severe disease can arise through an immune overreaction that leads to tissue scarring and “cytokine storms”, and influenza infections can also make the body more vulnerable to secondary bacterial infections that can cause pneumonia and death (18, 19, 24, 25).

Seasonally circulating influenza viruses can be divided into multiple genera and subtypes, which have major differences in both their genetics and appearance to host immunity. Influenza A and B are the two types that circulate seasonally in humans (26), of which influenza A causes the majority of disease burden (3, 27, 28). Influenza A viruses (IAV) are further divided into subtypes based on the identity of its two major surface proteins, hemagglutinin (HA) and neuraminidase (NA). There are 18 HA and 11 NA subtypes identified (29), of which two are currently circulating in humans in the form of H1N1 and H3N2. H1N1 and H3N2 cocirculate, and vary in prevalence depending on the season and geographic location; for example, in the United States H3N2 has predominated in the majority of recent seasons with H1N1 predominating during the 2019-20 season and the 2023-24 season (Figure 1.1A).

The COVID-19 pandemic and its associated public health intervention measures had a striking effect on the circulation of influenza. The commencement of lockdowns in March 2020 had the immediate effect of ending the ongoing 2019-20 influenza season (30, 31). Global influenza cases remained exceedingly low for the next two years, and it was not until the 2022-23 influenza season that circulation returned to its pre-pandemic

levels (31, 32). While H1N1 and H3N2 both survived the pandemic, the Yamagata strain of Influenza B has not been detected since March of 2020 and is presumed to have gone extinct (31). The rapid drop of influenza circulation can largely be attributed to the physical distancing, isolation, and masking protocols induced by SARS-CoV-2, and speak to the effectiveness of these measures at preventing the spread of respiratory diseases (30).

In addition to infecting humans, influenza A also circulates in a variety of animals, including birds, pigs, dogs, cats, and horses (33-36). Due to differences in host biology, the viruses that infect different animals have different adaptations to preferentially infect their respective hosts. For example, avian influenza has a different receptor binding specificity than human influenza due to differences in surface glycoprotein expression between these two hosts (discussed below) (37-39). Host specificity causes barriers between interspecies spread, and prevents continuous exchange of viruses between humans and animals. These barriers can be overcome, however, through the emergence of mutations as well as the reassortment of viral gene segments. When this happens animal influenza viruses can “spill over” into humans and cause pandemics. The past century has seen multiple pandemics caused by the spillover of animal influenza into humans; among the most significant are the 1918 “Spanish Flu” H1N1 pandemic, the H3N2 pandemic of 1968, and most recently the 2009 “Swine Flu” H1N1 pandemic (40-43). Viruses that spill over often continue to circulate in humans and cause the familiar seasonal influenza epidemics, and the currently circulating H3N2 and H1N1 viruses are descended from the 1968 and 2009 pandemic viruses, respectively (43). Influenza pandemics remain one of the largest public health threats from infectious

diseases. Currently, an H5N1 panzootic has spread in farmed poultry globally (44, 45) and has resulted in the culling of billions of birds and billions of dollars in lost revenue (45, 46). The prevalence of this virus and its close proximity to humans increases the chance that a sustained spillover will occur (47).

Life Cycle Overview

The influenza virus is a member of the *Orthomyxoviridae* family, which are enveloped viruses with segmented, negative-sense ribonucleic acid (RNA) genomes (48). Both IAV and IBV have eight separate gene segments that produce 10 conserved essential proteins and several subtype-specific accessory proteins (48), while influenza C and D viruses have seven gene segments including a hemagglutinin-esterase fusion (HEF) segment whose protein product plays the functional roles of both HA and NA (49). The accessory proteins of IAV include PB1-F2 and PA-X, which modulate host immune response, as well as several other less-well characterized viral proteins (48, 50, 51). The eight gene segments of IAV and IBV are organized as viral ribonucleoproteins (vRNPs) inside the virion, with viral RNA (vRNA) wrapped around viral nucleoproteins (NP) and held at its ends by the heterotrimeric viral polymerase (Figure 1.2A) (14, 51, 52). The eight vRNPs associate with the M1 proteins coating the virion interior, which are in turn associated with the phospholipid membrane encasing the virion (51, 53). There are three surface proteins on the IAV virion which are embedded in the phospholipid membrane envelop: HA, NA, and M2. These proteins each play a vital role in distinct stages of the influenza life cycle (48, 51, 54, 55).

Influenza virions may either be spherical or filamentous. The factors contributing to these morphologies is not completely understood, but clinically derived virions are frequently filamentous whereas viruses cultured in immortalized cells are predominantly spherical (56, 57). Spherical virions are generally between 80 and 120 nanometers, while filamentous virions may reach 20 micrometers in length (57). Viral binding to the host cell is mediated by HA, which recognizes polysaccharides (glycans) containing a sialic acid as the terminal sugar (37, 58). Upon receptor binding, the virion is internalized into endosomes mainly through clathrin-mediated endocytosis (51, 59, 60), though clathrin-independent endocytosis routes may also be employed (60, 61) especially for filamentous virions (62). The acidification of the endosome triggers a conformational change in HA that induces its membrane fusion function, joining the viral membrane the endosomal membrane (58). Simultaneously, the acidic environment prompts the M2 membrane protein to commence its ion channel activity, sending protons into the virion interior (51, 63, 64). The acidification inside the virion causes the disassembly of the M1 layer and the dissociation of the vRNPs from the M1 proteins, allowing them to exit the virion and the endosome (51, 65) (See Figure 2B for a depiction of the viral life cycle).

Nuclear localization signals (NLS) on the NPs cause the vRNPs to traffic to the nucleus, where transcription commences (6, 48). The heterotrimeric viral polymerase is responsible for both replication of the viral RNA (vRNA) as well as transcription of messenger RNA (mRNA) for protein expression (66), and transition between these two functions is thought to rely on the polymerase complex associating with either host polymerase II (transcription) or ANP32A (replication) (51, 67). Genome replication is

accomplished by the transcription of complementary viral RNA (cRNA), which is then used as a template to produce more vRNA. During genome replication, newly synthesized vRNA and cRNA immediately associate with viral polymerase and NP proteins, which stabilize the nascent RNA and protect it from host degradation (51). cRNA and vRNA transcription occurs *de novo* (“unprimed”) and relies on spontaneous association of free nucleotides, and is slow and inefficient compared with mRNA transcription (48, 68).

mRNA transcription is rapid and proceeds before significant vRNA replication has occurred (15, 67). mRNA production depends on the host mRNA binding function of the polymerase basic 2 protein (PB2) and the cap-snatching function of the polymerase acidic protein (PA), which cleaves the methylated 5’ end of host mRNA and uses it to initiate polymerase basic 1 protein (PB1)-mediated viral mRNA synthesis (66). Repeated “stuttering” of transcription at a poly-U region at the 5’ end of the vRNA produces a poly-A tail at the end of viral mRNA, resulting in a stable mRNA that will be targeted for export from the nucleus and towards translation machinery (51, 69). Viral mRNA enters the cytoplasm and is translated, and the polymerase and NP proteins traffic back to the nucleus via exposed NLS where they associate with and aid in the transcription of vRNA and cRNA (48, 51, 66), ultimately forming the vRNP segments that will be packaged into newly formed virions. These vRNPs are exported from the nucleus through association with both viral M1 proteins and nuclear export proteins (NEP) (also known as NS2, formed by the alternative splicing of viral gene segment 8) (70, 71).

Viral proteins are translated via either free ribosomes or endoplasmic reticulum (ER)-associated ribosomes, depending on whether they will be cytosolic or integral membrane proteins (HA, NA, M2) (48, 51). In addition to structural and polymerase proteins, the influenza genome also encodes nonstructural (NS) proteins which are implicated in both intracellular localization and innate immune suppressing functions (72). Translated proteins are trafficked to the cellular membrane, where they associate. HA and NA localize on lipid rafts on the apical cell membrane, causing these rafts to merge and concentrate viral membrane proteins (73). M2 gathers at the edges of these rafts and may play a role in membrane budding and scission (51, 74). The combination of several factors, including lipid recruitment and membrane “bending” functions of viral proteins, ultimately result in viral budding and detachment from the cell membrane (48, 73, 75, 76). Since the cell surface still contains an abundance of viral receptors, the HA protein will immediately bind to the cell, and the sialidase function of the NA protein removes sialic acid residues from glycans, precluding virus binding back to the infected cells and allowing the virion to spread and infect neighboring cells (77). NA also prevents virions from becoming trapped in the decoy receptors present on host mucins, virus particle aggregation and promotes viral movement towards new, susceptible cells (77, 78).

The Hemagglutinin Protein

Two of the first steps involved in influenza infection - receptor binding and membrane fusion - are accomplished by HA. HA is synthesized by ribosomes associated with the ER, where it is translated into the ER membrane and undergoes

folding as a monomer before quickly associating with two other monomeric HAs (48). Glycans are added to HA by host machinery in multiple locations corresponding to canonical glycosylation sites, and these glycans have been shown to affect viral fitness and antigenic profile (79-82). During trafficking to the cell membrane, each HA monomer is cleaved from the precursor HA0 into HA1 and HA2 by serine proteases that reside in the trans-Golgi network and at the cell membrane (83, 84). This proteolytic cleavage is essential for proper HA function, as HA0 is incapable of performing the membrane fusion function necessary for viral entry into host cells (4). This cleavage site can either be monobasic or polybasic; seasonal influenza viruses have monobasic sites and are cleaved by human airway trypsin-like (HAT) and transmembrane protease serine 2 (TMPRSS2) proteases, whose expression is limited to the upper airway (85, 86). The polybasic cleavage sites of highly pathogenic avian influenza (HPAI) can be cleaved by furin which is expressed throughout the body, and as a result HPAI can produce disseminated infection and greater pathology (51, 84, 86, 87). Upon being trafficked to the cell membrane, HA localizes with NA, M2, and M1 and viral budding is initiated. HA has been proposed to play a role in viral budding, as expression of HA and NA alone has been shown to be sufficient to induce budding and virus-like particle formation (88).

The canonical receptors for influenza are polysaccharides (glycans) that contain sialic acid as their terminal sugar. Glycans are ubiquitous cellular components, both intracellularly and on the cell surface. Functionally they serve as modifications to an abundance of cellular proteins, and play crucial physiological roles including signaling, trafficking, cell-cell interactions, immunity, and ligand binding (89, 90). Dysregulation of

glycosylation is associated with a variety of chronic diseases and cancers (89, 90). The ubiquity of these molecules makes them prime targets as viral receptors, and there are numerous glycans with which HA has been shown to interact. The receptor binding site (RBS) of HA is situated close to the head of the molecule, and is formed from several regions of the HA1 monomer including the 130 loop, 150 loop, 220 loop, and 190 helix, as well as several anchor residues (Y98, W153, H183, L194, and Y195, H3 numbering) (Figure 1.3B) that are highly conserved and essential for receptor binding (37, 58, 86, 91). Mutations to these regions have the potential to change the structure of the RBS, and can alter the receptor specificity of HA (37, 91-94).

Differences in receptor structure preferences contribute to the host preferences of human and avian influenza viruses. Canonically, viruses that preferentially infect humans bind strongly to glycans whose terminal sialic acid is linked in an α2,6 configuration with the penultimate galactose, while the HA of avian influenza viruses preferentially bind to α2,3 sialic acid (SA) glycans (86, 95, 96). These receptor binding differences represent a major barrier to viral spillover, and many of the mutations that allow avian influenza to infect humans are those that change receptor binding diversity (86, 96, 97). Differential glycan expression also occurs within hosts; while humans predominantly express α2,6 SA glycans in the upper respiratory tract, the lower respiratory tract has higher levels of α2,3 SA glycans and avian influenza can occasionally infect humans by replicating in this region (98). The location of these receptors greatly reduces the risk of human-to-human transmission of avian influenza, but also may result in severe lower respiratory tract pathology (98, 99). Receptor expression makes swine susceptible to both human and avian influenza viruses, as they

express both α2,6 SA and α2,3 SA glycans in their respiratory tracts (37, 100, 101). This also contributes to their traditionally defined status as “mixing vessels” for human and avian influenza, as coinfections of swine can lead to gene segment reassortments and the emergence of novel viruses that are able to infect humans (95, 101-103). It should be noted, however, that the importance of swine as intermediaries in past pandemics remains a matter of debate (104), and more recent analysis suggests that glycan expression on the swine respiratory tract closely resembles that of humans (105). Recent studies have also suggested that the 2,3/2,6 SA receptor binding dichotomy is not clear cut and that human influenza viruses may bind to and use α2,3 SA glycans for cell entry (106-110).

In addition to the receptor binding differences between avian and human influenza, receptor binding differences exist between circulating strains of human influenza viruses. Circulating H3N2 viruses have slowly lost receptor binding affinity (111) and diversity (112, 113) since their emergence in 1968, and H3N2 receptor binding diversity has even changed between seasons and between cocirculating clades (114, 115). Shortly after its emergence, the 2009 pandemic H1N1 was shown to have detectable levels of α2,3 SA binding (116) and reduced α2,6 SA receptor binding compared with the prior seasonal H1N1 (117-119), and modeling revealed differences in RBS structure within H1 viruses over the past century (120). Seasonal differences between circulating H1N1 receptor binding have also been observed (121, 122). These data point to receptor binding as an important and dynamic aspect of evolution of human influenza viruses. Changes to receptor binding can also emerge during the vaccine manufacturing process. Egg-based influenza vaccines are grown in chicken

eggs which have a different glycan profile than the human respiratory tract. As a result, mutations associated with adaptation to eggs can emerge and change the receptor binding profile of the resulting vaccine virus (123-125), which has been correlated with a reduction in vaccine effectiveness and viral *in vitro* fitness phenotype (92).

The receptor binding role of HA is balanced by the receptor cleaving function of NA (77, 78, 126). As new influenza virions bud on the cell surface, their HA molecules are immediately exposed to receptors on the cell membrane to which they can bind. NA is a sialidase enzyme, and it is able to cleave the terminal sialic acids from potential HA receptors (77). Without NA activity, nascent virions would remain stuck to the cells from which they emerged and infection of new cells could not occur (127, 128). NA is also able to cleave the decoy receptors present in mucus; heavily glycosylated mucins in the respiratory tract function as innate immune barriers to infection, attracting HA and preventing virions from binding to cells. NA allows the virus to escape from mucus traps and continue traveling towards cells (129). The combination of iterative viral attachment to glycans through HA and subsequent receptor removal by NA has been described as a mechanism of viral motility (129, 130). This presents another barrier to reassortment-induced viral spillover, as the HA-NA balance must be maintained (131). The 2009 H1N1 pandemic virus demonstrated a reduction in avidity to select glycan receptors compared with its precursor swine H1N1, which was accompanied by a reduction in NA enzymatic activity that balanced the HA-NA activity. This balance, not seen in the swine H1N1, likely contributed to viral spillover (132). The HA-NA balance also poses evolutionary limitations on seasonal influenza, as mutations that change the binding

activity of HA or the enzymatic activity of NA may offset this functional equilibrium (131, 133-135).

After receptor binding and virus internalization into the host endosome, hemagglutinin performs its role as the membrane fusion protein through an irreversible pH-induced conformational change. The first 20 residues on the N-terminal of the HA2 protein serve as the fusion peptide (136); before confirmation change this region is internal within HA, but it protrudes significantly on the tip of a coiled coil structure after rearrangement and is able to insert into the endosomal membrane and bring the viral and host membranes into close contact for hemifusion (48, 51, 58). A final fusion step, requiring anchoring by the HA2 transmembrane domain, causes HA to form a hairpin structure and bring the two membranes together sufficiently to form pores (48, 51, 58). vRNPs are then able to exit the endosome through these pores and traffic to the nucleus to begin replication.

Antibody Recognition of Hemagglutinin

In addition to its contribution to receptor binding and membrane fusion, HA is also a significant target for the host immune response. HA is the viral protein most heavily targeted by host antibody response, and it is the primary antigenic determinant of the virus (86, 137). Influenza viruses are classified into clades based on genetic differences between HA, and different influenza clades are often antigenically distinct from each other (114, 138-140). The process by which circulating influenza acquires mutations that change its antigenic profile is known as antigenic drift, and contributes to the need for annual vaccination with updated clades of contemporary influenza viruses

(141). These mutations can cause loss of antibody binding due to a reduction in antibody affinity to the targeted region, as well as the addition or removal of glycosylation sites that change HA structure and access to antigenic sites (86, 142). Many of the neutralizing antibodies produced against influenza are effective through blocking receptor binding (137, 139, 143), though antibodies targeting the HA stalk have been identified which serve to block membrane fusion (39, 143, 144). Antigenic sites have been mapped for HA, and antigenic drift mutations most commonly occur on these antigenic sites (Figure 1.3C).

While antigenic drift causes a gradual decrease in population immunity against circulating viruses, a much more drastic change known as antigenic shift can produce new viruses against which the population is completely immunologically naïve (145). Antigenic shift is the consequence of reassortment events between multiple IAV subtypes (145). This can include the reassortment of multiple animal viruses (e.g. a swine virus reassorting with an avian virus), as well as human and animal viral reassortment (41). In either case, the co-infection of a single host cell results in segments of multiple viruses being packaged in a single virion. Reassortments are not always advantageous from the perspective of the virus, but occasionally a reassorted virus gains the capacity to transmit between humans. Due to their animal origin they are typically antigenically distinct from currently circulating human influenza viruses, and can spread rapidly through the immunologically naïve human population. Reassortment events are often the cause of influenza pandemics; the 2009 H1N1 pandemic was the result of multiple reassortments between avian, swine, and human influenza (146, 147). The resulting virus was transmissible in humans and spread quickly as the population

had limited prior immunity to this virus, with the exception of elderly individuals whose prior exposure to the 1918-derived H1N1 likely contributed to cross-protection against the 2009 pandemic H1N1 (148). The pandemic only waned once the population gained a degree of immunity through natural infection or vaccination, and this virus continues to circulate as seasonal influenza (149, 150).

Influenza Vaccination

Due to the frequency of mutations and rate of antigenic drift in seasonal influenza, annual vaccinations are necessary to maintain immunity against currently circulating clades. The choice of influenza clades to be included in the vaccine is made by the World Health Organization (WHO) Global Influenza Surveillance and Response System (GISRS), whose recommendations are typically followed by countries and vaccine manufacturers (151). The WHO's vaccine recommendations are made twice a year: once in February for the following Northern Hemisphere season, and once in September for the following Southern Hemisphere season. These recommendations are based on two major factors: 1) which genotypes of each virus subtype are predominant (i.e. which are epidemiologically important), and 2) how well these viruses are neutralized by serum of vaccinated humans and ferrets (152-155). This neutralization assay data helps determine whether current clades are antigenically drifted from prior vaccine formulations, and whether the vaccine needs to be updated as a result.

There are several drawbacks to this system of vaccine selection. Recommendations for vaccine composition are made months in advance of the influenza season for which they are recommended. This is a necessary delay, as it

gives manufacturers, distributors, and doctors enough time to produce and administer the vaccine before the season starts (156), but significant changes in prevalence or genotypes of circulating viruses can occur in the interim and render the vaccine ineffective (157, 158). Additionally, each virus subtype can have multiple cocirculating clades with antigenic differences. The vaccine only includes one strain from each subtype, and this limitation prevents the vaccine from capturing the total antigenic diversity of circulating influenza (159, 160). Influenza vaccines generally range in effectiveness from 40% to 60% at preventing illness (161), though vaccine mismatches have lowered effectiveness to less than 20% in some years (162).

There are three major kinds of influenza vaccines: inactivated influenza vaccines (IIV), live attenuated influenza vaccines (LAIV), and recombinant influenza vaccines. Different IIV are manufactured using either hen eggs or cultured cells (163), but the only LAIV currently licensed in the United States is grown in hen eggs (156). For IIV and LAIV, the desired HA and NA chosen for the vaccine are typically combined with internal segments (PB2, PB1, PA, NP, M, NS) that have been specifically adapted for the culture system (163). The reassorted viruses generated will still have the antigenic profile of the currently circulating viruses, but with internal segments that allow the virus to replicate in eggs and/or provide an attenuated phenotype for LAIV (164). Unlike LAIV and IIV, recombinant influenza vaccines do not require an entire replicating influenza virus and instead involve the insertion of an HA gene into a baculovirus background. This baculovirus, when introduced to a cell culture, will induce the production of large quantities of HA protein which can be purified and used for vaccination (165).

Vaccine manufacturing processes can elicit genetic changes in the viruses. As previously mentioned, the growth of viruses in eggs can lead to adaptations that change receptor binding preferences and reduce vaccine effectiveness (92, 156, 163, 166, 167), and there is a significant push to transition away from egg-based vaccines. Cell-based and recombinant vaccine technologies offer the advantage of removing the risk of egg-induced viral changes, and recombinant vaccines have the added benefit of being a faster vaccine manufacturing platform than LAIV or IIV. Monitoring vaccine effectiveness and antigenic match to circulating viruses remains an important task for improving vaccine formulation and ensuring optimum protection.

Dual Evolutionary Pressures on Hemagglutinin

While the regions of HA associated with membrane fusion are well-conserved, there is significant amino acid variability of many of the residues around and within the receptor binding site (168). The frequency of mutations on the HA head can be understood by examining the regions of HA most heavily associated with antibody recognition and antigenicity. There are five canonical antigenic sites on H1 (Sa, Sb, Ca1, Ca2, and Cb) (Figure 1.3C), and three of these are adjacent to or overlap the RBS (86, 169). These sites are targeted by antibodies as a way of blocking receptor binding, and therefore viral infection of the host cell. This produces substantial evolutionary pressure on influenza in favor of mutating these regions to evade antibody recognition, and as a result the viral RBS undergoes frequent changes.

The consequence of this overlap between the RBS and antigenic sites is competing evolutionary pressures on influenza viruses. On one hand, mutations to

these antigenic sites are beneficial to the virus because they allow for the evasion of antibody-mediated virus neutralization. On the other hand, mutations to the RBS can easily be deleterious to viral fitness and there is evolutionary pressure to maintain certain residues that are necessary for receptor binding (135). This includes the anchor residues of the RBS whose mutations would be fatal to the virus (58, 91, 170), but it also includes a variety of other amino acids on or near the receptor binding site (171). There are many examples of mutations that contribute to antigenic drift also having an impact on receptor binding and viral replication (82, 112, 171-178), and conversely receptor binding mutations associated with egg adaptation of vaccine viruses have been shown to alter the antigenic profile of the virus (92).

Compensation for potentially deleterious antigenic drift mutations can be achieved through epistasis. Epistasis refers to the effect that one mutation or gene has on the phenotype of another mutation or gene. A mutation may be harmful to one virus but beneficial to another virus, depending on the presence of other mutations (179, 180). Epistatic interactions in viruses can be either intra-gene or inter-gene; the deleterious effects of a mutation on HA can be compensated for by additional HA mutations, or mutations on other gene segments (e.g. NA) (181-185). This phenomenon is crucial for influenza evolution. An antigenic drift mutation on HA that is harmful to viral fitness can be balanced by the emergence of additional “enabling mutations”, allowing the virus to maintain fitness while still evading antibody recognition.

A variety of methods have been employed to study the contribution of epistasis to influenza evolution. Large-scale statistical analyses of H1N1 and H3N2 have shown that the emergence of mutations is clustered, indicating positive epistasis (185-188), and

statistical models of influenza evolution are more accurate when accounting for epistasis (189). Broad mutational analyses have shown that epistasis in the 220-loop allows the RBS to accommodate significant changes while maintaining fitness, and that many of these mutations are able to confer antigenic drift (190). Detailed studies of specific residues have shown that combining two deleterious or neutral mutations can rescue receptor binding and fitness (191-193), and that the negative fitness effects of HA antigenic drift mutations can be compensated by secondary mutations (194). Passaging influenza in the presence of neutralizing antibodies has been shown to produce antibody escape mutations that change receptor binding properties (176), and in one study an antigenic drift mutation that reduced fitness was ameliorated by the emergence of secondary enabling mutations after serial passaging (175). The use of mass spectrometry has even been employed to study epistasis. The construction of phylogenetic “mass trees” using hundreds of H3 proteins revealed the coordinated evolution of mutational pairs, with “parent mutations” on antigenic sites emerging shortly before non-antigenic site “child mutations”, gradually altering the HA structure (195). Epistatic interactions extend to the HA glycosylation state; while the addition of glycosylation sites on the HA head can produce antigenic drift, it can cause a loss of receptor binding that is only compensated for by the acquisition of secondary enabling mutations (82). Conversely, changes to glycosylation can sometimes rescue viral fitness from harmful antigenic drift mutations (196). Similar to the epistatic evolution of HA antibody epitopes, it has been shown that antigenic drift mutations on NP segment T-cell epitopes reduced fitness on their own, but emerged in coordination with enabling mutations that rescued viral fitness (179). The epistatic interactions between antigenic

drift mutations and fitness-rescuing mutations clearly plays a major role in the ability for influenza to evade adaptive immunity.

Epistasis also imposes constraints on viral evolution. Given the frequency of mutations sustained during influenza replication, it has been questioned why new clades do not emerge more frequently than they do (197). One model suggests that multiple mutations need to emerge near-simultaneously to produce an antigenically drifted virus that has not incurred a fitness cost (171, 196, 197), and another suggests that the emergence of antigenic mutations is limited by the presence of harmful mutations in the viral background (198). Epistasis also limits mutational reversion to previous amino acids. For example, the E190 residue of human H3N2 used to be ubiquitous, but mutated to D190 in recent years as a putative antigenic drift mutation (199). In the following years other RBS mutations have emerged, and the reversion of this residue to D190 in a contemporary H3 genotype is now deleterious due to intervening changes in the H3 RBS (199). More broadly, antigenic drift of H3 from 1968 to 1997 produced irreversible structural changes to HA that resulted in differing abilities of these viruses to withstand point mutations (200). This structural change closed the door to previously available amino acid variation, but also opened the door to new, previously deleterious mutations. These findings point to significant antigenic site diversity that epistasis provides to influenza, given sufficient evolutionary time (200).

Considering the enormous role of epistasis in the evolution of seasonal influenza, the interactions between emergent mutations and the genetic background on which they appear should always be considered when studying changes to circulating viruses. Not only will this broaden our understanding of the phenotypic consequences of important

mutations and their mechanistic underpinnings, but it will advance our understanding of how epistatic interactions drive viral evolution. While influenza remains one of the most well-researched viruses, many of the studies investigating epistasis are over a decade old and focus on viruses that circulated in the distant past. Given the resurgence of seasonal influenza in the years following COVID-19 and the continued threat of avian influenza spillover, monitoring the evolution of contemporary viruses is a necessary undertaking. The research presented in the following chapters describes characteristics of recently circulating seasonal H1N1, and highlights the relationship between changes to antigenic profile and *in vitro* fitness that have been balanced during contemporary viral evolution.

Figures

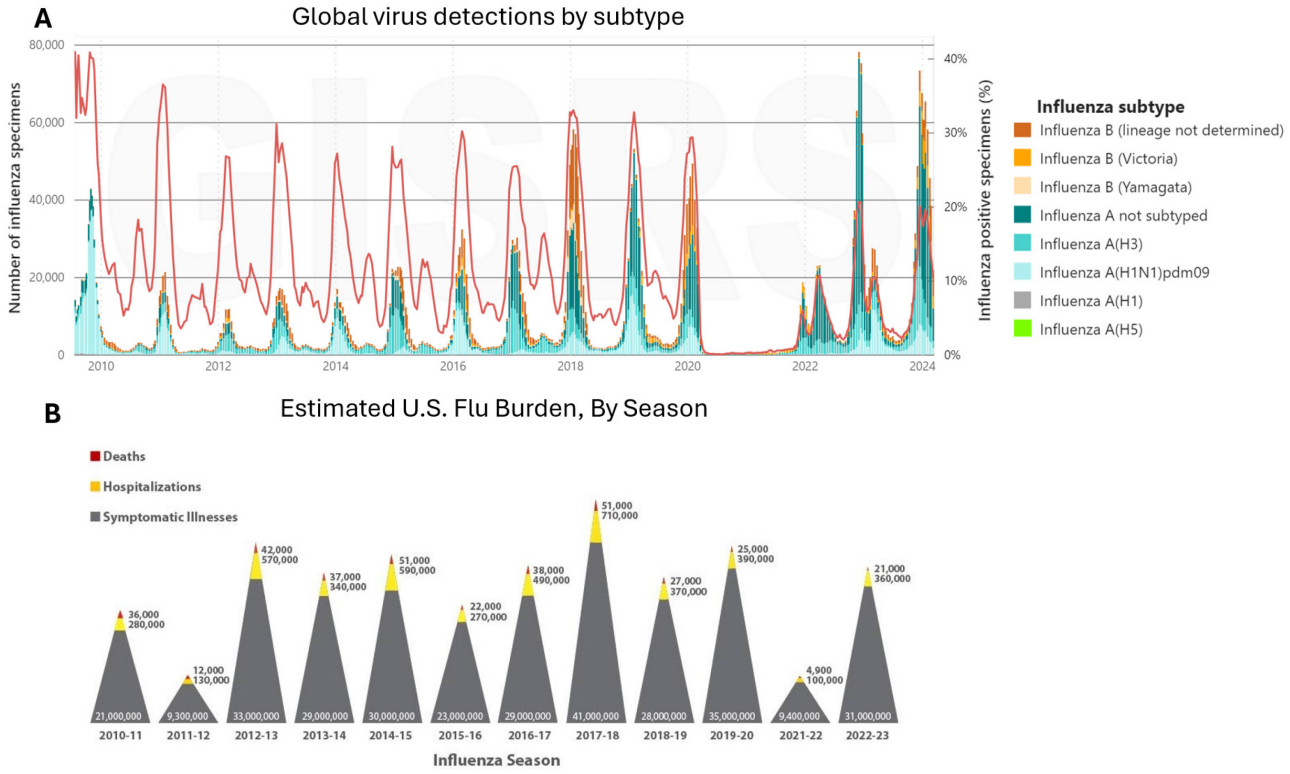


Figure 1.1: Influenza circulation since 2010

A) Global influenza virus detections by subtype, based on data reported to FluNet and curated by the World Health Organization Global Influenza Surveillance and Response System. Adapted from FluNet (201). B) Estimated burden of influenza in the United States by season. Adapted from the United States Centers for Disease Control and Prevention (CDC) (202).

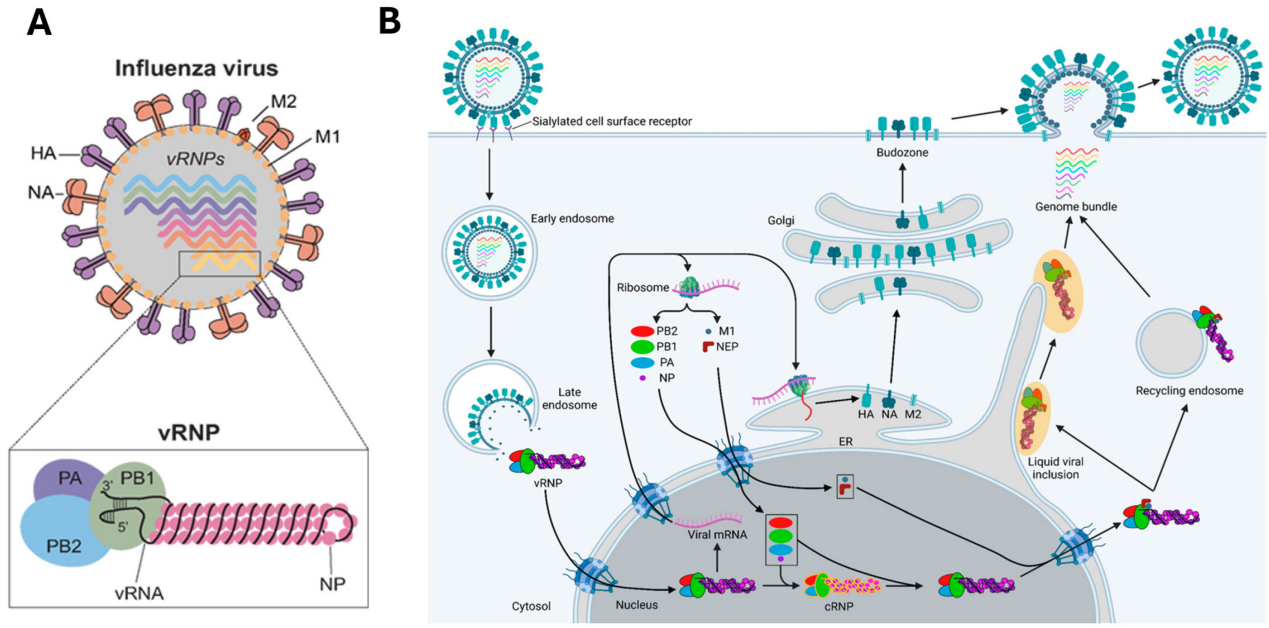


Figure 1.2: Influenza virology

A) Structure and organization of influenza virion. Adapted from (48). B) Influenza life cycle. Adapted from (51).

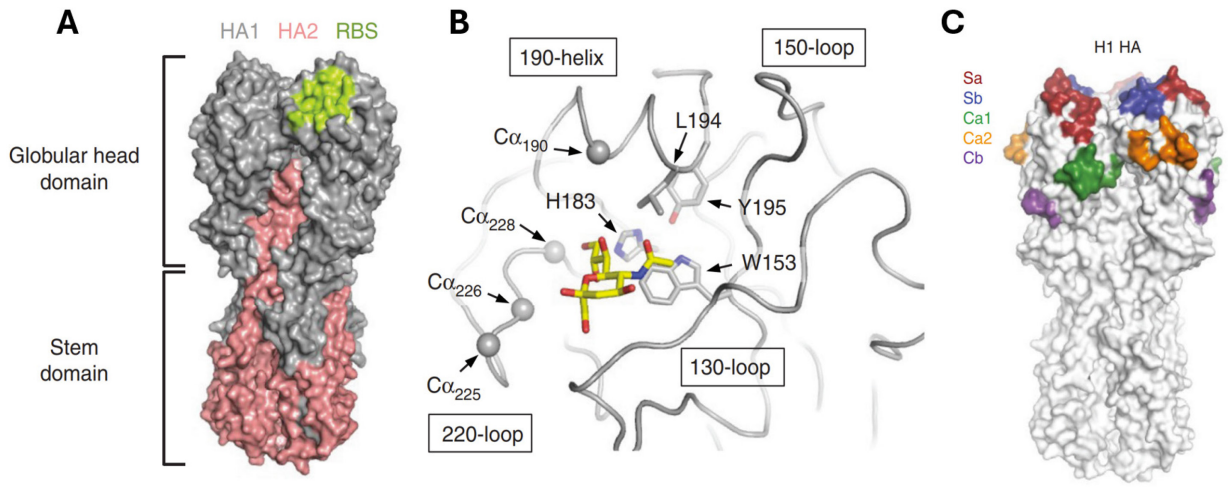


Figure 1.3: Influenza hemagglutinin structure and key regions

A) domains and receptor binding site of hemagglutinin. B) Detailed view of hemagglutinin receptor binding site. C) Map of canonical hemagglutinin H1 antigenic sites. There is significant overlap between the receptor binding site and antigenic sites. Adapted from (86).

CHAPTER 2: 2019-20 H1N1 clade A5a.1 viruses have better *in vitro* fitness compared with the co-circulating A5a.2 clade

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Abstract

Surveillance for emerging human influenza virus clades is important for identifying changes in viral fitness and assessing antigenic similarity to vaccine strains. While fitness and antigenic structure are both important aspects of virus success, they are distinct characteristics and do not always change in a complementary manner. The 2019-20 Northern Hemisphere influenza season saw the emergence of two H1N1 clades: A5a.1 and A5a.2. While several studies indicated that A5a.2 showed similar or even increased antigenic drift compared with A5a.1, the A5a.1 clade was still the predominant circulating clade that season. Clinical isolates of representative viruses from these clades were collected in Baltimore, Maryland during the 2019-20 season and multiple assays were performed to compare both antigenic drift and viral fitness between clades. Neutralization assays performed on serum from healthcare workers pre- and post-vaccination during the 2019-20 season show a comparable drop in neutralizing titers against both A5a.1 and A5a.2 viruses compared with the vaccine strain, indicating that A5a.1 did not have antigenic advantages over A5a.2 that would explain its predominance in this population. Plaque assays were performed to investigate fitness differences, and the A5a.2 virus produced significantly smaller plaques compared with viruses from A5a.1 or the parental A5a clade. To assess viral replication, low MOI growth curves were performed on both MDCK-SIAT and primary differentiated human nasal epithelial cell cultures. In both cell cultures, A5a.2 yielded significantly reduced viral titers at multiple timepoints post-infection compared with A5a.1 or A5a. Receptor binding was then investigated through glycan array experiments which showed a reduction in receptor binding diversity for A5a.2, with fewer glycans

bound and a higher percentage of total binding attributable to the top three highest bound glycans. Together these data indicate that the A5a.2 clade had a reduction in viral fitness, including reductions in receptor binding, that may have contributed to the limited prevalence observed after emergence.

Introduction

Seasonal influenza is a persistent contributor to global morbidity and mortality. While the burden of disease can vary significantly from year to year, overall influenza is responsible for millions of infections and hundreds of thousands of deaths annually (2, 204). There are several influenza virus strains and subtypes that vary in prevalence in different seasons. As these viruses circulate they accrue mutations, and some mutations contribute to antigenic drift and result in the need for regular updates to influenza vaccine formulations (141, 205). Mutations in the influenza virus can also result in changes to various aspects of viral fitness, including replication kinetics (131, 206), receptor binding (37, 92, 111-113), and viral budding (207). While antigenic structure and viral fitness are both important components of viral evolution, they are distinct phenomena; for example, a mutation mediating escape from preexisting immunity could simultaneously confer a detrimental effect on fitness. Studying how viruses balance these characteristics is crucial to improving our understanding of viral evolution.

The 2019-20 Northern Hemisphere (NH) influenza season was the third H1N1-predominant season since the 2009 H1N1pdm viruses emerged (208). At the start of

the season the 6B.1A.5a (A5a) clade constituted the majority of circulating H1N1, but it quickly dropped in prevalence as the A5a.1 and A5a.2 subclades emerged and began to cocirculate (150) (Figure 2.1A). These clades differ by several amino acids on the viral hemagglutinin protein; A5a.1 contains D187A and Q189E while A5a.2 is defined by K130N, N156K, L161I, and V250A (Figure 2.1B). The clade-defining mutations for both subclades include mutations in canonical antigenic sites (169), suggesting that they may have contributed to antigenic drift. Before its appearance in the 2019-20 season N156K had also emerged previously in regional circulation (209, 210), and had already been described as a potentially important mutation that changed antigenic structure in studies involving both ferret and human serum (209, 211). Both subclades also include mutations associated with the receptor binding site (RBS) of hemagglutinin, including the 130 loop (K130N) and the 190 helix (D187A, Q189E). Changes to the RBS can alter viral fitness by modifying the strength and diversity of receptor binding (92, 111). The N156K mutation, while not on the canonical RBS, has also been associated with changes in receptor binding (211).

The H1N1 vaccine component for 2019-20 NH season was A/Brisbane/2/2018 which belongs to clade 6B.1A.1 (150, 152), and a variety of studies have been performed to analyze the impact of the additional mutations on antigenic drift. Experiments using ferret serum post-infection with A5a or A5a.1 viruses showed a significant reduction in antibody recognition of the A5a.2 clade when compared with A5a or A5a.1, and ferret serum raised against A5a.2 viruses showed poor recognition of A5a.1 clade viruses (153, 212, 213). Ferret antibodies have been noted to preferentially recognize antigenic site Sa and may therefore disproportionately respond to the A5a.2

mutations (214); further antigenic characterization by the WHO using human serum post-vaccination with several prior vaccine strains showed a reduction in antibody recognition against both A5a.1 and A5a.2 viruses compared with A5a (153). Vaccine effectiveness studies for the 2019-20 season, however, report a reduction in effectiveness against infection and hospitalization with A5a.2 viruses compared with A5a.1 and A5a (159, 160, 215). Taken together, these data indicate that A5a.2 may have increased antigenic drift that allowed it to escape existing immunity.

Despite the potential antigenic advantages of the A5a.2 clade, A5a.1 predominated globally over both A5a.2 and A5a after its emergence and constituted over 90% of total circulating H1N1 by 2021 (150) (though it is noted that global circulation of influenza was greatly reduced during the 2020-2021 Northern Hemisphere influenza season due to the continuing SARS-CoV-2 pandemic). This suggests that factors of A5a.2 other than escape from preexisting immunity may have contributed to its lack of circulation. To investigate the phenotypic consequences of the clade-defining mutations of A5a.1 and A5a.2, representative viruses were isolated from influenza-positive patients in Baltimore, Maryland during the 2019-20 NH influenza season. Growth curves, plaque assays, glycan arrays, and neutralization assays were performed, and A5a.2 was shown to have reduced replication, plaque formation, and receptor binding diversity compared with both A5a.1 and A5a. These data indicate that A5a.2 suffered a reduction in fitness that may have prevented it from becoming the predominant H1N1 clade in the 2019-20 Northern Hemisphere season.

Results

A5a.1 and A5a.2 clade H1N1 viruses cocirculated in the 2019-20 influenza season in the Northern Hemisphere, with clade A5a.1 predominating (Figure 2.1A). A5a.1 and A5a.2 are distinguished by several distinct clade-defining mutations on the hemagglutinin protein (Figure 2.1B). A5a.1 contains D187A and Q189E mutations which occupy the 190 helix of the receptor binding site and antigenic site Sb. The A5a.2 clade is defined by K130N, N156K, L161I, and V250A. K130N is located on the 130-loop of the receptor binding site, while N156K and L161I occupy antigenic site Sa (Figure 2.1B). A5a.2 also contains E179D on the HA2 subunit of hemagglutinin. To characterize these clades, representative viruses were chosen from clades A5a.1, A5a.2, and the ancestral A5a clade. These viruses were collected from influenza-positive patients in Baltimore, Maryland during the 2018-19 and 2019-20 influenza seasons. For A5a.2 the virus A/Baltimore/R0675/2019 was chosen. For A5a.1 the viruses chosen were A/Baltimore/R0686/2019 and A/Baltimore/R0688/2019, with A/Baltimore/R0688/2019 containing the T72N mutation on the neuraminidase protein which resulted in an additional putative glycosylation site. A/Baltimore/R0496/2018 was chosen to represent the A5a clade. Table 2.1 depicts the complete set of amino acid differences between these viruses, with A/Baltimore/R0496/2018 as a reference.

Vaccinations in the 2019-20 NH influenza season were shown to offer reduced protection against infection and hospitalization with A5a.2 compared with A5a.1 (159, 160, 215). Studies using ferret serum indicated that the N156K mutation resulted in antigenic drift and reduced recognition by antibodies raised by infection with the 2019-20 NH vaccine strain (153, 212, 213), and experiments with human serum post-

vaccination indicate a reduction in neutralizing titers against both A5a.1 and A5a.2 (153). Experiments using human serum are an important component of antigenic characterization, as ferret antibodies have been shown to preferentially recognize antigenic site Sa compared with human antibodies (214). To assess the antigenic consequences of clade-defining mutations on viral recognition by serum antibodies, neutralization assays were performed using serum from healthcare workers before and after receiving the 2019-20 NH influenza vaccine (Figure 2.2). Representative viruses were used from both the A5a.2 clade (A/Baltimore/R0675/2019) and the A5a.1 clade (A/Baltimore/R0688/2019). For pre- and post-vaccination timepoints, serum showed reduced neutralizing antibody titers against both circulating clades compared with the vaccine strain (A/Brisbane/02/2018) (Figure 2.2A). No significant differences in neutralizing titers were seen between A5a.1 and A5a.2 at either timepoint. Fold change in neutralizing antibodies were similar between all three viruses, with a comparable percent of participants seroconverting post-vaccination (Figure 2.2B). These data suggest that the clade-defining mutations of A5a.1 and A5a.2 showed similar escape from pre- and post-vaccination serum in this population that has a high annual acceptance rate of influenza vaccines.

To assess differences in viral replication between these clades, growth curves and plaque assays were performed on the representative viruses from the A5a.1 and A5a.2 clades, as well as the ancestral A5a clade. Viral replication was assessed through low-multiplicity of infection (MOI) growth curves on Madin-Darby Canine Kidney sialyltransferase-1 (MDCK-SIAT) cells. These are Madin-Darby Canine Kidney (MDCK) cells stably transfected with human 2,6 sialtransferase, resulting in an overexpression of

the canonical human influenza receptor (2,6 sialic acid) and a reduction in 2,3 sialic acid expression (216). These growth curves revealed a reduction in A5a.2 viral titer at multiple timepoints compared with A5a and A5a.1 (Figure 2.3A) and significantly lower total virus production as measured by area under the curve (Figure 2.3C). Growth curves were also performed on human nasal epithelial cell (hNEC) cultures, a primary cell culture model that is physiologically similar to the upper respiratory tract that seasonal influenza preferentially infect (206, 217-222). Growth curves on hNEC cultures reflected MDCK-SIAT growth curves, with A5a.2 producing significantly reduced titers at multiple timepoints (Figure 2.3B) and reduced total virus production (Figure 2.3D). A5a and A5a.1 viruses both produced similar plaque sizes on MDCK cells, while A5a.2 formed significantly smaller plaques (Figure 2.3E and 2.3F). Together these data indicate that the A5a.2 clade has reduced viral fitness compared with the cocirculating A5a.1 clade and its A5a precursor, shown through a reduction in plaque size and replication on multiple cell types.

Both A5a.1 and A5a.2 contain clade-defining mutations on the RBS of the hemagglutinin protein. Receptor binding is a crucial component of viral fitness and a reduction in receptor binding diversity or avidity could result in the reduced plaque sizes and replication seen for A5a.2. To assess whether these mutations resulted in phenotypic changes, receptor binding was assessed using the Consortium for Functional Glycomics (CFG) glycan microarray (version 5.5) as previously described (92, 93, 223). See Table 2.3 for a complete list of glycans included in the final graphs. For all clades, viruses predominantly bound to glycans with alpha 2,6 linkages of the sialic acid with the penultimate sugar (Figure 2.4), which is the preferential linkage

recognized by human influenza viruses (96, 224). To investigate differences in receptor binding diversity, percent of total binding was calculated for each glycan that bound with at least 5% of the relative fluorescence units (RFUs) of the highest bound glycan (Table 2.2). Percent of total binding is frequently used in the analysis of virus/glycan interactions (93, 225-229) and is recommended for the interpretation of CFG microarray data (229). Cutoffs range from 1% (226) to 10% (225), with 4% and 5% being used in similar analyses(93, 229). 5% was chosen here to exclude background fluorescence while still including lower-strength virus/receptor interactions which have been shown to contribute to virus entry (108).

The three highest bound glycans constituted over 35% of total binding for the A5a.2 virus, while the top three glycans account for 21% or less of total binding for A5a.1 and A5a viruses. Additionally, A5a and A5a.1 viruses bound to 32 or more glycans with at least 5% of the RFUs of the highest bound glycan, while A5a.2 only bound 22 (Table 2.2). Taken together these data indicate that viruses from the A5a.2 clade have reduced receptor binding diversity compared with the ancestral A5a and co-circulating A5a.1 clades. Both A5a (Figure 2.5) and A5a.1 (Figure 2.6) also demonstrated clade-specific binding to glycans not bound by the other clades; A5a bound to several branched alpha 2,3 SA glycans as well as N-Glycolylneuraminic acid (Neu5Gc) glycans not expressed by human cells (Figure 2.5), while A5a.1 bound to several single chain 2,6 SA glycans and one single chain 2,3 SA glycan (Figure 2.6). These data confirm that mutations in the receptor binding site can result in viral clades with distinct receptor binding profiles, and suggest that differences in receptor binding may have contributed to the fitness differences observed for A5a.2.

Discussion

The accumulation of mutations in circulating seasonal influenza results in antigenic drift and the need for annual vaccinations, as well as changes to viral fitness. Mutations in individual influenza proteins have been shown to affect plaque size (230), viral replication (131, 206, 219, 230), and receptor binding preferences and diversity (37, 92, 111-113). Importantly, there is overlap between functional and antigenic regions of both the neuraminidase (NA) and hemagglutinin (HA) proteins (37, 131, 169, 231), and individual mutations have the potential to alter both antigenic structure and fitness simultaneously (131, 173, 176, 232, 233). Antigenic differences in circulating clades have been associated with changes to viral receptor binding profiles (234), and egg-adaptation mutations that changed receptor preferences in a live attenuated influenza virus also contributed to antigenic drift and reduced replication in human cells (230, 235).

The overlap between antigenic and functional regions may result in tradeoffs between novel mutations that improve one trait at the expense of the other. An example found in H3N2 is the emergence of a glycosylation site on the NA protein, which led to antigenic drift at the expense of viral replication and NA enzymatic activity (131, 236). This mutation increased in prevalence after its emergence, suggesting that the antigenic advantages may have offset any detriments to fitness (219). Here we investigate a potential tradeoff between antigenic structure and fitness that led the A5a.1 clade of H1N1 to predominant in the 2019-20 Northern Hemisphere influenza season despite the apparent antigenic advantages of the cocirculating A5a.2 clade. We show that A5a.2

had deficits in plaque size, replication, and receptor binding compared to A5a and A5a.1 that may have contributed to its limited prevalence in the 2019-20 season.

A5a.1 and A5a.2 are distinguished by several important mutations to the viral hemagglutinin segment. A5a.1 is defined by D187A and Q189E, while A5a.2 viruses have the mutations K130N, N156K, L161I, and V250A as well as the HA2 subunit mutation E179D. The N156K mutation of A5a.2 has emerged independently in earlier seasons and has been shown to cause antigenic drift in studies using both ferret and human serum (209, 211), and previous antigenic characterization of the 2019-20 NH clades has shown that A5a.2 is drifted from both A5a.1 and the 2019-20 NH vaccine strain in ferret serum and vaccine effectiveness studies (153, 159, 160, 212, 213, 215). Experiments with human serum show a similar degree of antigenic drift in both A5a.1 and A5a.2 compared with preceding clades (153).

To assess antigenic differences between the 2019-20 NH H1N1 clades, neutralization assays were performed on representative A5a.1 and A5a.2 viruses using serum from healthcare workers from the 2019-20 NH influenza season. These experiments revealed a similar reduction in neutralization of both A5a.1 and A5a.2 viruses compared with the vaccine strain (Figure 2.2), both pre- and post-vaccination. Fold-increase in neutralizing titers after vaccination were similar for all three viruses. It is important to note that healthcare workers represent a highly vaccinated population, with mandatory vaccination leading to over 90% vaccine coverage (237). Consecutive yearly vaccination has been shown to increase pre-vaccination antibody titers (238), and likely contributed to the high pre-vaccination neutralizing antibody levels observed here. High pre-vaccination antibody titers have previously been inversely correlated with

seroconversion after vaccination (239, 240), and it is possible that the increased antigenic drift of A5a.2 reported elsewhere may have been masked in our study by high baseline antibody levels.

While antigenic comparisons of these clades have revealed similar or increased antigenic drift in A5a.2, the predominance of A5a.1 in the 2019-20 NH influenza season suggests a potential fitness advantage over A5a.2. Here we show that a representative A5a.2 virus produces significantly smaller plaques in MDCK cells compared with viruses from both A5a.1 and the ancestral A5a clade. Small viral plaque size can be caused by multiple factors (241), and may be influenced by many aspects of the viral life cycle including replication, receptor binding, or cell-to-cell spread. To assess differences in viral replication, growth curves were performed in MDCK-SIAT and human nasal epithelial cell (hNEC) cultures. MDCK-SIAT cells are an immortalized cell line that overexpress alpha 2,6 sialic acids, the canonical human influenza receptor (216), and hNEC cultures are a primary cell culture system that models the upper respiratory tract (242). hNEC cultures are grown at an air-liquid interface (ALI) and differentiate into multiple epithelial cell types including ciliated and mucus-producing cells (217, 242). While MDCK-SIAT cells are engineered to be easily infected with human influenza, hNEC cultures are considered to be more physiologically relevant and may reveal differences not seen in MDCK-SIAT growth curves (131, 218, 219). Both growth curves were performed at 33°C to reflect the temperature of the upper respiratory tract. In both MDCK-SIAT and hNEC growth curves A5a.2 consistently demonstrated a significant reduction in replication compared to A5a.1 or A5a (Figure 2.3). This reduced replication could have contributed to the small plaque sizes of A5a.2.

To further explore the potential mechanisms of reduced fitness in A5a.2, glycan arrays were performed on the representative viruses from the three clades. These experiments measure viral binding to a variety of polysaccharides, and can reveal differences in both receptor preferences and binding diversity. They can also provide insight into host specificity; the canonical influenza receptors are glycans with a sialic acid (SA) as their terminal sugar, and different SA linkages to the penultimate sugar are differentially preferred by avian and human influenza viruses. Human influenza preferentially binds to alpha 2,6 SA linkages while avian influenza recognizes alpha 2,3 SA (96, 224). Glycan arrays on the representative viruses confirm a preference for alpha 2,6 SA glycans for all viruses tested, but there was considerable difference in binding diversity between clades. A5a.1 and A5a viruses bound to over 45% more glycans than A5a.2 with at least 5% of the relative fluorescence units of the highest bound glycan (Table 2.2). Additionally, the top three bound glycans represent over 35% of total binding for A5a.2, while the top three only constituted 21% or less of total binding for A5a.1 and A5a viruses.

Differences in receptor binding have previously been associated with reduced replication kinetics (92) and virus propagation (111), though this correlation does not appear in every context (93, 106, 225, 243). Diversity of receptor binding and low-affinity receptor interactions have also been shown to be important for viral entry into cells (108), and it is possible that the reduction in binding diversity seen in the A5a.2 clade could impact its fitness through impaired cell binding and entry. Additionally, viruses from both A5a and A5a.1 clades each bound to unique glycans not bound by the other clades (Figures 2.5 and 2.6). This confirms that different clades can have unique

binding profiles that distinguish them from each other, and further demonstrates the reduced binding diversity of A5a.2 which had no uniquely bound glycans.

Interestingly, as A5a.2 has continued to circulate it quickly gained a series of mutations on the hemagglutinin protein, including K54Q, A186T, Q189E, E224A, R259K, and K308R, and has subsequently become the predominant circulating clade of H1N1 (150). Several of these mutations occupy both antigenic and receptor binding sites, suggesting that they may have served some compensatory function that allowed the virus to improve its fitness and/or antigenic structure and outcompete A5a.1. Further investigation will be needed to identify the impact of these more recent mutations on the viral phenotype. It is also important to note that this study characterized clinical isolates of viruses representing the various clades, and differences between these viruses are not limited to the hemagglutinin (HA) protein (Table 2.1). Additional research is needed to identify the role of individual clade-defining HA mutations on the overall reduction in fitness of A5a.2. These findings nevertheless highlight the deficiencies in receptor binding and viral fitness that may have contributed to the limited spread of the A5a.2 clade in the 2019-20 Northern Hemisphere influenza season.

Methods

Ethics statement and human subjects

Serologic samples for this study were obtained from healthcare workers (HCWs) recruited from the Johns Hopkins Center for Excellence in Influenza Research and Surveillance (JH-CEIRS) during the annual Johns Hopkins Hospital (JHH) employee

influenza vaccination campaign in the Fall of 2019. Pre- and post-vaccination (~28 day) human serum were collected from subjects, who provided written informed consent prior to participation. The JHU School of Medicine Institutional Review Board approved this study, IRB00288258. Virus was isolated for this study from deidentified influenza A virus H1N1 positive samples, collected from patients who provided written informed consent during the 2019-20 influenza season at the Johns Hopkins Hospital, under the Johns Hopkins University (JHU) School of Medicine Institutional Review Board approved protocol, IRB00091667.

Cell Cultures

Madin-Darby canine kidney (MDCK) cells (provided by Dr. Robert A. Lamb) and MDCK-SIAT cells (provided by Dr. Scott Hensley) were maintained in complete medium (CM) consisting of Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum, 100 units/ml penicillin/streptomycin (Life Technologies) and 2mM Glutamax (Gibco) at 37°C and 5% CO_2. Human nasal epithelial cells (PromoCell) were seeded on 24-well Corning transwell plates with PneumaCult Ex-Plus media on apical and basolateral sides. After cells reached confluence (approximately 10 days, determined by a trans-epithelial electrical resistance reading of >300 ohms), media was switched to PneumaCult ALI media and the apical surface was left at an air-liquid interface (ALI). hNEC cultures were considered fully differentiated when mucus and beating cilia were visible, approximately three weeks after transition to ALI.

Viral Isolation and Sequencing

Nasopharyngeal swabs or nasal wash from individuals who were influenza A positive during the 2019-20 Northern Hemisphere influenza season were used for virus isolation on primary cells. The apical side of hNEC wells were washed twice with 300ul of phosphate buffered saline (PBS) and 100ul of sample was added to the cells and incubated for two hours. The sample was then aspirated and cells were washed twice with 300ul of PBS. At three, five, and seven days post-infection 300ul of hNEC infection media (DMEM supplemented with .3% BSA (Sigma), 100 units/ml pen/strep (Life Technologies), 2mM Glutamax) was added to the well and incubated for ten minutes. TCID50 was performed on collected media and stocks were made from the collected media when virus was detected at concentrations greater than 10^4 TCID50/mL.

To generate viral stocks, T75 flasks (Corning) were seeded with MDCK-SIAT cells and grown to confluence in complete media. Viral isolates from the previous step were diluted to a multiplicity of infection (MOI) of 0.001 in infection media (hNEC infection media plus 5μg/ml N-acetyl trypsin). Cells were washed twice with PBS plus 100mg/L each of anhydrous calcium chloride and magnesium chloride hexahydrate (PBS +/+) and inoculum was added to the flasks and incubated at 33°C, rocking every 15 minutes. Inoculum was then aspirated, and 13mL of infection media was added to the flask. After flasks showed 75% cell death (~3 days post infection), media was collected and centrifuged at 500g for 10 minutes to remove cell debris. Stocks were aliquoted and stored at -80 ^{o}C.

Viral RNA was extracted using the QIAamp viral RNA mini extraction kit, and Illumina RNA Prep with Enrichment(L) Tagmentation with Respiratory Virus Oligo Panel

v2 (20044311) was used for library preparation. Quality was checked using the Qubit and Agilent Bioanalyzer, and samples were sequenced using a MiSeq Illumina sequencer (v3 2X 300bp). Consensus sequences were generated using the DRAGEN RNA Pathogen Detection pipeline using custom .bed files and FASTA files for IAV. Sequences are accessible through GISAID with the following IDs: EPI_ISL_17614868 for A/Baltimore/R0496/2018, EPI_ISL_17617226 for A/Baltimore/R0675/2019, EPI_ISL_17614713 for A/Baltimore/R0686/2019, and EPI_ISL_17617227 for A/Baltimore/R0688/2019.

Tissue Culture Infections Dose 50 (TCID50)

96-well plates were seeded with MDCK-SIAT cells. After reaching confluence, plates were washed twice with 100ul per well of PBS +/+ and 180ul of infection media (DMEM supplemented with .3% BSA (Sigma), 100 units/ml pen/strep (Life Technologies), 2mM Glutamax, and 5µg/ml N-acetyl trypsin) was added to each well. Ten-fold serial dilutions were performed on viruses and 20ul of the dilution was added to its corresponding well in the cell plate. Each virus dilution was added to cell plates in sextuplicate, plates were incubated for six days at 33ºC before being fixed with 4% paraformaldehyde for >3 hours and stained overnight with naphthol blue-black. 50% tissue culture infectious dose was calculated as previously described(244, 245).

Plaque Assay

MDCK cells were grown in complete medium to confluence in 6-well plates. After reaching confluence, media was removed and cells were washed twice with PBS +/+. Ten-fold serial dilutions of viruses were made in infection media, and 250uL of the virus dilutions were added to the wells. Plates were incubated for one hour at 33ºC and rocked every 15 minutes to ensure even distribution of inoculum. After one hour, the virus inoculum was removed and phenol-red free DMEM supplemented with .3% BSA (Sigma), 100U/ml pen/strep (Life Technologies), 2mM Glutamax (Gibco), 5mM HEPES buffer (Gibco) 5µg/ml N-acetyl trypsin (Sigma) and 1% agarose was added. Cells were incubated at 33ºC for two days and then fixed with 4% paraformaldehyde overnight. After fixing, the agarose overlay was removed and cells were stained with naphthol-blue black. Plaque area was analyzed in ImageJ.

Viral Growth Curves

For MDCK-SIAT growth curves, cells were seeded in a 24-well plate. After cell plates reached confluence, they were washed twice with PBS +/+ (described above). Virus inoculum was made by diluting virus stock to an MOI of 0.001 in infection media (described above). Inoculum was added to the cell wells and incubated at 33ºC for one hour with plates being rocked every 15 minutes. After one hour, inoculum was removed and plates were washed twice with PBS +/+ and 500ul of infection media was added to each well. For each timepoint, infection media was collected from the wells and replaced with 500ul of fresh infection media. Viral titer in collected media was determined by TCID50 (described above).

hNEC growth curves were performed on fully differentiated hNECs grown at an air-liquid interface. Virus inoculum was made by diluting virus stocks to an MOI of 0.01 in hNEC infection media. Wells were washed three times with hNEC infection media and inoculum was added to the wells. Inoculum was incubated on the cells for two hours, after which it was removed and cells were washed three times with PBS and left at an air-liquid interface. For each timepoint, 100ul of hNEC infection media was added to the wells and incubated at 33°C for 10 minutes. Media was then collected and cells were left at ALI. Virus concentration in collected media was determined through TCID50.

Partial virus Purification and Labeling with Alexa Fluor 488

T150 flasks were seeded with MDCK-SIAT cells and grown to confluence in complete media. Virus stocks were diluted to 0.001 MOI in infection media. Cells were washed twice with PBS +/+ and inoculum was added to the flasks and incubated at 33°C, rocking every 15 minutes. Inoculum was then aspirated, and 18mL of infection media was added to the flasks. After cells showed 75% death (~3 days post infection), media was collected and centrifuged at 500g for 10 minutes to remove cell debris.

5mL of 20% sucrose in PBS was added to Ultra-Clear ultracentrifuge tubes (Beckman Coulter), and the remaining volume of the tube was filled with the collected infection media from the previous step. Samples were centrifuged at 24,800 RPM for 1 hour, after which the liquid was aspirated and the pellet was resuspended in 100ul of PBS +/+. TCID50 was performed on purified viruses to confirm activity and concentration.

1mg of Alexa Fluor 488 (Invitrogen A20000) was resuspended in 1mL of diH_2O. 25ul of Alexa Fluor solution was added to 200ul of purified virus and 20ul of 1M NaHCO3 (pH 9.0), mixed by pipetting, and incubated in the dark at room temperature (23 ^oC) for 1 hour. Solution was added to Slide-A-Lyzer™ 7K MWCO MINI Dialysis Devices (Thermo 69560), devices were loaded onto dialysis floats (Thermo 69588) and dialyzed while stirring at 4 ^oC in 1L PBS. After 1 hour, PBS was replaced with 1L of fresh pre-chilled PBS and the dialysis was continued overnight. The next morning, the PBS was again replaced and the dialysis was continued for another hour. Dialyzed labeled virus was collected and stored at -80 ^oC, and TCID50 was performed to confirm virus activity and concentration. Titer of labeled viruses were: 2.32E+10 TCID50/mL (A/Baltimore/R0496/2018), 5.00E+09 TCID50/mL (A/Baltimore/R0675/2019), 2.32E+10 (A/Baltimore/R0686/2019), 7.34E+10 TCID50/mL (A/Baltimore/R0688/2019).

Glycan Array

The microarray chosen was version 5.5 of the CFG glycan microarray, which includes 562 glycans in replicates of 6. Glycan microarray slides were submerged in 100 milliliters of PBS wash buffer (phosphate-buffered saline containing 0.005% Tween-20) for five minutes to re-hydrate. Samples were diluted 1:100 and in PBS binding buffer (PBS with 0.05% Tween-20 and 1% bovine serum albumin), and 70 microliters of diluted sample were loaded onto the slide. A cover slip was placed over the slide and incubated at room temperature for one hour in darkness, after which the cover slip was removed and the slide was washed four times in 100 milliliters of PBS wash buffer, then PBS, then

deionized water. Slides were then dried before being read in a GenePix microarray scanner.

For each set of six replicates, only the median four values were used in analysis. A graph of the raw data was visually analyzed to confirm selective binding of terminally sialylated glycans, after which data was filtered to only include these glycans for a total of 139 glycans (see Table 2.3) and organized by structure using the Glycan Array Dashboard (GLAD)(246). Glycans were considered in final analysis if virus bound to them with at least 5% of the relative fluorescence units (RFUs) of the highest bound glycan; total binding was found by summing the RFUs of all of these glycans, and percent of total binding was found by dividing the RFUs of individual glycans by total binding. Structures of glycans uniquely bound by one clade were identified using GLAD.

Serum Neutralization Assay

Pre- and post-vaccination human serum obtained through the Johns Hopkins Center for Excellence in Influenza Research and Surveillance (JH-CEIRS) study (HHSN272201400007C) were used in this study. Lyophilized Receptor Destroying Enzyme II (RDE, Hardy Diagnostics) was dissolved into 20 mL of saline (0.9% NaCl in H_2O) and a 1:3 ratio of serum and RDE was incubated at 37°C overnight followed by RDE inactivation at 57°C for 35 minutes. MDCK cells were seeded on 96-well plates. When cells reached confluence plates were washed twice with 100ul per well of PBS +/+. Virus inoculum was prepared by diluting virus stock in infection media to a concentration of 1,000 TCID50/ml. RDE-treated serum was serially diluted twofold in 96-

well U bottom plates (Thermo) using infection media yielding 55ul per dilution, and 55ul of virus inoculum was added to each dilution. The serum/virus mixture was incubated at room temperature for one hour, and then 50ul of serum/virus mixture was transferred in duplicate onto the cell plates, yielding 25 TCID50 per well. After 24 hours, the serum/virus mixture was removed and plates were washed once with PBS +/+ before adding 100ul per well of infection media. Six days post-infection the plates were fixed with 4% paraformaldehyde for >3 hours and stained overnight with naphthol blue-black.

Statistical Analyses

All statistical analyses were performed in GraphPad Prism 9.1.0. Growth curves were analyzed using 2-way ANOVA with Tukey post-hoc test. Area under the curve (AUC) was calculated for each growth curve replicate and AUC values were compared using one-way ANOVA for differences in total virus production. Plaque assays were analyzed using Kruskal-Wallis ANOVA with Dunn's multiple comparison test. Serology was analyzed using either 2-way ANOVA or 1-way ANOVA, both with Tukey post-hoc test.

Availability of Data and Materials

The datasets used and/or analyzed during the current study are available through the Johns Hopkins Research Data Repository at https://doi.org/10.7281/T1/H1ZPLI.

Acknowledgements

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Figures

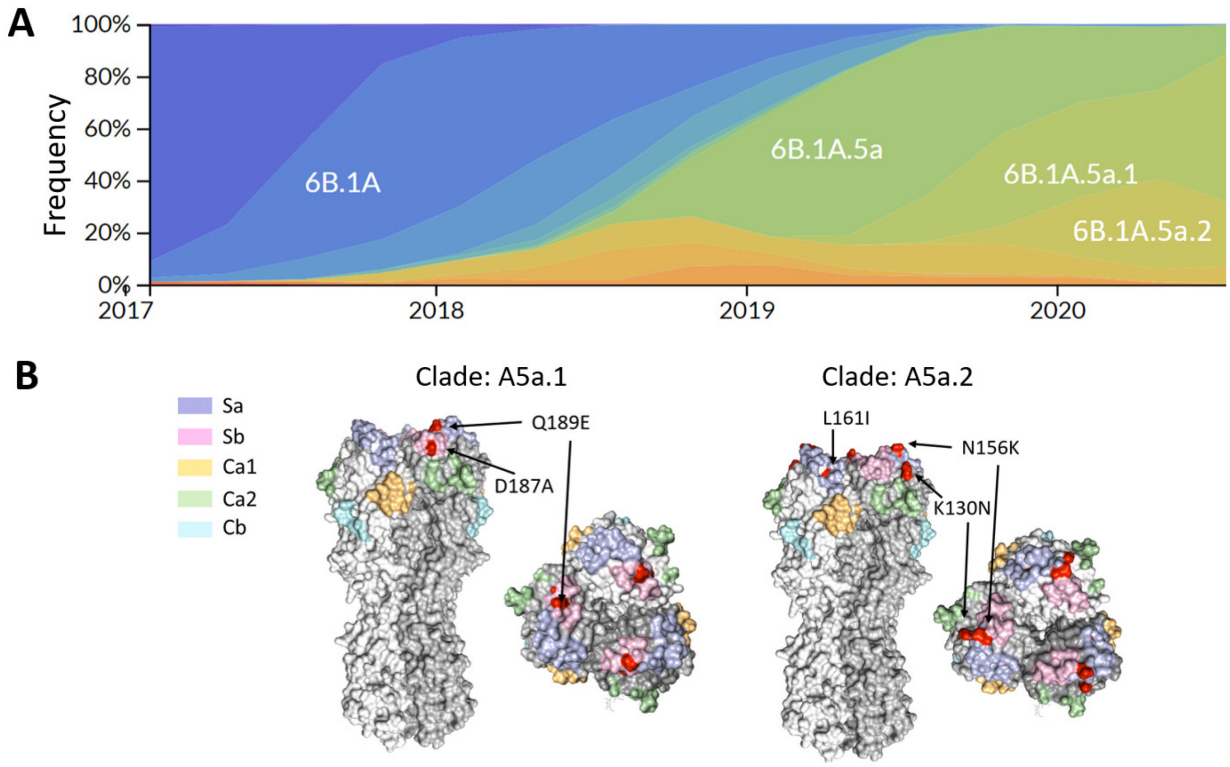


Figure 2.1: Emergence of the 6B.1A.5a.1 (A5a.1) and 6B.1A.5a.2 (A5a.2) clades of H1N1 and their clade-defining mutations

(a) Frequencies of H1N1 clades from January 2017-June 2020, generated through the NextStrain pipeline (150). (b) Side and top view of hemagglutinin trimers from A5a.1 and A5a.2 clades are displayed with antigenic sites shaded using *PyMOL* (PDB: 3LZG). Clade-defining amino acid mutations are shown in red (not shown are V250A in HA1 and E179D in HA2 for the A5a.2 clade).

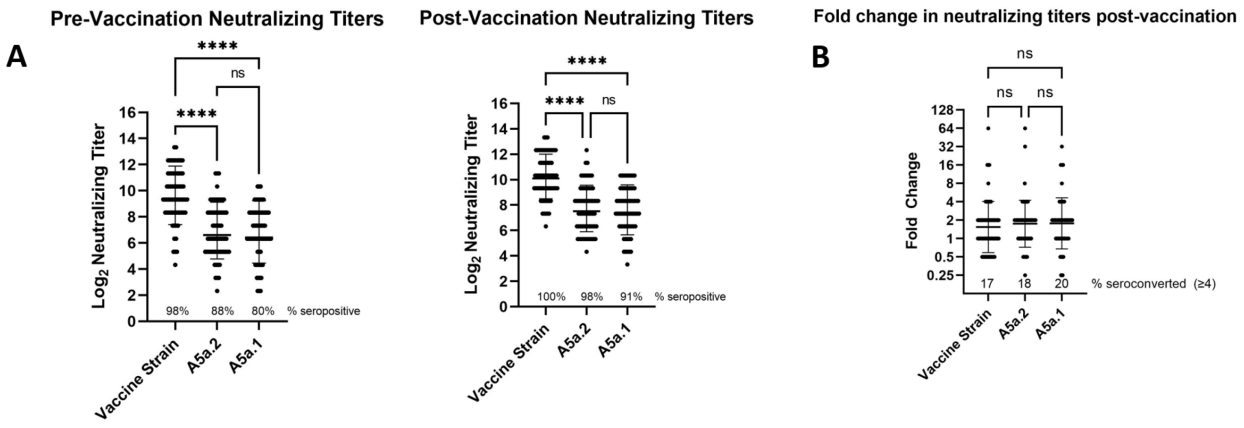


Figure 2.2: Comparison of neutralizing antibody titers in serum of healthcare workers pre- and post-vaccination

(a) Neutralizing titers of antibodies against the H1N1 vaccine strain (A/Brisbane/02/2018), A5a.2 (A/Baltimore/R0675/2019), and A5a.1 (A/Baltimore/R0688/2019). Neutralizing titers were significantly higher against the vaccine strain than the two circulating clades, and there were no significant differences between A5a.1 and A5a.2 pre- or post-vaccination. 1-way ANOVA with Tukey post-hoc, **** $p<0.0001$ (b) Fold-change of neutralizing titers post-vaccination show a similar seroconversion rate between all three viruses. Bars represent geometric mean, intervals are geometric standard deviation. One-way ANOVA with Tukey post-hoc. N=66

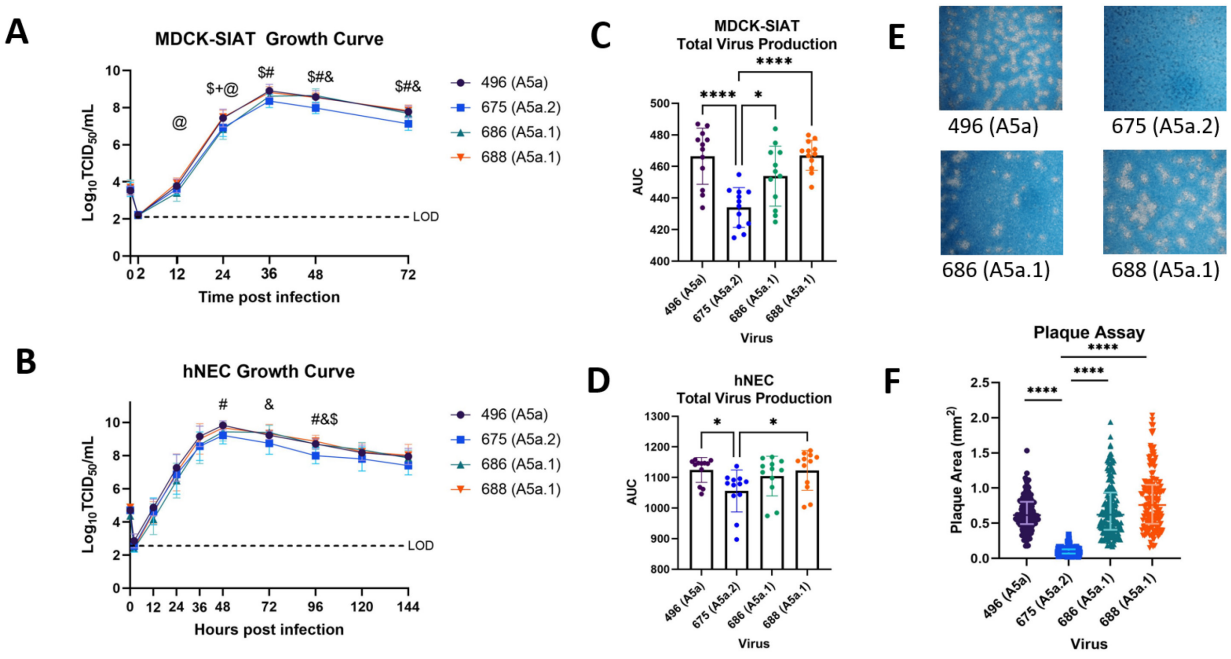


Figure 2.3: Characterization of Viral Fitness

Growth curves on MDCK-SIAT cells (MOI 0.001) show a reduction in growth (a) and total virus production (c) of A5a.2 viruses compared to A5a and A5a.1. Growth curves on human nasal epithelial cells (MOI 0.01) show a similar phenotype (b,d).1-way (AUC) or 2-way (growth curves) ANOVA with Tukey post hoc test. Comparisons shown if $p<0.05$: # 675 versus 496, & 675 versus 686, $ 675 versus 688, @ 686 versus 688, + 496 versus 686. Each graph shows three experiments with four replicates each. (e) representative images of plaque sizes. (f) Comparison of plaque areas reveal significantly reduced plaque sizes for the A5a.2 virus versus A5a and A5a.1 viruses. N=3, >50 plaques per virus. Kruskal-Wallis ANOVA with Dunn's multiple comparisons test, **** $p<0.0001$.

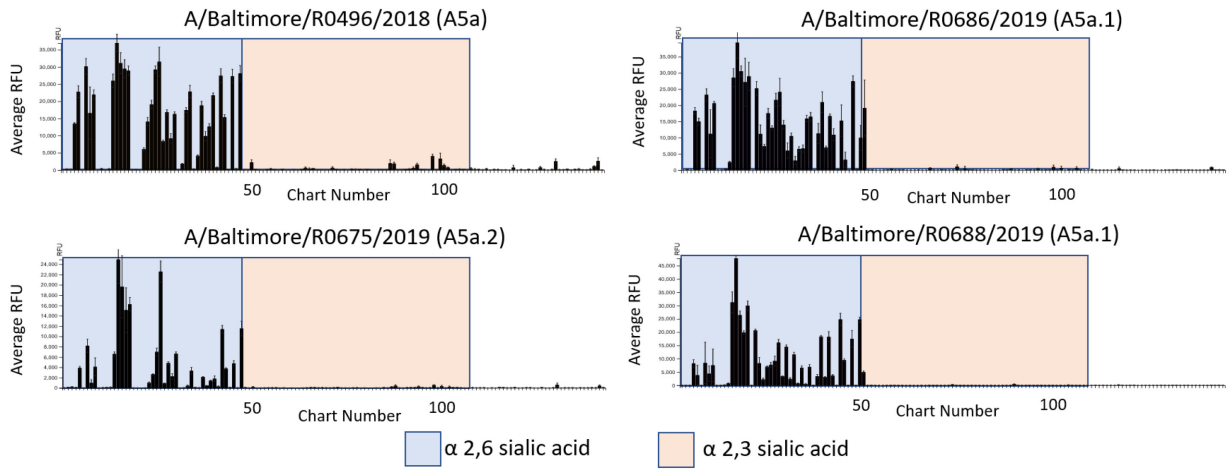


Figure 2.4: Consortium for Functional Glycomics (CFG version 5.5) glycan microarray profile of viral receptor binding of terminally sialylated glycans

Y axis depicts relative fluorescence units (RFUs), and X axis depicts individual glycans from the array. Unshaded regions contain sialylated glycans with linkages other than 2,3 or 2,6, and Neu5Gc glycans.

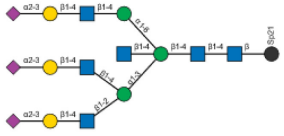
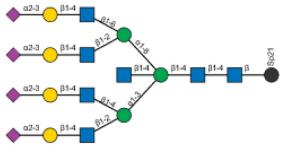
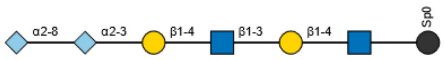
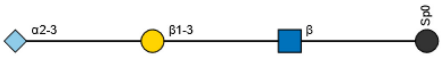
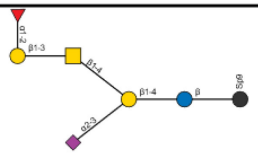
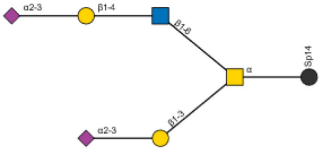
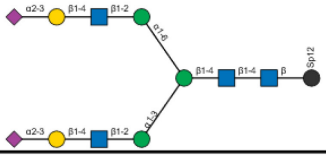
Glycan ID	Glycan Name	Glycan Structure	% of total binding 496 (A5a)
448	Neu5Aca2-3Galb1-4GlcNAcb1-4Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21		0.6
450	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21		0.5
536	Neu5Gca2-8Neu5Gca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAc-Sp0		0.4
274	Neu5Gca2-3Galb1-3GlcNAcb-Sp0		0.4
64	Fuca1-2Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp9		0.3
309	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14		0.3
315	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12		0.3

Figure 2.5: Glycans bound only by A5a virus

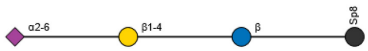
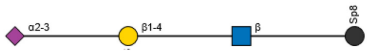
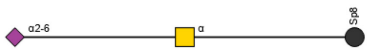
			% of total binding	
Glycan ID	Glycan Name	Glycan Structure	686 (A5a.1)	688 (A5a.1)
266	Neu5Aca2-6Galb1-4Glcb-Sp8		4.3	4.8
46	Neu5Aca2-3(6S)Galb1-4GlcNAcb-Sp8		3.3	1.1
267	Neu5Aca2-6Galb-Sp8		1.9	1.9
355	Neu5Aca2-6GlcNAcb1-4GlcNAc-Sp21		1.1	-
259	Neu5Aca2-6GalNAca-Sp8		0.4	-

Figure 2.6: Glycans bound only by A5a.1 viruses

	A/Baltimore/R0675/2019 (A5a.2)	A/Baltimore/R0686/2019 (A5a.1)	A/Baltimore/R0688/2019 (A5a.1)
HA1	K130N/N156K/L161I/V250A	R45K/T120I/D187A/Q189E	D187A/Q189E
HA2	E179D		
NA	M19T/Y66F/N222K	S52N/I223V	I40T/S52N/T72N
M1			
M2	H81Q	E70D/H81Q	E70D/H81Q
NP		A70S/V444I	A70S/V444I
NS1	D189G	D189G	I145V/D189G
NS2	I32V	I32V	I32V
PA	V63L/S225C/I505V	M336R/M374I	K213R/I465M/T528I
PA-X	V63L		S213G
PB1	M174V/I728V	I728V	M646L/I728V
PB1-F2			
PB2	Q183L/R630K	Q183L	Q183L

Table 2.1: Amino acid differences between the 2019-20 viruses chosen for characterization compared with the 2018-19 A5a virus (A/Baltimore/R0496/2018)

NP segment numbering was based on the established start codon for H3N2 and H1N1, instead of the recently emerged upstream start codon (220).

Glycan ID	Glycan Name	Percent of total binding for each glycan			
		496 (A5a)	675 (A5a.2)	686 (A5a.1)	688 (A5a.1)
261	Neu5Aca2-6Galb1-4(6S)GlcNAcb-Sp8	5.8	13.5	6.7	11.1
321	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	5.0	12.2	4.1	3.7
262	Neu5Aca2-6Galb1-4GlcNAcb-Sp8	4.9	10.6	5.2	6.2
55	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	4.8	4.4	4.0	2.0
263	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0	4.6	8.2	4.6	4.6
319	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-3GlcNAcb-Sp0	4.6	3.8	3.7	2.1
264	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0	4.6	8.8	4.9	7.0
560	Neu5Aca2-6Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14	4.4	6.2	1.7	5.8
468	Neu5Aca2-6Galb1-4 GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-3)GalNAca-Sp14	4.3	6.2	2.6	5.8
489	Neu5Aca2-6GalNAcb1-4(6S)GlcNAcb-Sp8	4.3	2.6	4.7	4.1
260	Neu5Aca2-6GalNAcb1-4GlcNAcb-Sp0	4.1	3.6	4.9	7.3
398	Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21	3.6	1.8	2.8	1.6
316	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	3.6	2.1	2.6	0.9
57	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp24	3.5	2.2	3.5	1.8
464	Neu5Aca2-6Galb1-4GlcNAcb1-6(Galb1-3GlcNAcb1-3)Galb1-4Glcb-Sp21	3.4	1.0	1.9	0.8
311	Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	3.0	1.4	2.2	1.8
452	Neu5Aca2-6Galb1-4GlcNAcb1-4Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-4(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	3.0	1.1	3.6	4.3
335	Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12	2.7	2.6	1.0	3.4
337	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3Manb1-4GlcNAcb1-4GlcNAc-Sp12	2.6	3.6	0.5	2.7
469	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24	2.4	2.0	0.5	2.2
454	Neu5Aca2-6Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-4(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	2.0	0.8	2.8	4.2
336	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6Manb1-4GlcNAcb1-4GlcNAc-Sp12	1.4	1.2	1.8	0.6
56	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Man-a1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	2.6		1.9	1.0
300	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	2.2		3.0	1.6
310	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	2.1		3.1	1.9
453	Neu5Aca2-6Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	1.6		1.2	0.7
334	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12	1.3		2.4	0.8
451	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	0.7		1.9	0.8
356	Neu5Aca2-6GlcNAcb1-4GlcNAcb1-4GlcNAc-Sp21	0.3		1.1	1.5
366	Neu5Aca2-6Galb1-4GlcNAcb1-3GalNAc-Sp14	2.8		2.7	
293	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	1.0		1.3	
448	Neu5Aca2-3Galb1-4GlcNAcb1-4Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	0.6			
450	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21	0.5			
536	Neu5Gca2-8Neu5Gca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAc-Sp0	0.4			
274	Neu5Gca2-3Galb1-3GlcNAcb-Sp0	0.4			
64	Fuca1-2Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp9	0.3			
309	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14	0.3			
315	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12	0.3			
266	Neu5Aca2-6Galb1-4Glcb-Sp8			4.3	4.8
46	Neu5Aca2-3(6S)Galb1-4GlcNAcb-Sp8			3.3	1.1
267	Neu5Aca2-6Galb-Sp8			1.9	1.9
355	Neu5Aca2-6GlcNAcb1-4GlcNAc-Sp21			1.1	
259	Neu5Aca2-6GalNAca-Sp8			0.4	
	Total Number of Glycans Bound	**38**	**22**	**36**	**32**

Table 2.2: Percent of total glycan binding

Included in this list are all glycans with ≥ 5% of the RFU value of the highest bound glycan for that virus. For each virus, percent of total binding is listed for each glycan,

and the total number of glycans bound for that virus is included at the bottom. Glycans highlighted in blue are bound only by the A5a clade virus, while glycans in green were bound only by viruses from the A5a.1 clade. Glycans highlighted in yellow were bound by every virus except for A5a.2.

Chart Number	Glycan ID	Glycan Structure
1	230	Neu5Aca2-6(Neu5Aca2-3)GalNAca-Sp8
2	238	Neu5Aca2-6(Neu5Aca2-3Galb1-3)GalNAca-Sp8
3	239	Neu5Aca2-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14
4	316	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12
5	480	Galb1-4(Fuca1-3)GlcNAcb1-6(Neu5Aca2-6(Neu5Aca2-3Galb1-3)GlcNAcb1-3)Galb1-4Glc-Sp21
6	55	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12
7	56	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Man-a1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21
8	57	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp24
9	124	Gala1-6Glcb-Sp8
10	125	Galb1-2Galb-Sp8
11	127	Galb1-3GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0
12	259	Neu5Aca2-6GalNAca-Sp8
13	260	Neu5Aca2-6GalNAcb1-4GlcNAcb-Sp0
14	261	Neu5Aca2-6Galb1-4(6S)GlcNAcb-Sp8
15	262	Neu5Aca2-6Galb1-4GlcNAcb-Sp8
16	263	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0
17	264	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0
18	265	Neu5Aca2-6Galb1-4Glcb-Sp0
19	266	Neu5Aca2-6Galb1-4Glcb-Sp8
20	267	Neu5Aca2-6Galb-Sp8
21	293	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12
22	300	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12
23	311	Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12
24	319	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-3GlcNAcb-Sp0
25	321	Neu5Aca2-6Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0
26	334	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12
27	335	Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp12
28	336	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6Manb1-4GlcNAcb1-4GlcNAc-Sp12
29	337	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3Manb1-4GlcNAcb1-4GlcNAc-Sp12
30	355	Neu5Aca2-6GlcNAcb1-4GlcNAc-Sp21
31	356	Neu5Aca2-6GlcNAcb1-4GlcNAcb1-4GlcNAc-Sp21
32	366	Neu5Aca2-6Galb1-4GlcNAcb1-3GalNAc-Sp14
33	398	Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21
34	444	Neu5Aca2-6Galb1-4GlcNAcb1-6(Fuca1-2Galb1-3GlcNAcb1-3)Galb1-4Glc-Sp21
35	451	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21

36	452	Neu5Aca2-6Galb1-4GlcNAcb1-4Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-4(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21
37	453	Neu5Aca2-6Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21
38	454	Neu5Aca2-6Galb1-4GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-6Galb1-4GlcNAcb1-4(Neu5Aca2-6Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21
39	464	Neu5Aca2-6Galb1-4GlcNAcb1-6(Galb1-3GlcNAcb1-3)Galb1-4Glcb-Sp21
40	467	Neu5Aca2-6Galb1-4GlcNAcb1-6GalNAca-Sp14
41	468	Neu5Aca2-6Galb1-4 GlcNAcb1-6(Neu5Aca2-6Galb1-4GlcNAcb1-3)GalNAca-Sp14
42	469	Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-6Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4(Fuca1-6)GlcNAcb-Sp24
43	474	Neu5Aca2-6Galb1-4GlcNAcb1-6(Fuca1-2Galb1-4(Fuca1-3)GlcNAcb1-3)Galb1-4Glc-Sp21
44	489	Neu5Aca2-6GalNAcb1-4(6S)GlcNAcb-Sp8
45	507	Neu5Aca2-6Galb1-4GlcNAcb1-2Man-Sp0
46	560	Neu5Aca2-6Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14
47	46	Neu5Aca2-3(6S)Galb1-4GlcNAcb-Sp8
48	63	Fuca1-2Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0
49	64	Fuca1-2Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp9
50	143	Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0
51	214	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0
52	219	Neu5Aca2-3Galb1-3GalNAca-Sp8
53	220	Neu5Aca2-3Galb1-3GalNAca-Sp14
54	221	GalNAcb1-4(Neu5Aca2-8Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0
55	222	GalNAcb1-4(Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0
56	223	Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3Galb1-4Glcb-Sp0
57	224	GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0
58	226	GalNAcb1-4(Neu5Aca2-3)Galb1-4GlcNAcb-Sp0
59	227	GalNAcb1-4(Neu5Aca2-3)Galb1-4GlcNAcb-Sp8
60	228	GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0
61	229	Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-3)Galb1-4Glcb-Sp0
62	231	Neu5Aca2-3GalNAca-Sp8
63	232	Neu5Aca2-3GalNAcb1-4GlcNAcb-Sp0
64	233	Neu5Aca2-3Galb1-3(6S)GlcNAc-Sp8
65	234	Neu5Aca2-3Galb1-3(Fuca1-4)GlcNAcb-Sp8
66	235	Neu5Aca2-3Galb1-3(Fuca1-4)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0
67	236	Neu5Aca2-3Galb1-4(Neu5Aca2-3Galb1-3)GlcNAcb-Sp8
68	237	Neu5Aca2-3Galb1-3(6S)GalNAca-Sp8
69	240	Neu5Aca2-3Galb-Sp8
70	241	Neu5Aca2-3Galb1-3GalNAcb1-3Gala1-4Galb1-4Glcb-Sp0

71	242	Neu5Aca2-3Galb1-3GlcNAcb1-3Galb1-4GlcNAcb-Sp0
72	244	Neu5Aca2-3Galb1-3GlcNAcb-Sp0
73	245	Neu5Aca2-3Galb1-4(6S)GlcNAcb-Sp8
74	246	Neu5Aca2-3Galb1-4(Fuca1-3)(6S)GlcNAcb-Sp8
75	247	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4(Fuca1-3)GlcNAcb-Sp0
76	248	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb-Sp0
77	249	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb-Sp8
78	250	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb-Sp8
79	251	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3Galb1-4GlcNAcb-Sp8
80	252	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0
81	253	Neu5Aca2-3Galb1-4GlcNAcb-Sp0
82	254	Neu5Aca2-3Galb1-4GlcNAcb-Sp8
83	255	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAcb-Sp0
84	257	Neu5Aca2-3Galb1-4Glcb-Sp0
85	258	Neu5Aca2-3Galb1-4Glcb-Sp8
86	269	Neu5Aca2-8Neu5Aca2-3Galb1-4Glcb-Sp0
87	281	Neu5Aca2-3Galb1-4GlcNAcb1-6(Galb1-3)GalNAca-Sp14
88	287	Neu5Aca2-3Galb1-3GlcNAcb1-3Galb1-3GlcNAcb-Sp0
89	288	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb1-3GlcNAcb-Sp0
90	309	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14
91	315	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp12
92	320	Neu5Aca2-3Galb1-3(Fuca1-4)GlcNAcb1-3Galb1-3(Fuca1-4)GlcNAcb-Sp0
93	326	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAc-Sp14
94	365	Neu5Aca2-3Galb1-4GlcNAcb1-3GalNAc-Sp14
95	367	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-3GalNAca-Sp14
96	381	GalNAcb1-4(Neu5Aca2-3)Galb1-4GlcNAcb1-3GalNAca-Sp14
97	387	Neu5Aca2-3Galb1-3GlcNAcb1-3GalNAca-Sp14
98	399	Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0
99	400	Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0
100	429	Neu5Aca2-3Galb1-4GlcNAcb1-3Galb-Sp8
101	440	Neu5Aca2-8Neu5Aca2-3Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp0
102	447	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21
103	448	Neu5Aca2-3Galb1-4GlcNAcb1-4Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21
104	449	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21
105	450	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-4GlcNAcb1-4(Neu5Aca2-3Galb1-4GlcNAcb1-2)Mana1-3)Manb1-4GlcNAcb1-4GlcNAcb-Sp21

106	459	Neu5Aca2-3Galb1-4GlcNAcb1-6(Neu5Aca2-3Galb1-4GlcNAcb1-3)GalNAca-Sp14
107	465	Neu5Aca2-3Galb1-4GlcNAcb1-2Mana-Sp0
108	366	Neu5Aca2-6Galb1-4GlcNAcb1-3GalNAc-Sp14
109	370	Galb1-3GalNAca1-3(Fuca1-2)Galb1-4GlcNAc-Sp0
110	473	Neu5Aca2-3Galb1-3GlcNAcb1-2Mana1-6(GlcNAcb1-4)(Neu5Aca2-3Galb1-3GlcNAcb1-2Mana1-3)Manb1-4GlcNAcb1-4GlcNAc-Sp21
111	478	Neu5Aca2-3Galb1-3GlcNAcb1-6GalNAca-Sp14
112	513	Neu5Aca2-3Galb1-3GlcNAcb1-2Mana-Sp0
113	516	Neu5Aca2-3Galb1-3GalNAcb1-4Galb1-4Glcb-Sp0
114	521	Neu5Aca2-3Galb1-4(Fuca1-3)GlcNAcb1-2Mana-Sp0
115	535	Neu5Gca2-8Neu5Aca2-3Galb1-4GlcNAc-Sp0
116	538	Neu5Aca2-8Neu5Aca2-3Galb1-4GlcNAc-Sp0
117	561	GlcNAcb1-6(Neu5Aca2-3Galb1-3)GalNAca-Sp14
118	562	Galb1-3GalNAcb1-4(Neu5Aca2-8Neu5Aca2-8Neu5Aca2-3)Galb1-4Glcb-Sp21
119	9	Neu5Aca-Sp8
120	10	Neu5Aca-Sp11
121	11	Neu5Acb-Sp8
122	136	Neu5Acb2-6(Galb1-3)GalNAca-Sp8
123	225	Neu5Aca2-8Neu5Aca2-8Neu5Aca-Sp8
124	268	Neu5Aca2-8Neu5Aca-Sp8
125	271	Neu5Acb2-6GalNAca-Sp8
126	272	Neu5Acb2-6Galb1-4GlcNAcb-Sp8
127	273	Neu5Gca2-3Galb1-3(Fuca1-4)GlcNAcb-Sp0
128	274	Neu5Gca2-3Galb1-3GlcNAcb-Sp0
129	276	Neu5Gca2-3Galb1-4GlcNAcb-Sp0
130	277	Neu5Gca2-3Galb1-4Glcb-Sp0
131	278	Neu5Gca2-6GalNAca-Sp0
132	279	Neu5Gca2-6Galb1-4GlcNAcb-Sp0
133	280	Neu5Gca-Sp8
134	312	Neu5Aca2-8Neu5Aca2-8Neu5Acb-Sp8
135	313	Neu5Gcb2-6Galb1-4GlcNAc-Sp8
136	533	Neu5Gca2-8Neu5Gca2-3Galb1-4GlcNAc-Sp0
137	534	Neu5Aca2-8Neu5Gca2-3Galb1-4GlcNAc-Sp0
138	536	Neu5Gca2-8Neu5Gca2-3Galb1-4GlcNAcb1-3Galb1-4GlcNAc-Sp0
139	537	Neu5Gca2-8Neu5Gca2-6Galb1-4GlcNAc-Sp0

Table 2.3: **List of all glycans included in Figure 2.4 charts**

CHAPTER 3: Clade-defining mutations in human H1N1 hemagglutinin protein from 2021-2023 have opposing effects on *in vitro* fitness and antigenic drift

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Abstract

Seasonal influenza viruses frequently acquire mutations that have the potential to alter both virus replication and antigenic profile. Recent seasonal H1N1 viruses have acquired mutations to their hemagglutinin (HA) protein receptor binding site (RBS) and antigenic sites, and have branched into the clades 5a.2a and 5a.2a.1. Both clades demonstrated improved *in vitro* fitness compared with the parental 5a.2 clade as measured through plaque size, infectious virus production in human nasal epithelial cells, and receptor binding diversity. Both clades also showed reduced neutralization by serum from healthcare workers vaccinated in the 2022-23 Northern Hemisphere influenza season compared to the vaccine strain. To investigate the phenotypic impact of individual clade-defining mutations, recombinant viruses containing single HA mutations were generated on a 5a.2 genetic background. The 5a.2a mutation Q189E improved plaque formation and virus replication, but was more efficiently neutralized by serum from individuals vaccinated in 2022-23. In contrast, the 5a.2a mutation E224A and both 5a.2a.1 mutations P137S and K142R impaired aspects of *in vitro* fitness but contributed to antigenic drift. Surprisingly, the E224A mutation and not Q189E caused broader receptor binding diversity seen in clinical isolates of 5a.2a and 5a.2a.1, suggesting that receptor binding diversity alone may not be responsible for the phenotypic effects of the Q189E mutation. These data document an evolutionary trade-off between mutations that improve viral fitness and those that allow for the evasion of existing host immunity.

Introduction

Seasonal influenza viruses are among the largest causes of global respiratory infections. Interventions including antivirals and vaccinations have helped reduce the burden of disease, but influenza continues to infect roughly one billion people and cause the deaths of hundreds of thousands each year (2). One of the major challenges to combatting this disease is the genetic diversity present within these viruses, and the rapid pace of continued diversification. “Seasonal influenza” refers to two co-circulating types of viruses from the *Orthomyxoviridae* family, influenza A virus (IAV) and influenza B virus (IBV), which can each be further subdivided into strains and subtypes. IAV is the predominant contributor to annual disease burden (247), and within IAVs both H1N1 and H3N2 subtypes co-circulate with relative prevalence changing from season to season (248).

Each IAV subtype consists of multiple clades as defined by the hemagglutinin (HA) sequence, with distinct genotypes and antigenic profiles (249). Mutations can alter numerous aspects of viral phenotype, including changes to both fitness and antigenic profile. Changes to fitness can include differences in receptor binding (37, 92, 113, 203), virus replication (131, 203, 206), interactions with innate immunity (250), transmissibility (251-253), and other characteristics (254). Antigenic profile refers to the regions of a virus which are recognized by host antibodies, and changes to this profile through antigenic drift can alter how the virus appears to the immune system and result in a loss of antibody binding. Frequent mutations to viral antigenic sites contribute to our need for annual vaccination with updated strains whose antigenic profiles match those of currently circulating viruses (255). HA functions as the primary antigenic determinant of

influenza and is also responsible for receptor binding and virus-host membrane fusion (58). These roles result in dual selective pressures on influenza to both evade existing host immunity and maintain or improve virological fitness by optimizing cell attachment and entry.

The H1N1 subtype has seen significant genetic diversification over the past several seasons, with multiple clades emerging and achieving varying degrees of prevalence. In the 2019-20 Northern Hemisphere (NH) influenza season the 5a.1 and 5a.2 clades both emerged from the 5a parental clade (154). The 5a.1 clade quickly predominated over 5a.2, and the 5a.2 clade was shown to have impaired *in vitro* fitness compared with 5a.1 which may have contributed to its limited spread (203).The 5a.2 clade rapidly acquired a series of HA mutations at the start of 2021 and branched into a novel subclade, 5a.2a, which quickly became the predominant H1N1 clade. 5a.2a is defined by mutations in the receptor binding site and antigenic site mutations A186T, Q189E, and E224A (H1 numbering). The A186T mutation has emerged previously in regionally circulating H1N1, including Ukraine (256), Tunisia (257), Turkey (258), Spain (259), Cuba (210), and the Czech Republic (260). Notably, Q189E had appeared as a clade-defining mutation in 5a.1 (203) but is new to the 5a.2 lineage. The E224A mutation has been shown to affect viral pathogenesis (192), cell tropism (261), and receptor binding (262, 263), and alanine already exists as the predominant residue at site 224 in swine H1 HA (192). Initial serological characterization indicates that 5a.2a is antigenically drifted from the 5a.2 clade (155, 264).

As the 5a.2a clade circulated in 2021 it continued to acquire mutations, and by 2022 the subclade 5a.2a.1 had emerged but remained a minority clade. It is defined by

mutations including P137S and K142R, which occupy both the receptor binding site and antigenic site Ca2 . Both of these mutations have previously emerged in regional circulation (259, 265-267), and have been identified repeatedly in circulating swine H1 HA (268-271). Serological data suggests that this clade may be antigenically drifted from both the 2022-23 NH influenza vaccine strain and the 5a.2a clade (155).

To evaluate phenotypic changes to these recent H1N1 clades, experiments were performed to identify both antigenic and *in vitro* fitness differences between representative virus genotypes. Both 5a.2a and 5a.2a.1 clades were antigenically drifted from 5a.2 and demonstrated improved *in vitro* fitness. The mutations contributing to escape from vaccine induced immunity did so at the expense of infectious virus production, and a mutation that improved virus replication increased virus recognition by serum from vaccinated individuals. These data highlight the evolutionary trade-off between acquiring mutations that allow for the evasion of existing immunity, while maintaining adequate virological fitness.

Results

Emergence of novel 5a.2 subclades

After the emergence of the 5a.1 and 5a.2 clades in 2019, 5a.1 quickly predominated and almost completely overtook 5a.2 by the end of 2020 (Figure 3.1A). Characterization of these viruses revealed a reduction in *in vitro* fitness for 5a.2 compared with 5a.1 (203), which may have contributed to the success of the latter. In 2021, a series of mutations emerged within the 5a.2 clade which were subsequently

classified as a new clade, 5a.2a. 5a.2a spread rapidly and by 2022 overtook 5a.1 to become the primary H1N1 clade (Figure 3.1A). The mutations defining this new 5a.2a clade include multiple changes to antigenic sites, as well as several differences in the receptor binding site of the HA protein (Figure 3.1B). This suggests potential adaptations to the viral phenotype which may have improved fitness or antigenic profile, allowing it to circulate more widely than its 5a.2 parental clade.

The 5a.2a.1 clade, an offshoot of 5a.2a, also appeared around this time and slowly rose in prevalence as a sizable minority within H1N1 (Figure 3.1A). In addition to the mutations that define the 5a.2a clade, 5a.2a.1 contains several changes to both the receptor binding site and antigenic sites (Figure 3.1C), and initial serological data indicate that this clade may be antigenically drifted from 5a.2a (155), though its limited initial success suggests that any phenotypic changes acquired by this clade did not produce overwhelming advantages over 5a.2a.

To identify phenotypic differences between the 5a.2a and 5a.2a.1 clades, representative viruses were isolated from clinical samples of H1N1 collected from the Johns Hopkins Hospital during the 2022-23 Northern Hemisphere (NH) influenza season. These include two 5a.2a viruses, A/Baltimore/JH-22363/2022 and A/Baltimore/JH-22377/2022, and one 5a.2a.1 virus, A/Baltimore/JH-22400/2022. Two 5a.2a viruses were chosen to reflect intra-clade genetic diversity; A/Baltimore/JH-22377/2022 contains the T216A hemagglutinin mutation which appeared with regularity in this clade in 2022 and which has increased in prevalence in H1N1 since the 2022-23 season. In addition, a 5a.2 virus from the 2019-20 NH season, A/Baltimore/R0675/2019, was included as a comparison group for the parental 5a.2 clade.

Recent H1N1 clades show improved in vitro fitness compared with the parental 5a.2 clade

To compare *in vitro* fitness between these clades, plaque formation and viral replication were assessed. Both 5a.2a and 5a.2a.1 viruses produced significantly larger plaques on MDCK cells compared with the 5a.2 parental clade (Figure 3.2A, 3.2B). Additionally, the 5a.2a virus A/Baltimore/JH-22363/2022 produced significantly larger plaques than the 5a.2a.1 virus A/Baltimore/JH-22400/2022, though neither were significantly different from the 5a.2a virus A/Baltimore/JH-22377/2022 which contains the T216A mutation (Figure 3.2B). Low-MOI growth curves were then performed on human nasal epithelial cell (hNEC) cultures, which are primary differentiated cell cultures grown at an air-liquid interface, and are an *in vitro* model of the upper respiratory tract (217, 220, 221, 272, 273). Both 5a.2a viruses, but not 5a.2a.1, replicated to significantly higher titers at multiple timepoints compared with 5a.2 (Figure 3.2C). To compare total infectious virus production, the area under the curve was calculated for each growth curve, which confirmed that 5a.2a, but not 5a.2a.1, produced significantly more infectious virus compared with 5a.2 (Figure 3.2D). Differences in replication were observed primarily at timepoints after peak virus titer had been reached, so to investigate this phenotype, virus production was separated into pre-peak and post-peak sections. Total virus production pre-peak was not significantly different between any of the clades (Figure 3.2E), however post-peak virus production was significantly higher for both recent H1N1 clades compared with the 5a.2 clade (Figure 3.2F).

Recent H1N1 clades are antigenically drifted from their ancestral clade

Antigenic changes were assessed through virus neutralization assays using serum from the Johns Hopkins Centers for Excellence in Influenza Research and Response (JH-CEIRR) 2022 cohort. This cohort includes serum samples from healthcare workers both pre- and post-vaccination with the annual influenza vaccine, and participants include both men and women with ages ranging from 22-75. At both pre- and post-vaccination timepoints, the 5a.2a and 5a.2a.1 clades were less well neutralized by serum antibodies compared with the vaccine strain for that season, a virus from the 5a.2 clade (Figure 3.3A, 3.3B). The 5a.2a.1 clade also appeared to be antigenically drifted from 5a.2a, showing a significant reduction in neutralization both pre- and post-vaccination compared with at least one of the representative 5a.2a viruses. Interestingly, vaccination produced a comparable boost in neutralizing antibody titers across the virus panel; the fold change in neutralizing titers was similar for all four viruses tested (Figure 3.3C), and seroconversion rates were not significantly different (Figure 3.3D).

Identifying the phenotypic contributions of individual clade-defining mutations

While the initial characterization of representative clinical isolates from these clades suggests important phenotypic changes associated with recent clade-defining mutations, the impact of individual gene segments and mutations cannot be determined by characterizing clinical virus isolates alone. To elucidate the contribution of individual

genes and mutations, an infectious clone of the A/Baltimore/R0675/2019 5a.2 virus was generated as previously described (274) by creating eight plasmids containing individual gene segments of the virus (92, 275, 276). To confirm that the virus cloning and rescue process alone did not alter viral phenotype, a rescued A/Baltimore/R0675/2019 virus was compared with the genetically identical wildtype virus and both demonstrated similar *in vitro* fitness as judged by plaque sizes (Figure 3.4A and 3.4B) and replication on hNEC cultures (Figure 3.4C to 3.4F).

To confirm that changes to the hemagglutinin protein were responsible for the observed differences between 5a.2 and 5a.2a, a reassortant virus was generated containing seven gene segments from the 5a.2 virus and an HA segment from 5a.2a (A/Sydney/5/2021). This reassorted virus generated significantly larger plaques on MDCK cells (Figure 3.4A and 3.4B) and replicated to significantly higher titers on hNEC cultures compared with the 5a.2 virus (Figure 3.4C and 3.4D). Virus replication reflected the trend seen in clinical isolates, with differences in total virus production being driven by increases in post-peak virus production (Figure 3.4E and 3.4F).

After confirming that the HA protein was driving differences between clades, the contributions of individual clade-defining mutations were assessed. Viruses were generated using the seven segments from the 5a.2 virus along with 5a.2 HA segments containing individual point mutations from either the 5a.2a (A186T, Q189E, and E224A) or 5a.2a.1 (P137S and K142R) clades. The Q189E mutation from the 5a.2a clade significantly increased plaque size on MDCK cells compared with the 5a.2 virus, while the E224A mutation from 5a.2a and the P137S mutation from 5a.2a.1 both reduced plaque size (Figure 3.5A, 3.5B).

Growth curves on human nasal epithelial cells showed that every point mutant resulted in significantly different viral titers at multiple timepoints post-infection compared with the 5a.2 virus (Figure 3.5C). Area under the curve analysis revealed that total virus production was reduced for all viruses except for the Q189E mutant (Figure 3.5D), with every single mutant virus producing significantly less virus than the 5a.2 virus before peak titer (Figure 3.5E). Post-peak production, however, was increased for Q189E compared with 5a.2 (Figure 3.5F), mirroring the trend seen in both the clinical isolates and the 5a.2a reassorted virus. Both E224A and the two 5a.2a.1 mutations (P137S and K142R) produced significantly less virus post-peak compared with 5a.2 (Figure 3.5F).

Differences in receptor binding between recent H1N1 clades

The changes observed to *in vitro* fitness of recent clades, combined with the location of clade-defining mutations on the receptor binding site, suggested a role for receptor binding as a mediator for fitness changes. To investigate receptor binding diversity, Consortium for Functional Glycomics glycan microarrays were performed on both the clinical isolate panel and the mutant virus panel. These glycan arrays contain over 100 terminally sialylated glycans including ones with either alpha 2,3 or alpha 2,6 linkages. Receptor binding profiles for both clinical viruses and point mutants demonstrate a preference for alpha 2,6 terminally sialylated glycans, the canonical human influenza receptor, and show differences in both quantity and identity of bound glycans (Figure 3.6A, 3.6B). To visualize these differences, a heat map was generated of all alpha 2,6 glycans included in the array, showing relative binding intensity for each

virus. While there was some overlap in bound glycans among all viruses (indicated by a blue circle), both 5a.2a and 5a.2a.1 viruses bound to additional glycans not bound by the parental 5a.2 clade (indicated by a green circle), and these glycans were also bound by the E224A mutant virus (Figure 3.6C, 3.6D). This suggests that the E224A mutation is responsible for this increase in receptor binding diversity of recent H1N1 clades. Additionally, the K142R mutation from the 5a.2a.1 clade was shown to abrogate binding to a specific glycan (ID number 560), which was strongly bound by all other viruses (indicated by a purple circle). The Q189E mutation that increased *in vitro* replication fitness did not show major changes in glycan binding pattern, suggesting that altered receptor binding diversity is most likely not the reason for the improved fitness of viruses containing this mutation. These results show that recent H1N1 mutations have produced altered receptor binding diversity, but that receptor binding diversity alone does not correlate with overall viral fitness.

Individual HA mutations can reduce or increase neutralization by serum from vaccinated individuals

To assess the contributions of individual clade-defining mutations on antigenic drift, neutralization assays were performed on the point mutant virus panel using post-vaccination serum from the 2022 JH-CEIRR healthcare worker cohort. The E224A, P137S, and K142R mutations all significantly reduced serum neutralization compared with the vaccine strain, while the addition of the Q189E mutation actually increased virus neutralization by serum antibodies (Figure 3.7). These data indicate that the E224A and 5a.2a.1 mutations, while potentially deleterious to the virus from an *in vitro*

fitness perspective, may have been selected due to their contribution to antigenic drift and evasion of existing immunity. Conversely, the Q189E mutation increases neutralization by serum compared to the vaccine strain whereas the same mutation improved virus fitness *in vitro*. Taken together, these data indicate that the presence of mutations at multiple antigenic sites allows for the tolerance of a mutation, Q189E, that increases serum neutralization, particularly when that mutation can improve *in vitro* virus fitness.

Discussion

The emergence of the 5a.2a clade of H1N1, and the speed with which it predominated, was surprising considering the failure of its parental clade, 5a.2 (Figure 3.1A). The 5a.2 clade had significantly reduced *in vitro* fitness compared with the cocirculating 5a.1 clade and their parental 5a clade, including reduced receptor binding diversity, replication, and plaque formation (203). The contrasting success of its progeny 5a.2a clade suggested that there were phenotypic advantages conferred by the 5a.2a clade-defining mutations. Indeed, *in vitro* fitness assays reveal that 5a.2a forms significantly larger plaques (Figure 3.2A, 3.2B), replicates significantly better in human nasal epithelial cell (hNEC) cultures (Figure 3.2C-3.2F), and has increased receptor binding diversity (Figure 3.6C) compared with 5a.2. The 5a.2a clade was also antigenically drifted from 5a.2, as demonstrated by neutralization assays using serum from healthcare workers (Figure 3.3A, 3.3B). These advantages likely contributed to the ability of 5a.2a to overtake the other co-circulating H1N1 clades.

Three clade-defining mutations for 5a.2a, A186T, Q189E, and E224A, occupy both the receptor binding site (RBS) and antigenic site Sb (Figure 3.1B), suggesting a key role in fitness differences. When characterized, it was shown that only Q189E was able to independently increase *in vitro* fitness, while E224A actually reduced plaque size and total virus production in hNEC cultures (Figure 3.5A-3.5F). Interestingly, E224A was the only 5a.2a mutation that contributed to antigenic drift, while Q189E actually resulted in increased virus neutralization (Figure 3.7). Q189E also appeared as a clade-defining mutation of the 5a.1 clade that predominated from 2020-2021 (Figure 3.1A) and was present in the 2021-22 Northern Hemisphere (NH) influenza vaccine (277). Exposure to 5a.1 through either natural infection or vaccination may have generated antibody responses against this epitope, and the healthcare workers whose serum was tested above (Figures 3.3 and 3.7) would have received the 2021-22 NH vaccine due to workplace vaccination requirements. The 224 residue, on the other hand, has been glutamic acid since the emergence of 2009pdm H1N1 (150), and its switch to alanine in 2021 would have been the first population-level exposure to a new amino acid at this site. The phenotypic contributions of Q189E and E224A highlight the evolutionary balance between mutations that alter fitness and those that contribute to antigenic drift. Given the overlap between the RBS and key antigenic sites, a benefit to one component may come at a cost to the other. The accrual of multiple mutations with contrasting advantages and disadvantages may allow the virus to improve its phenotype while mitigating the deleterious effects of any one change.

Receptor binding specificity is a major factor by which hemagglutinin contributes to virus replication and it has been shown to correlate with viral fitness (37, 92, 113,

203). The reduction of *in vitro* fitness of the 5a.2 clade was associated with a loss of receptor binding diversity (203), suggesting that differences in cell attachment may have contributed to the poor circulation of 5a.2. It was therefore hypothesized that the improved *in vitro* fitness of the 5a.2a clade would be associated with an increase in receptor binding diversity, and this was confirmed by glycan microarray experiments (Figure 3.6). Since the Q189E mutation was shown to be the driver of the improved *in vitro* fitness phenotype, this was the mutation that was expected to contribute to the increased receptor binding diversity. Surprisingly, the effect of Q189E on receptor binding was negligible, while the E224A mutation appears to be the primary cause of the increased diversity (Figure 3.6). Since the stark effect of Q189E on viral replication and plaque formation was not correlated with an increase in receptor binding diversity, its mechanism for producing the observed *in vitro* fitness changes is not yet completely understood. The glycan microarray used for measuring binding diversity consists of synthetic glycans; the mutations tested above may produce differences in receptor binding that are not able to be captured using a synthetic microarray. An array using naturally-derived glycans may reveal differences more specific to the human respiratory tract (112, 278). Glycan microarrays may also overlook differences in low-affinity and non-canonical receptor interactions which have been shown to contribute to viral entry (108, 109). It is also possible that receptor affinity might be altered by one or more of these mutations without producing a significant broadening of receptor binding diversity. Changes in affinity may also explain the differences in replication seen pre-peak versus post-peak for these viruses.

The 224 residue, and specifically the E224A mutation, has previously been studied for its effect on RBS confirmation. It was noted shortly after its emergence that the 2009 pandemic H1N1 may have reduced stability in its interactions between the 190-helix and 220-loop compared with the 1918 H1N1 due to the amino acids at positions 216 and 224 (117). Shang *et al.* modeled differences between these viruses and found that the E224A mutation, in combination with S183P, stabilized interactions between these two regions (262). The addition of E224A to the pdm2009 genetic background was shown to increase binding affinity to fetuin (193) and alter cell tropism (261), though it replicated to lower titers on human airway epithelial cells and had no impact on transmission in ferrets (220). Jayaraman *et al* predicted in 2011 that changing key RBS residues, including E224, may improve receptor binding affinity by creating a more hydrophobic interaction network in this region of the RBS (192, 193, 279). They generated a pdm2009 virus with S183P, A186T, and E224A and demonstrated that it had improved receptor affinity compared with the original pdm2009. Interestingly, all three of these mutations have since become the predominant residues at their respective sites, indicating that this constellation of amino acids is advantageous to the virus. E224A only emerged once these other mutations were also present in circulating H1, and it is possible that there are fitness advantages of E224A that only manifest in the context of these other mutations.

Similarly, while A186T did not appear to produce any apparent *in vitro* fitness advantages on its own, it may produce a synergistic effect when combined with E224A. It has been shown that some mutations only contribute to phenotypic changes when combined with a second mutation, and that the receptor binding phenotype of E224A

can be enhanced by additional mutations (192, 193). Experiments that test combinations of recent clade-defining mutations are needed to elucidate any synergistic effects of these amino acid changes.

In addition to assessing differences between the 5a.2a and 5a.2 clades, this study also investigated characteristics of the 5a.2a.1 clade that emerged from 5a.2a. Unlike 5a.2a, 5a.2a.1 did not rapidly predominate within circulating H1N1, but rather gradually increased in prevalence (Figure 3.1A). The 5a.2a.1 clade was shown to replicate at least as well as 5a.2, and it produced significantly larger plaques and had increased receptor binding diversity (Figures 3.2 and 3.6). Both of its major clade-defining mutations, however, resulted in a reduction in performance on one or more *in vitro* fitness assays (Figures 3.5 and 3.6). The benefits of these mutations appear to be largely antigenic, as they both significantly reduced serum neutralization when added to the 5a.2 genetic background (Figure 3.7), and likely contributed to the overall antigenic drift seen for 5a.2a.1 (Figure 3.3A, 3.3B).

While the phenotypes of individual 5a.2a.1 mutations were characterized here by adding them to a 5a.2 genetic background, circulating 5a.2a.1 viruses also contain all of the clade-defining mutations associated with 5a.2a. This likely explains the difference in *in vitro* fitness observed between the clinical 5a.2a.1 virus and its individual clade-defining mutations. The deleterious effects of the 5a.2a.1 mutations are likely ameliorated by advantages conferred by one or more 5a.2a mutations, allowing this virus to have adequate fitness while preserving the antigenic changes resulting from the new mutations. This exemplifies the evolutionary balancing act between mutations conferring fitness benefits and those that allow the virus to evade adaptive immunity.

One model of influenza evolution proposes the cyclical emergence of mutations with varying levels of receptor avidity influenced by pressure to escape immunity (173), and earlier surveillance of circulating H1N1 since 2009 has shown that mutations increasing receptor avidity were followed by those that reduced avidity (263). An analogous process may be driving recent H1N1 evolution, with the Q189E mutation of 5a.2a increasing viral fitness sufficiently to allow for subsequent 5a.2a.1 mutations driving antigenic drift.

This process can be understood through epistasis, or the impact of secondary mutations on an individual mutant phenotype. Epistatic processes are thought to be a major component of influenza evolution, due in part to overlapping functional regions of hemagglutinin and the frequency of pleiotropic mutations (187, 280). Epistasis in the RBS specifically has been shown to allow for significant sequence diversity and antigenic escape while maintaining viral fitness (190). The striking difference between individual and combined mutations shown here, as well as the recent temporal clustering of RBS mutations, points to a significant epistatic component to contemporary H1N1 clade-defining mutations. The synergistic effect of these mutations, including their contribution to receptor avidity and affinity, should be investigated to determine how the E224A mutation is impacted by the 190-helix mutations, and whether positive epistasis is observed between Q189E and the 5a.2a.1 mutations.

Methods

Ethics statement and human subjects

Serum used in this study was obtained from healthcare workers recruited during the annual Johns Hopkins Hospital employee influenza vaccination campaign in the Fall of 2022 by the Johns Hopkins Centers for Influenza Research and Response (JH-CEIRR). Pre- and post-vaccination (~ 28 day) human serum were collected from subjects, who provided written informed consent prior to participation. The JHU School of Medicine Institutional Review Board approved this study, IRB00288258. Virus was isolated for this study from deidentified influenza A virus H1N1 positive samples, collected from patients who provided written informed consent during the 2022-23 influenza season at the Johns Hopkins Hospital, under the JHU School of Medicine Institutional Review Board approved protocol, IRB00091667.

Cell cultures

Madin-Darby canine kidney (MDCK) cells (provided by Dr. Robert A. Lamb) and MDCK-SIAT cells (provided by Dr. Scott Hensley) were maintained in complete medium (CM) consisting of Dulbecco's Modified Eagle Medium (DMEM) supplemented with 10% fetal bovine serum, 100 units/ml penicillin/streptomycin (Life Technologies) and 2 mM Glutamax (Gibco) at 37 °C and 5% CO_2. Human nasal epithelial cells (hNEC) (PromoCell) were cultivated as previously described (92, 203).

Viral RNA Extraction and Multi-segment Amplification by Reverse-Transcriptase Polymerase Chain Reaction

Viral nucleic acid was extracted from samples as previously described (281) using the Chemagic Viral RNA/DNA Kit following the manufacturer's instructions (Revvity). Samples were eluted into 60µL of elution buffer and used directly after extraction in PCR. The whole genome was amplified using a single multi-segment 1-step reverse-transcription polymerase chain reaction (RT-PCR) with the MBTuni-12 [5'-ACGCGTGATCAGCAAAAGCAGG] and MBTuni-13 [5'-ACGCGTGATCAGTAGAAACAAGG] primer sets for Influenza A (281, 282) with the SuperScript III One-Step RT-PCR kit as per manufacturers guidelines (Invitrogen).

Library Preparation and Next-Generation Sequencing and Virus Genome Assembly

After amplification, concentrations were checked using the Qubit 2.0 fluorometer (ThermoFisher Scientific). 2µL of PCR product was prepared for sequencing using the native barcoding genomic DNA kit (EXP-NBD196) and the NEBNext ARTIC Library Prep kit, following the manufacturer's instructions (New England BioLabs / Oxford Nanopore Technologies). 50ng of the barcoded library was sequenced using the R9.4.1 flow cell on a GridION (Oxford Nanopore Technologies). Resulting fastq files were demultiplexed using the artic_guppyplex tool (Artic version 1.2.2). Nucleotide sequence assembly was performed using the FLU module of the Iterative Refinement Meta-Assembler (IRMA version 1.0.2), using IRMA's default settings, which include a minimum average quality score of 24 and a site depth of 100 (281, 283).

Clinical virus sequences are accessible through GISAID with the following IDs: EPI_ISL_17617226 for A/Baltimore/R0675/2019, EPI_ISL_17626342 for A/Baltimore/JH-22363/2022, EPI_ISL_17626298 for A/Baltimore/JH-22377/2022, and EPI_ISL_17626340 for A/Baltimore/JH-22400/2022.

Viral isolation

Viral stock preparation was performed as described previously (203, 219) using nasopharyngeal swabs from influenza H1N1-positive individuals during the 2022–2023 Northern Hemisphere influenza season at JHH.

Virus Cloning and Plasmids

Virus cloning was performed as previously described (92, 131, 284). Viral RNA was extracted using the QIAamp viral RNA mini extraction kit and converted to cDNA using universal influenza A primers. Individual segments were amplified from cDNA with Phusion High-Fidelity DNA Polymerase (New England BioLabs) using segment-specific primers, which also added restriction enzyme sites to the ends of the segments. For all segments, the pHH21 plasmid was used (245, 285). Both the plasmid and individual segments were digested separately with restriction enzymes (either BsmBI or BsaI, New England BioLabs) and then combined and ligated together with T4 DNA ligase (New England BioLabs). Ligated plasmids were transformed into DH5α bacteria (Thermo Fisher) and grown overnight on LB plates containing carbenicillin to select for colonies that successfully received the plasmid. Individual colonies were isolated and grown, and plasmid was extracted using a Qiagen QIAprep Spin Miniprep Kit. Plasmid

sequences were verified using restriction enzyme digestion and whole plasmid sequencing, which was performed by Plasmidsaurus using Oxford Nanopore Technology with custom analysis and annotation. Bacteria expressing the correct plasmid were grown in bulk and high-purity plasmid was extracted using ZymoPURE Plasmid Midiprep kit (Zymo research) and verified by sequencing of the coding region of the influenza virus segment.

For generating plasmids containing specific point mutations, mutagenic primers were designed using the New England Biolabs online software NEBaseChanger (primer sequences are available upon reasonable request). The Q5 Site-Directed Mutagenesis Kit (New England BioLabs) was then used with the A/Baltimore/R0675/2019 hemagglutinin plasmid to produce plasmids with the desired mutations. Plasmids were transformed, extracted, and confirmed as described above.

Recombinant Virus Production

Recombinant viruses were generated using the 12 plasmid reverse genetics system described previously (92, 275, 276).

Tissue culture infectious dose 50 (TCID50)

TCID50 assays were performed on MDCK-SIAT cells as previously described (131, 203, 286). The 50% tissue culture infectious dose was calculated as previously described (244, 245).

Plaque assay

Plaque assays were performed and analyzed as previously described (131, 203) using MDCK cells grown in complete medium to confluence in 6-well plates. Overlay media was phenol-red free DMEM supplemented with .3% BSA (Sigma), 100U/ml pen/strep (Life Technologies), 2mM Glutamax (Gibco), 5mM HEPES buffer (Gibco) 5µg/ml N-acetyl trypsin (Sigma) and 1% cellulose.

Viral growth curves

hNEC growth curves were performed at an MOI of 0.01 TCID50 units per cell on fully differentiated hNECs grown at an air–liquid interface as described previously (131, 203, 286). Infectious virus concentration in collected media was determined through TCID50.

Partial virus purification and labeling with Alexa Fluor 488

For each virus analyzed using glycan arrays, 100 mL of virus stock was grown using the method referenced above, and viruses were partially purified through ultracentrifugation and labeled with Alexa Fluor 488 as previously described (203). Labeled virus was stored at − 80 °C, and TCID50 was performed to confirm virus activity and concentration. Titer of labeled viruses were: 4.20E+8 TCID50/mL (A/Baltimore/R0675/2019), 1.33E+8 TCID50/mL (A/Baltimore/JH-22363/2022), 1.70E+8 TCID50/mL (A/Baltimore/JH-22377/2022), 5.37E+8 TCID50/mL (A/Baltimore/JH-22400), 1.58E+8 TCID50/mL (A/Baltimore/R0675/2019 with A186T), 1.95E+8

TCID50/mL (A/Baltimore/R0675/2019 with Q189E), 4.20E+7 TCID50/mL (A/Baltimore/R0675/2019 with E224A), 6.17E+7 TCID50/mL (A/Baltimore/R0675/2019 with P137S), and 2.86E+7 TCID50/mL (A/Baltimore/R0675/2019 with K142R).

Glycan array

The Consortium for Functional Glycomics version 5.5 microarray, containing 562 glycans in replicates of 6, was used for glycan arrays as previously described (203, 287). Microarray slides were submerged in 100 ml of PBS wash buffer (phosphate-buffered saline containing 0.005% Tween-20) for five minutes to re-hydrate. 70 µl of undiluted sample were loaded onto the slide. A cover slip was placed over the slide and incubated at room temperature for one hour in darkness, after which the cover slip was removed and the slide was washed four times in 100 ml of PBS wash buffer, then PBS, then deionized water. Slides were then dried before being read in a GenePix microarray scanner.

For each set of six replicates, only the median four values were used in analysis. A graph of the raw data was visually analyzed to confirm selective binding of terminally sialylated glycans, after which data was filtered to only include these glycans for a total of 139 glycans. Data was then normalized to a maximum value of 100, and organized by structure using the Glycan Array Dashboard (GLAD) (246). Both the heat map and structures of individual glycans were generated through GLAD.

Serum neutralization assay

Pre- and post-vaccination human serum was treated with Receptor Destroying Enzyme II (RDE, Hardy Diagnostics) and neutralization assays were performed on MDCK-SIAT cells as previously described (203, 288), performed in quadruplicate and using 100 TCID50 of virus per well.

Statistical analyses

All statistical analyses were performed in GraphPad Prism 9.1.0. Growth curves were analyzed using 2-way ANOVA with Tukey post-hoc test. Area under the curve (AUC) was calculated for each growth curve replicate and AUC values were compared using one-way ANOVA for differences in total virus production. Plaque assays were analyzed using Kruskal–Wallis ANOVA with Dunn's multiple comparison test. Neutralization assays and fold-change in neutralizing titers were analyzed using repeated-measures one-way ANOVA with Tukey post-hoc, and differences in seroconversion between clades were analyzed using a Chi-square test.

Data availability

The datasets used and/or analyzed in the current study are available upon reasonable request.

Acknowledgements

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Figures

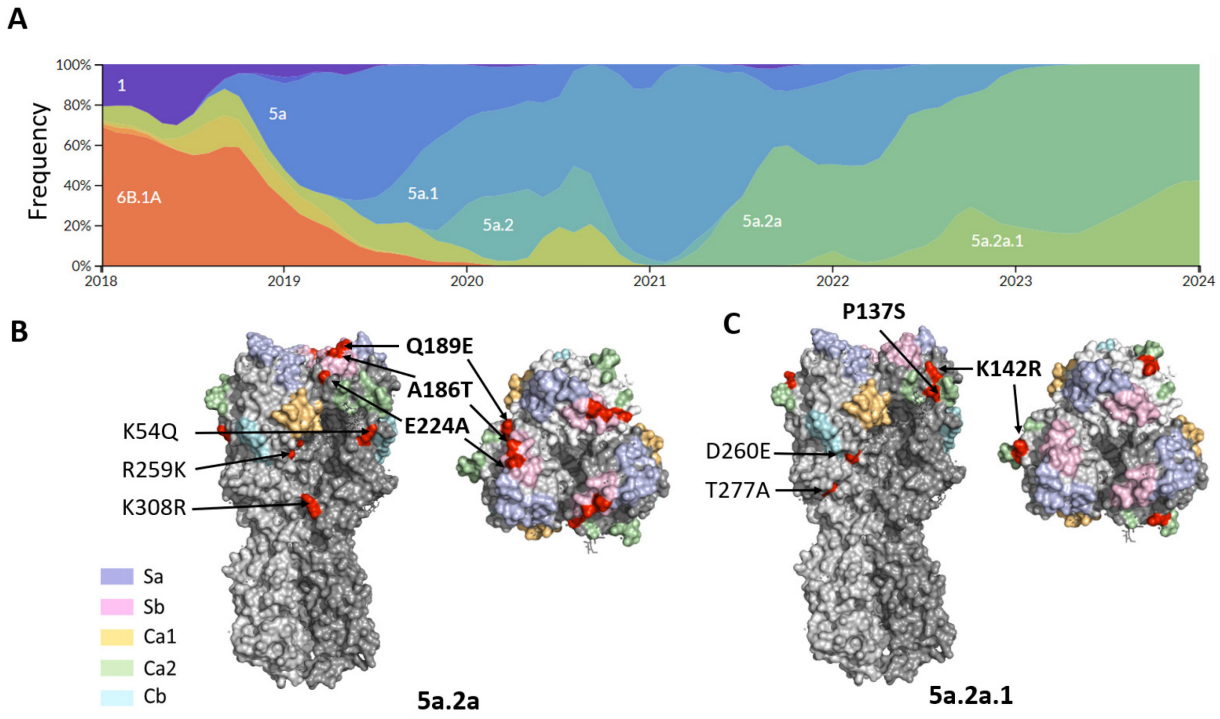


Figure 3.1: Emergence of seasonal H1N1 clades

A) Frequencies of circulating clades of H1N1 since 2018, generated using the NextStrain pipeline (150). Clade 5a.2a emerged in 2021 and quickly predominated, followed by 5a.2a.1 which slowly gained prevalence. Clade-defining mutations of B) 5a.2a and C) 5a.2a.1 mapped on the hemagglutinin trimer using PyMOL (PDB: 3LZG), with antigenic sites shaded. In addition to the mutations shown, 5a.2a.1 also contains all the clade-defining mutations of 5a.2a. Mutations surrounding the receptor binding site are in bold.

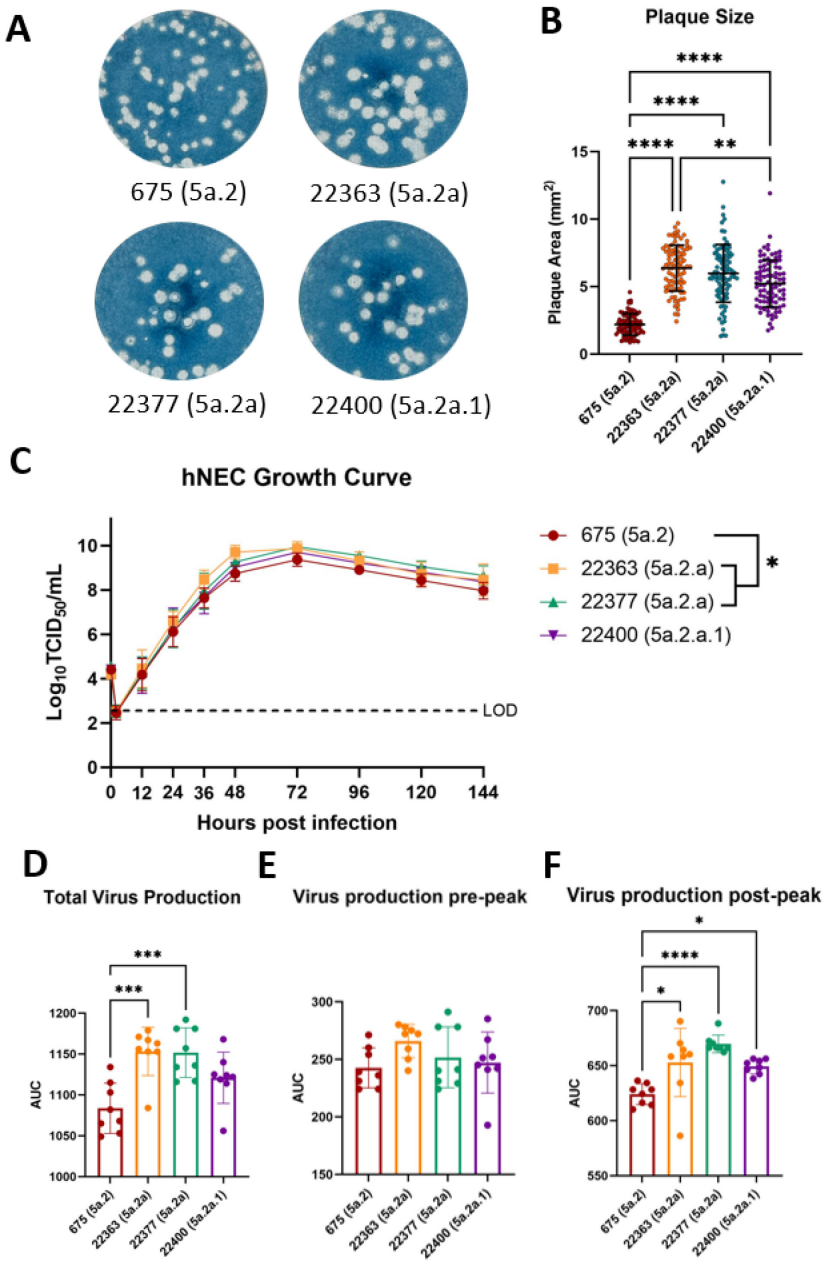


Figure 3.2: Characterization of *in vitro* replication fitness of circulating H1N1 viruses

A) Plaque formation on MDCK cells of representative viruses from circulating clades compared with their parental clade (5a.2). B) Recently circulating clades formed significantly larger plaques than their parental clade (Kruskal–Wallis ANOVA with Dunn's multiple comparisons test). Growth curves on human nasal epithelial cells reveal an

increase in C) the kinetics of infectious virus production and D) total virus production for the 5a.2a clade compared with 5a.2. Infectious virus production differences are not present in E) pre-peak titers (peak=72 hpi) but are driven by F) post-peak viral titers. One-way (AUC) or two-way (growth curve) ANOVA with Tukey post-hoc. * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$

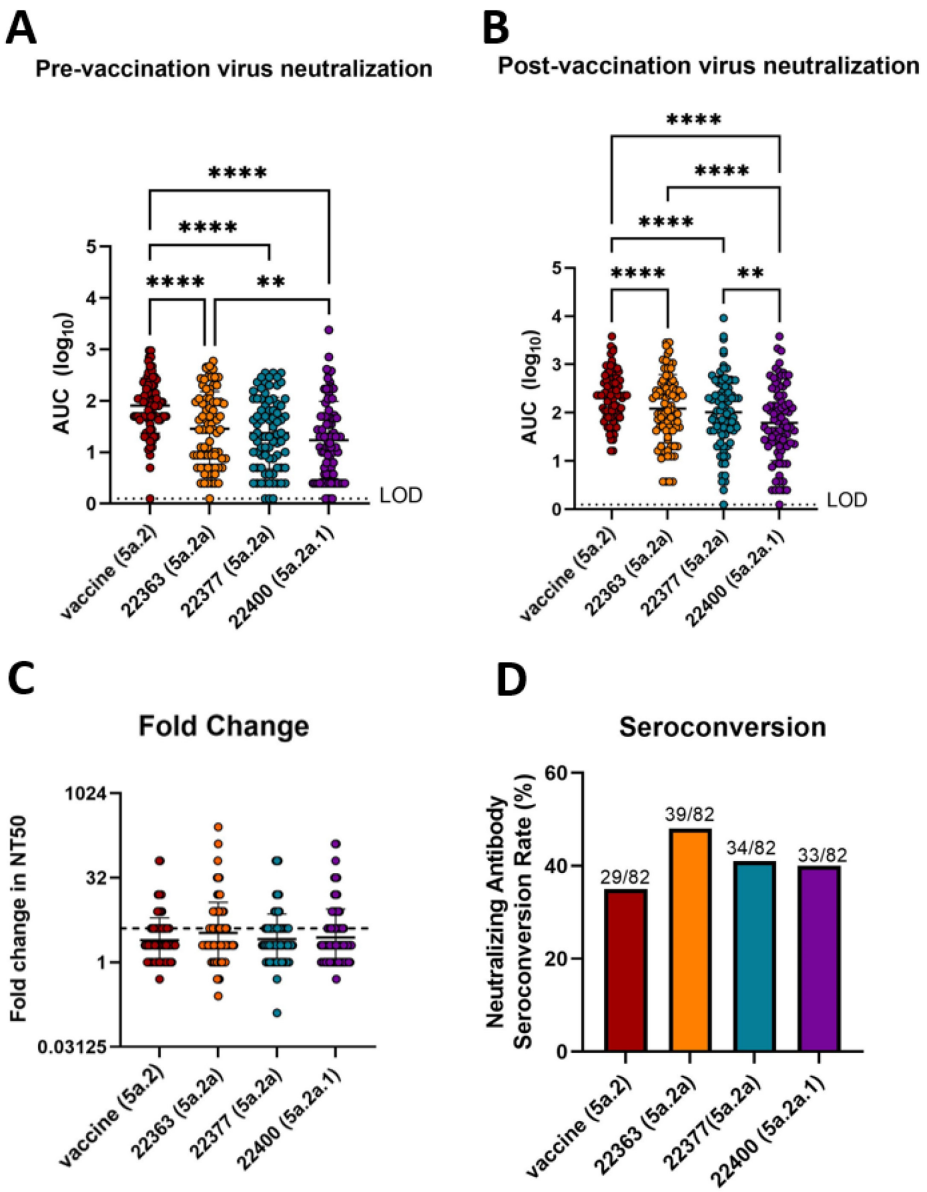


Figure 3.3: Neutralization of vaccine and circulating viruses by human serum

Virus neutralization was assessed using serum from a cohort of healthcare workers both A) pre- and B) post-vaccination with the 2022-23 Northern Hemisphere influenza vaccine. The virus panel included representative viruses from 5a.2a and 5a.2a.1 clades and the H1N1 vaccine strain, a 5a.2-clade virus (repeated measures one-way ANOVA with Tukey post-hoc). 5a.2a viruses were significantly less well neutralized than the

vaccine virus, and 5a.2a.1 had a further reduction from 5a.2a in both serum panels. C) Fold change in neutralizing antibody titers after vaccination was similar for all viruses (repeated measures one-way ANOVA with Tukey post-hoc). D) Percent of serum donors who seroconverted (showed at least a fourfold increase in neutralizing titers post-vaccination) was similar for each virus (Chi-square test). N=82. ** $p<0.01$, **** $p<0.0001$

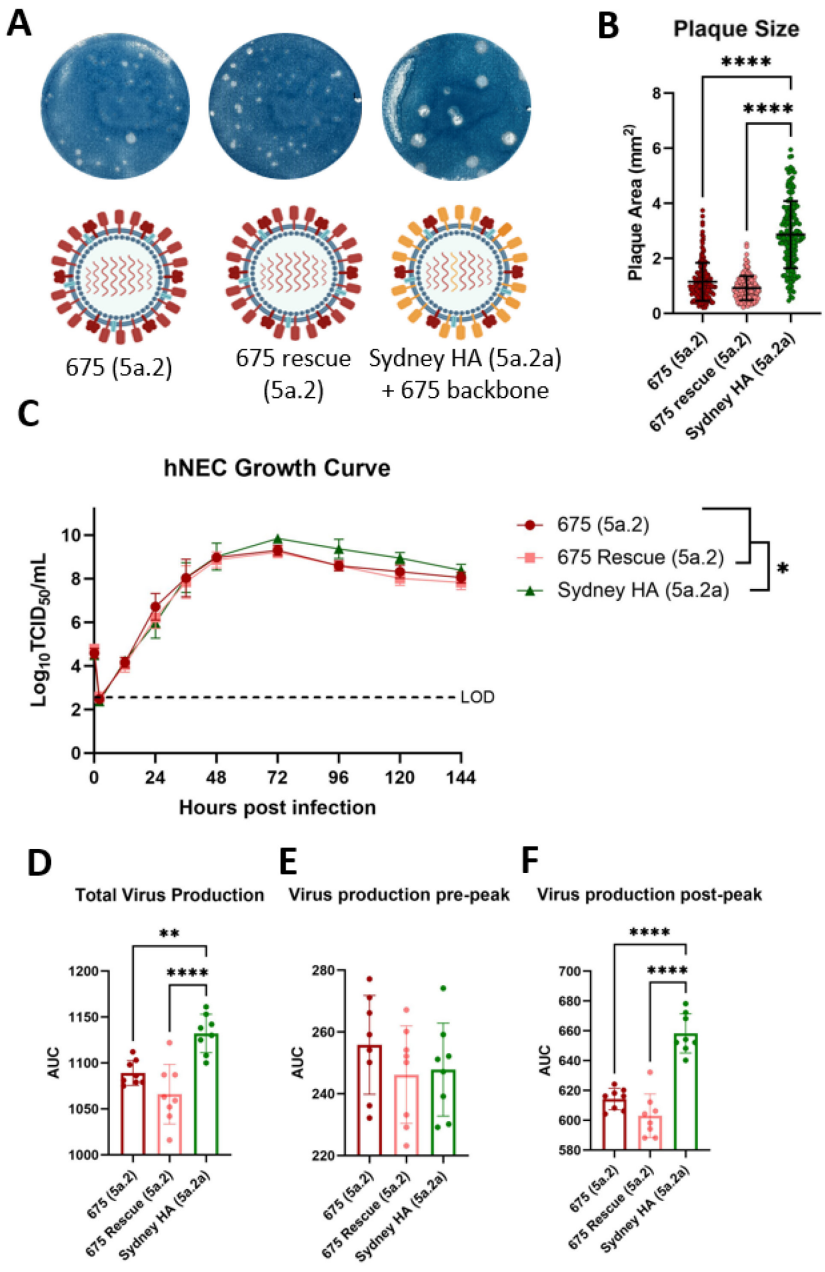


Figure 3.4: The contribution of the 5a.2a HA segment on *in vitro* replication fitness

A) A 1:7 reassortant virus was generated containing the HA from a 5a.2a virus (A/Sydney/5/2021) with all seven remaining segments from a 5a.2 virus (A/Baltimore/R0675/2019), and plaque assays were performed on MDCK cells. B) The reassortant virus produced significantly larger plaques than the wild type 5a.2 virus and

a control rescue 5a.2 virus (Kruskal–Wallis ANOVA with Dunn's multiple comparisons test). The reassortant virus replicated to a greater extent in hNEC cultures C), producing a greater amount of D) total infectious virus particles compared to the parental virus. Infectious virus production E) pre-peak was not different (peak=72 hpi) but more infectious virus was produced F) post-peak with the 5a.2a HA segment containing virus. One-way (AUC) or two-way (growth curve) ANOVA with Tukey post-hoc. ** $p<0.01$, **** $p<0.0001$

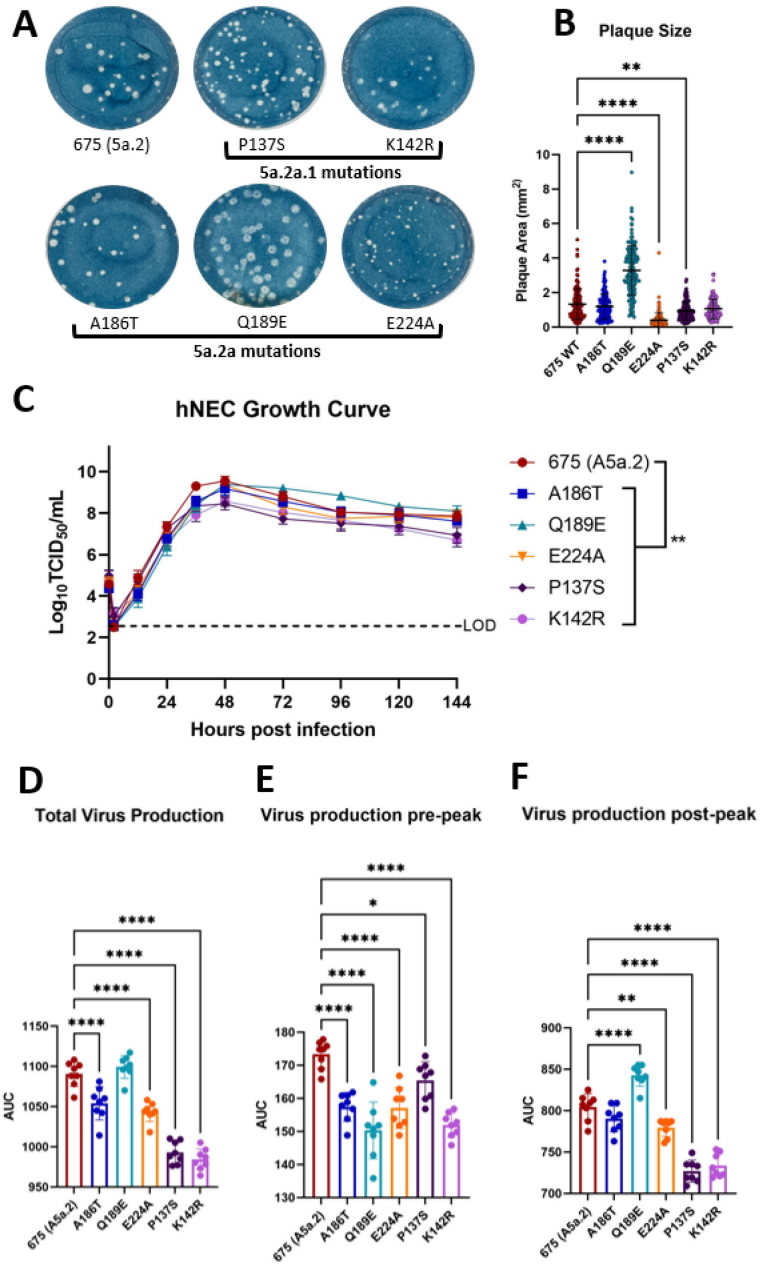


Figure 3.5: Determining the impact of individual clade-defining mutations in HA on *in vitro* replication fitness

Viruses were generated using a 5a.2 background (A/Baltimore/R0675/2019) containing individual point mutations from recent H1N1 clades (5a.2a and 5a.2a.1). A) Representative images of MDCK plaque assays. B) The Q189E mutation from 5a.2a significantly increased plaque area, while E224A and P137S reduced plaque area

(Kruskal–Wallis ANOVA with Dunn's multiple comparisons test). C) Viral growth curves on human nasal epithelial cells reveal significant temporal differences in replication between the 5a.2 virus and all point mutant viruses (Two-way ANOVA with Tukey post-hoc). D) Total virus production is reduced with all mutations except for Q189E. E) Pre-peak infectious virus titer is reduced for all point mutant viruses (peak=48 hpi), however virus production F) post-peak is only reduced for E224A, P137S, and K142R, while the Q189E mutation increases total virus production (One-way ANOVA with Tukey post-hoc). * $p<0.05$, ** $p<0.01$, *** $p<0.001$, **** $p<0.0001$

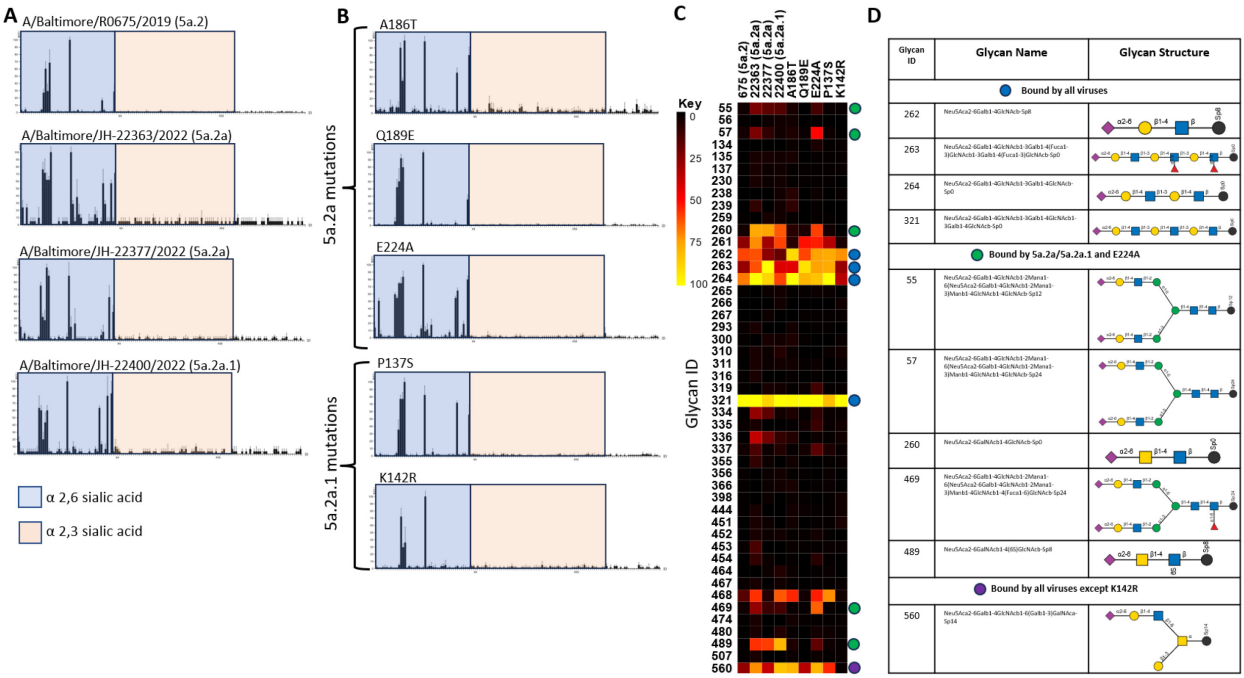


Figure 3.6: Comparison of receptor binding activity via the Consortium for Functional Glycomics (CFG version 5.5) glycan microarray

Receptor binding profiles of H1N1 clinical isolates A) and point mutants on a 5a.2-clade background B) show preference for alpha 2,6 terminally sialylated glycans, and differences in individual glycans bound. X axis depicts individual glycans included in the array, and Y axis depicts the relative fluorescence units (RFUs), a metric of virus binding, normalized to 100. C) heat map of normalized RFU values for each glycan, for each virus. A yellow square indicates maximum binding by that virus. Glycans of note are marked with colored circles; blue are glycans that were well-bound by all viruses, green are glycans bound well by viruses from the 5a.2a and 5a.2a.1 clades as well as the E224A point mutant virus, and purple is a glycan that was well-bound by all viruses except the K142R point mutant. D) Structures and names of glycans of note.

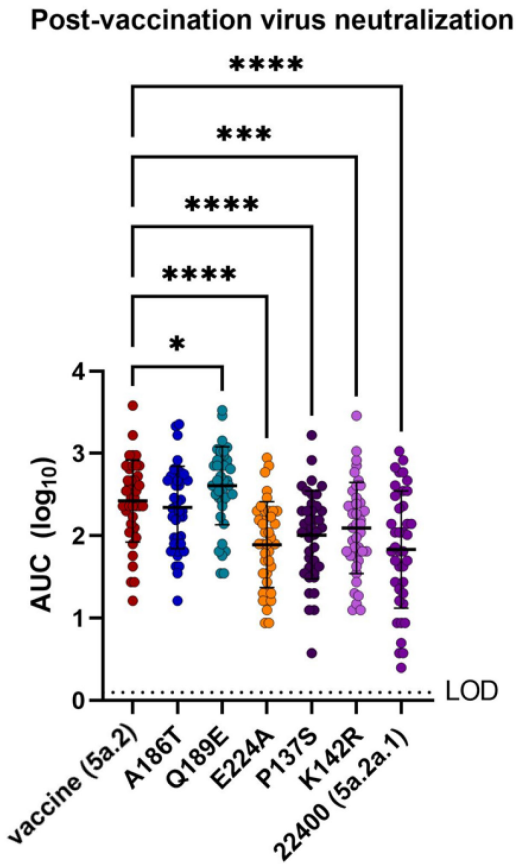


Figure 3.7: The impact of individual mutations on antigenic drift was assessed on a subset of post-vaccination serum from the 2022-23 Northern Hemisphere influenza vaccine healthcare worker cohort

The E224A, P137S, and K142R mutations reduced virus neutralization by serum antibodies, while the Q189E mutation increased neutralization and A186T had no discernable effect (repeated measures one-way ANOVA with Dunnett post-hoc using A/Victoria as comparison group). N=40. * $p<0.05$, *** $p<0.001$, **** $p<0.0001$

CHAPTER 4: General Discussion

Conclusion

The research presented in this dissertation identifies genetic and phenotypic differences between seasonal H1N1 virus clades that have circulated since the 2019-20 Northern Hemisphere (NH) Influenza Season, and correlates phenotypic advantages with epidemiologic success and clade predominance. Through investigating *in vitro* viral fitness differences and antigenic changes between recent H1N1, this dissertation contributes significantly to our understanding of the evolution of seasonal influenza. This includes identifying factors that may contribute to the differential success of cocirculating virus clades, as well as exploring the relationship between and genetic basis of viral fitness and antigenic drift in circulating influenza.

In chapter 2, the emergence of two novel clades of seasonal H1N1 was described. Representative viruses from these clades were characterized, and it was found that the clade with the lower prevalence (5a.2) also had reduced *in vitro* fitness compared with the parental 5a clade, as measured by plaque formation, viral replication, and receptor binding diversity. The predominant clade, 5a.1, was shown to be equally antigenically drifted from the vaccine strain as 5a.2 but maintained comparable levels of *in vitro* fitness compared with the parental 5a clade. These advantages likely contributed to the success of 5a.1 in the 2019-20 NH influenza season, and these findings point to the importance of viral fitness as a necessary precondition for the predominance of emergent influenza clades.

In chapter 3, the progression of these clades through the COVID-19 pandemic and into 2023 was described, including the emergence of the 5a.2a clade from 5a.2 in early 2021 and its subsequent predominance. The 5a.2a.1 clade, which was the result

of genetic diversification of 5a.2a, emerged at the end of 2021 and cocirculated as a minor clade until gaining prevalence in the 2023-24 NH Influenza Season. Representative viruses from the 2022-23 NH Influenza Season were chosen and characterized, and it was found that both 5a.2a and 5a.2a.1 had improved *in vitro* fitness and antigenic drift compared with the 5a.2 parental clade, and these phenotypic advantages likely contributed to the success of these clades whereas 5a.2 had failed to predominate. The rapid emergence of multiple mutations suggests that there was significant evolutionary pressure on the 5a.2 clade to improve its fitness. To investigate the contributions of individual clade-defining mutations on viral phenotype, an infectious clone system of H1N1 was produced. It was found that the mutation that contributed to improved *in vitro* fitness also increased virus neutralization by human serum antibodies. Conversely, the mutations that were found to reduce *in vitro* fitness also reduced virus neutralization, indicating that they contributed to viral antigenic drift.

These findings confirm the pleiotropic nature of many of the mutations that arise on influenza hemagglutinin (HA) during circulation. Since there is overlap between functional regions and antigenic regions on viral proteins, dual evolutionary pressures exist to evade adaptive immunity and to optimize viral fitness. The data presented here confirms that these pressures may produce mutations that benefit one component of viral phenotype at the expense of another, and suggests that the rapid appearance of multiple mutations on circulating viruses produces a positive epistatic effect, optimizing viral phenotype while mitigating the deleterious effects of individual mutations.

The Need for the Inclusion of Viral Fitness in Seasonal Influenza Surveillance

Seasonal influenza viruses are a persistent contributor to global morbidity and mortality, and influenza spillovers are a constant threat to global health security. Optimizing our preparedness and response to these viruses necessitates a massive, global surveillance effort to track cases, identify genetic variants, and investigate virological properties of emergent clades and/or subtypes. The World Health Organization's Global Influenza Surveillance and Response System (GISRS) is one crucial solution to this need, and over its 70 year history the GISRS has played a prominent role in leading, coordinating, and analyzing global influenza surveillance and data collection (151). The success of the GISRS has depended on the support and contribution of hundreds of countries, states, and regions, whose local influenza surveillance networks capture and communicate data to the GISRS (289). This includes the surveillance of the United States Centers for Disease Control and Prevention (CDC), which captures important data about influenza circulation within the country and performs their own analyses and virus characterization (290).

These surveillance efforts generally involve several parallel approaches. Data reflecting influenza-like illnesses and confirmed cases of influenza infection are reported, offering important context about influenza season severity and informing public health interventions. Genetic data about circulating viruses is analyzed to identify novel genotypes and virus clades, which provides information about viral evolution and diversity (208, 290). Antigenic changes to circulating viruses are then assessed by hemagglutinin inhibition (HAI) or virus neutralization assays, which determine whether influenza antibodies raised against past clades still recognize circulating viruses (290).

These data are used to determine degree of antigenic drift and can be used to make vaccine recommendations (151-155, 277). Finally, antiviral resistance is sometimes measured through assays like neuraminidase inhibition or PA endonuclease inhibition (290).

These approaches are necessary and valuable, but they do not offer a complete picture of the characteristics and trends of circulating influenza. A major component of virus success, virological fitness, is rarely included in routine surveillance. This includes research into viral replication, receptor binding, evasion of innate immunity, enzymatic activity, transmission, and a host of other virological phenotypes. It was shown in Chapter 2 that a reduction in *in vitro* fitness of a circulating clade of H1N1 correlated with its inability to predominate compared to a cocirculating clade with improved fitness, and Chapter 3 documented the rise of two novel clades whose improved *in vitro* fitness correlated with their epidemiologic success where their parental clade failed. These data confirm that changes to fitness of circulating viruses are a significant component of viral success. The WHO, in its own strategic agenda for influenza research, lists the analysis of virological properties like replication and transmission as a “highest priority” recommendation for filling knowledge gaps in influenza surveillance (291).

In addition to routine characterization of the fitness of clinical virus isolates, studying the contributions of individual mutations to viral phenotypes can fortify influenza surveillance. Mutations are often identified in subpopulations multiple times before they persist in a predominant phenotype; for example, both P137S and K142R were identified in regional circulation of H1N1 multiple times before emerging as the clade-defining mutations of 5a.2a.1 (259, 265-267). Similarly, Q189E appeared as a

clade-defining mutation of the 2019 5a.1 clade (Figure 2.1B) before it became a clade-defining 5a.2a mutation (Figure 3.1B). Characterization of this mutation in Chapter 3 showed that it imparted a significant improvement in multiple *in vitro* fitness assays, highlighting the importance of this mutation in the improved phenotype of the 5a.2a clade. Considering that the same mutations often appear multiple times before predominating, identifying the phenotypic consequences of mutations as they emerge allows us to compile knowledge about key residues and improves our ability to flag concerning mutations in emergent genotypes.

Structural Analysis Should Play a Larger Role in Seasonal Influenza Surveillance

Many of the mutations associated with changes to viral fitness, including those described here, appear on the RBS of the viral HA protein. The enormous role that receptor binding plays in viral fitness necessitates making RBS analysis a core pillar of fitness surveillance. HA modeling and RBS structure analysis are an important way to screen emergent mutations for ones that induce structural changes, and allows researchers to flag new mutations that may cause changes to receptor binding. These mutations can then be studied in more detail using *in vitro* receptor binding assays.

There are many examples of HA modeling guiding hypothesis generation, and of RBS structural changes correlating with viral phenotype. Modeling of S183P and E224A reveal that these residues form the junction between the 190 helix and 220 loop, and that the mutations change the RBS structure and contribute to binding pocket flexibility (262). This was confirmed in another study and the S183P mutation was shown to

correlate with an increase in receptor binding affinity (263), increased replication (292), and increased virulence in mice (293, 294). Structural modeling of the 2009 pandemic H1 hemagglutinin showed instability in the RBS compared with that of the prior seasonal H1N1 and of the 1918 pandemic H1N1, and several mutations including S183P, A186T, and E224A were predicted to stabilize the viral RBS through improved interactions between the 190 helix and 220 loop (279). Site-directed mutagenesis of these residues confirmed that they increased receptor affinity. Another mutant that improved stability in the RBS model, I216K, demonstrated improved receptor binding diversity and affinity as well as increased binding to human tracheal tissue sections. These results suggest that stabilizing this region of the RBS indeed improves receptor binding fitness. Interestingly, many of these mutations have since emerged in recently circulating H1N1 and form the majority of the clade-defining mutations of the 5a.2a clade (Figure 3.1B). These data confirm that structural modeling of the RBS can predict changes to viral receptor binding, and can play an important role in influenza fitness surveillance.

The inclusion of structural modeling in influenza surveillance is not without precedent. The location of emergent mutations is often discussed in surveillance reports, especially when they overlap known functional or antigenic regions. Retrospective analyses have studied long-term changes in receptor binding of seasonally circulating influenza (54, 111, 295). Changes to the RBS are especially reported in the context of avian and swine influenza surveillance, where mutations may confer increased transmissibility in humans (296, 297). Groups and organizations that produce influenza surveillance reports should make an effort to include structural

analyses along with the genetic and antigenic information commonly reported. Several publications have demonstrated the utility of this approach. In one study, phylogenetics and mutational frequency of H1N1 circulating in India was reported in conjunction with HA structural analysis, allowing for the assessment of both antigenic site and receptor binding site differences (298). Another study reported mutational differences between clades in H1N1 for several years after its emergence, modeled how these mutations impacted receptor binding site interactions, and explored the effects of these structural changes using receptor binding assays (263). Regional investigations like these, if reported seasonally, would greatly expand the body of information available to us about receptor binding site evolution and improve our ability to predict which mutations will likely drive fitness changes in future seasons.

The expansion of structural analysis in influenza surveillance should also extend to other influenza proteins. The NA protein also frequently mutates to evade immunity, and its functional regions can sustain frequent mutations. One recent study combined phylogenetic and structural analysis for recently circulating N1, and revealed clusters of NA mutations that corresponded with changes to the NA enzymatic site and secondary sialic acid binding site. These changes were shown to impact enzymatic activity, substrate specificity, and optimum pH (299), confirming the predictive power of modeling. Mutations to other proteins have also been shown to affect viral fitness, and an investment in the understanding of their functional regions may open the door to structure-based surveillance of more influenza proteins (179, 300, 301).

Diversifying the Tools Used to Study Receptor Binding

To study receptor binding diversity of influenza viruses, glycan microarrays are one of the most commonly used assays (93, 111, 118, 223, 228, 229, 278, 302, 303). These arrays contain a variety of terminally sialylated glycans with various structures and branches, and the binding of labeled virus or recombinant HA molecules to these glycans can easily be measured. The Consortium for Functional Glycomics glycan (CFG) microarray is the largest synthetic microarray available currently, and includes over 500 glycans (of which approximately 140 are terminally sialylated). Complete structures are available for every glycan on the array, and the breadth and depth of information available through the CFG microarray system is unparalleled.

Despite the quality of the CFG microarray system, however, there remain limitations to glycan microarrays that are necessary to discuss when interpretating their data. Synthetic microarrays are useful because of the level of control they provide researchers over the glycan composition, but they do not necessarily reflect the human lung glycome (278). One study demonstrated that currently available arrays, including the CFG array, did not reflect human respiratory tract glycans and was not predictive of virus replication (106). This paper called for a new generation of glycan microarrays that better reflect human glycomics based on more recent research.

One option for a more physiologically relevant array is a shotgun glycomics microarray. This technique involves harvesting glycans from the tissue of an organism of choice (e.g. human or swine trachea/lung) and mounting them on a microarray instead of using synthetic glycans (304). Shotgun glycomics-based arrays have been tested in several contexts. Using human lung glycans, shotgun glycomic arrays have revealed

that human and avian influenza was able to bind to phosphorylated, nonsialylated glycans (305). This is a noncanonical influenza receptor; it would likely not have been included in traditional glycan microarrays, and highlights the utility of using unbiased human lung-derived glycans to find unexpected virus/host interactions. When applied to pig lung-derived glycans, this technique revealed detailed information on the binding patterns of human and animal influenza in swine tissue (306). This technique is limited by both tissue availability and methodological complexity. Generating a tagged glycan library requires access to sufficient tissue from the organism of choice, which in some cases (especially for humans) may be challenging. Additionally, since these glycans are extracted from tissue they need to be processed before use. High performance liquid chromatography (HPLC) is needed to separate the mixed glycome into individual components, which then need to be labeled. Once these libraries are generated, mass spectrometry (MS) is required to elucidate glycan structures, a major disadvantage compared with the complete availability of structures used on synthetic microarrays. Nevertheless, the advantages of shotgun arrays have led to their inclusion in the list of arrays offered by the Consortium for Functional Glycomics (307).

While glycan arrays can reveal extensive information about receptor binding diversity and specificity of influenza viruses, they have limited utility in measuring receptor binding affinity (119). Other methods are needed to probe this aspect of viral receptor binding. These methods typically involve variations on a fetuin-binding assay (308, 309), where receptor analogues are mixed with soluble or surface-bound virions and their interaction is measured. Both glycan arrays and fetuin-binding assays are endpoint experiments, however, and do not provide information about binding dynamics.

A more advanced method of measuring receptor binding affinity is biolayer interferometry (BLI), which measures light interference to detect changes in binding of molecules (or virions) on a substrate-coated sensor. This method does not require a label and can measure real-time kinetics of virus/receptor interactions (78, 108, 111, 310, 311). As speed of binding is important for rapid viral association and infection of cells, BLI is a valuable tool for measuring the receptor binding fitness of influenza viruses.

The use of naturally derived glycan arrays and the development physiologically accurate synthetic arrays will improve the information that glycan arrays can provide about virus/receptor interactions, and affinity/avidity assays can provide another layer of receptor interaction data, but these methods still have key limitations. The first is the contribution of noncanonical receptors to which influenza may bind. IAV has been suggested to bind to and infect cells through a variety of nonsialylated surface proteins (312), including dendritic cell-specific intercellular adhesion molecule-3-grabbing nonintegrin (DC-SIGN) (313) and macrophage mannose receptor (MMR) (314), and bat influenza has been shown to infect cells via major histocompatibility complex (MHC) molecules (315, 316). While naturally derived arrays can reveal some non-canonical interactions, the sample preparation process does not capture the total extracellular environment and may miss noncanonical receptors. The second limitation is the contribution of low-affinity interactions in the binding and internalization of influenza viruses. The past decade has seen more information emerge about the importance of low-affinity receptor interactions. Mice lacking α2,6 sialic acid glycans and desialylated cells were shown to still be infectable with influenza (109, 317), and low-affinity α2,3

sialic acid interactions were shown to contribute to human IAV binding and entry through multivalent interactions (108). A model has been proposed by which influenza “rolls” on the cell surface through low-affinity interactions, providing time for the virus to form multiple interactions or bind with higher-affinity receptors (108, 110). The contribution of noncanonical receptors and low-affinity receptor interactions to virus entry therefore needs to be studied by additional assays besides glycan arrays and affinity assays.

Cell binding and tropism assays are a way of exploring receptor binding interactions in a more physiologically relevant setting. Instead of isolating specific receptors and measuring their interactions with viruses, cell-based assays involve exposing cell cultures to infectious viruses and measuring which cells retain virus and in what quantity. Infection of differentiated primary human airway cells can reveal differences in cell tropism that reflect receptor binding changes between viruses. Human airway cell cultures infected with several strains of human influenza viruses revealed differential infection of ciliated and nonsialylated cells based on the virus subtype in studies with both IAV and IBV (217, 288). Single mutation differences have also been shown to alter cell tropism in respiratory tissue and cultures (261, 318, 319). Optical tweezers and atomic force microscopy (AFM)-based single-molecule force spectroscopy have even been used to measure the dynamics of virus-cell interactions (320). While these cell-based assays do not offer the same degree of receptor-level granularity as other assays, the holistic nature of these studies helps uncover virus-cell interactions that cannot be measured through other receptor binding assays, and

accounts for the low-affinity and noncanonical receptor interactions that are often overlooked.

The techniques described above can be applied to the H1N1 viruses discussed in this dissertation to expand our understanding of the receptor binding differences between them. In Chapter 2, the 5a.2 clade showed a reduction in receptor binding diversity compared with both 5a.1 and the ancestral 5a clade. The use of BLI would reveal whether this loss of diversity correlated with a loss of receptor binding affinity, which could further explain the overall fitness deficit of this virus. Similarly, BLI could show if the gain in receptor binding diversity seen by the 5a.2a and 5a.2a.1 clades in Chapter 3 is correlated with an increase in binding affinity or a change in binding kinetics. Differences in replication on human nasal epithelial cells were also seen between these clades, and a cell tropism assay could reveal whether there were differences in the subsets of cells infected with the different clades. If fewer cell types were infected with certain viruses it would offer an explanation for the reduced total virus production seen for 5a.2 and for the E224A, P137S, and K142R point mutants.

An outstanding question from the data presented in chapter 3 is the contribution of the Q189E mutation to receptor binding phenotype. This mutation was sufficient to significantly increase plaque formation (Figure 3.5A,B) and total virus production post-peak timepoint (Figure 3.5F), but did not increase receptor binding diversity on the glycan array. Given this mutation's location on the receptor binding site it is still hypothesized to confer the observed phenotypes through changes to receptor binding, but additional assays are needed to identify these changes. A shotgun glycomics array may reveal differences in receptor binding on human respiratory tract glycans (as

opposed to the synthetic array used here), and a cell-based assay may reveal differences in tropism. Alternatively, the change to a charged residue may have increased receptor binding affinity without altering overall diversity, and BLI analysis would reveal these differences. A diversification of the assays used to measure influenza receptor binding will significantly improve our ability to understand the phenotypic consequences of emergent mutations on this important aspect of virological fitness.

Investigating Epistasis through Combined Point Mutant Assays and Structural Modeling

Chapter 3 demonstrated that clinical viral isolates of the 5a.2a and 5a.2a.1 H1N1 clades had significantly improved *in vitro* fitness and were antigenically drifted compared with the 5a.2 clade. When individual clade-defining mutations were examined, however, it was found that several mutations actually reduced *in vitro* fitness. These include the E224A mutation of 5a.2a, and the P137S and K142R mutations of 5a.2a.1. This poses an important question: how do these clades show improved fitness overall when many of their clade-defining mutations reduced fitness?

The prevailing hypothesis concerns the Q189E mutation. On its own, this mutation was shown to increase plaque formation and viral replication, and was the only clade-defining mutation tested that actually improved *in vitro* fitness. It is proposed that the Q189E mutation rescues these clades from the fitness costs associated with the other mutations. While E224A, P137S, and K142R reduce *in vitro* fitness on their own,

the Q189E mutation may interact epistatically with these mutations to ameliorate their fitness costs. To test this hypothesis, viruses need to be generated using the infectious clone system described above which contain combinations of Q189E and E224A, P137S, or K142R to determine whether the addition of Q189E is able to rescue viral fitness. These combined point mutant viruses should be tested on the assays used above and compared with single point mutants; if this hypothesis is correct, the addition of Q189E (Q189E + E224A, Q189E + P137S, Q189E + K142R) will significantly increase plaque size and total virus production compared with the single mutants (E224A, P137S, K142R). These viruses should also be characterized using the additional receptor binding methods described above, which will reveal more information about receptor binding differences produced by the combination of these RBS clade-defining mutations. Considering the Q189E also emerged in the context of the 5a.1 clade (Chapter 2), it would also be beneficial to study the effects of this mutation on the genetic background of that clade to determine whether its phenotypic contributions were different between 5a.1, 5a.2a, and 5a.2a.1.

The other question that arises from the individual point mutant data concerns the reason for their emergence. If E224A, P137S, and K142R all reduced *in vitro* fitness on their own then why did they predominate; i.e. what was the evolutionary advantage they conferred? The answer likely lies in their contribution to antigenic drift. These mutations were all shown to reduce virus neutralization by serum antibodies (Figure 3.7), indicating that they altered the antigenic profile of the virus. While viral fitness is an important component of viral success, antigenic drift and the evasion of existing immunity is also crucial for the virus. These mutations did not confer fitness advantages,

but they may have yielded antigenic advantages that made them valuable and evolutionarily favorable. This is especially true if the Q189E mutation is able to mitigate their negative fitness consequences. Conversely, the Q189E mutation was actually shown to increase virus neutralization (Figure 3.7) and likely emerged because of its contribution to fitness. It is hypothesized that the combination of putative antigenic drift mutations (E224A, P137S, and K142R) and fitness-improving mutations (Q189E) resulted in a virus clade that retained the positive contributions of each mutation while mitigating their negative effects. This is corroborated by the clinical virus isolate data for these clades, which demonstrated both improved *in vitro* fitness and increased antigenic drift. The combined point mutant viruses described above should also be tested for antigenic drift through neutralization assays, which will confirm whether the addition of E224A, P137S, and K142R reduces virus neutralization in the presence of the Q189E mutation.

There are mechanistic explanations for how these mutations can interact epistatically to produce different phenotypes when present in combination versus on their own. The secondary and tertiary structure of the influenza hemagglutinin determines the shape and flexibility of the RBS, and this structure is determined by interactions between amino acids in the RBS, including the 130 loop, 190 helix, and 220 loop (37). Changing these residues can modify this structure, leading to altered receptor binding properties (92-94, 113, 123, 177, 192, 193, 263, 270, 279, 292, 318, 319). Since there is extensive interaction between many residues in the RBS, the effect that one mutation has depends on the identity of the residues with which it interacts. The phenotype of the E224A mutation has been shown to change depending on the

presence of a secondary mutation, T197A, and their combination greatly increased receptor binding affinity and diversity compared to individually (193). Interestingly, the 197 residue is not directly on the receptor binding site, demonstrating the dependence of the RBS conformation on a variety of residues within HA. Similarly, another study demonstrated that the sequential addition of circulating RBS mutations modulated the receptor binding properties of a recombinant virus, confirming that epistatic interactions are involved in the mutations that appear in the wild (263). A structural analysis revealed that the combination of S183P and E224A induces a more open and flexible RBS due to changing interactions between the 190 helix and 220 loop (262). Jayaraman *et al*. predicted that the combination of S183P, A186T, and E224A stabilized the RBS by producing a more hydrophobic interaction network, and confirmed that these mutations increased receptor binding affinity (279). This ample evidence suggests that E224A might actually be beneficial to receptor binding in the context of the other clade-defining mutations of 5a.2a. The impact of P137S and K142R on RBS structure and function may similarly be dependent on the presence of these other mutations. The Q189E mutation is a change from a polar to a charged residue and may have significant ramifications for the interactions between the 190 helix and the 130 and 220 loops, on which E224A, P137S, and K142R are situated. In addition to the combined point mutant assays proposed here, modeling of the RBS with these mutations may offer mechanistic insight into the structural changes induced by these mutations on their own versus combined.

Inter-Gene Epistasis and Viral Evolution

The above discussion, and the scope of this dissertation, is limited to intra-gene epistasis of the HA protein. It is important to note that epistatic interactions are not limited to HA, and have been shown to extend to other viral proteins, including NS1 (321), NP (179, 280), and the polymerase proteins (322). In addition, interactions between influenza proteins play a major role in overall viral phenotype. The most striking example is pandemic influenza, whose emergence is often facilitated by reassortment events between viruses that produce a combination of segments with newfound ability to transmit between humans (43, 147). Less dramatic changes can also take place between circulating seasonal influenza through the appearance of reassorted viruses with altered fitness. Statistical analyses of seasonal influenza has shown that reassortment events are associated with increased viral fitness and strain longevity (323). A reassortment event between seasonal H3N2 was associated with clade success independent of HA background, suggesting epistatic interactions between non-HA segments improving viral phenotype (324).

In addition to reassortment events driving epistatic phenotypes, inter-gene epistasis can influence the mutations that emerge on influenza proteins. Phylogenetic analysis has revealed that HA and NA evolution is coordinated, and that mutations in one protein can influence the acquisition of mutations on the other (185). The appearance of HA mutations has also been shown to be dependent on NA fitness, corroborating the theory that the balance of HA receptor binding and NA receptor cleaving must be maintained during viral evolution (181). Beyond HA and NA, epistatic

interactions have also been demonstrated between polymerase segments, which are able to modulate replication fidelity and viral growth (322).

This evidence confirms the need for studying inter-gene interactions in circulating influenza clades. Using the reverse genetics system described above, reassortant viruses can be generated with desired gene segment swaps to identify how the phenotype of one gene segment is influenced by the genetic background of the other sequences. Reassortant viruses can be characterized using the replication, receptor binding, and virus neutralization assays described in this dissertation, as well as other targeted assays to measure various viral properties (e.g. genome replication, virion motility, particle formation, innate immune activation). Considering that there have been mutations to many gene segments beyond HA in circulating H1N1, it is necessary to understand whether these mutations confer fitness changes to the virus. One example is investigating HA-NA balance; if the receptor affinity experiments proposed above reveal differences between circulating clades, neuraminidase activity should be investigate to see if there is a corresponding change in enzymatic activity, and if so the causative mutation should be identified through site-directed mutagenesis. These experiments could further our understanding of the evolutionary mechanisms behind HA-NA balancing.

Transmission as a Component of Viral Fitness

The experiments described and proposed above have been limited to *in vitro* or *in silico* approaches. *In vitro* methods are useful due to their ability to target specific

components of viral fitness, from receptor binding to viral replication, membrane fusion, or neuraminidase activity. *In silico* experiments allow for the detailed examination of both structural and large-scale evolutionary aspects of viral fitness. What they lack, however, is their ability to model the course of an actual influenza human in a mammalian host. *In vivo* experiments are therefore a valuable tool in the virologist's toolbox, and can link mechanistic *in vitro* experiments with a more holistic viral phenotype.

There are several model organisms that are used to study influenza infection. Mice are the most common, as they are a very well established model system and provide logistic benefits over other organisms (325). They are typically not readily infectable with human influenza, however, and often the virus needs to be adapted to a mouse model before it is usable (325). This is acceptable for many applications, including studying host response to infection, but limits the utility of mice in studying unadapted human clinical isolates. Several other organisms including guinea pigs and hamsters have also been used, but the preferred model organism for studying influenza infection and transmission *in vivo* are ferrets. Ferrets have a similar distribution of receptors on their respiratory tract (37) and are easily infectable with human influenza viruses, producing a respiratory pathology that can be readily measured (325). Transmission studies using ferrets typically involves either co-caging or housing in separate but adjacent cages. The latter offers the benefit of specifically measuring airborne transmission (aerosol and/or respiratory droplets), while the former can produce transmission through other routes including touch (325).

In addition to the abundance of research on infection and transmission of animal viruses in ferrets (326-330), ferrets have been used to study the different *in vivo* properties of human virus clades. One study looked at the evolution of circulating H1N1 and the differing abilities of strains to infect ferrets and transmit via aerosols (331), and different H1N1 viruses were shown to have differential transmissibility and competitive fitness depending on the presence of NA resistance mutations (332). Single mutation differences have also been studied using ferrets, and have the ability to changes in a variety of infection and transmission properties (279, 333-335). Ferret infection studies would provide useful insight into the effect of the 5a.2a and 5a.2a.1 clade mutations on *in vivo* viral properties if applied to the viruses discussed in this dissertation. Specifically, it would be useful to see how the Q189E mutation affects tissue tropism, pathology, and transmissibility in ferrets, given its enhanced *in vitro* fitness properties. Additionally, it would be helpful to see whether the E224A, P137S, and K142R mutations actually produce an attenuated fitness phenotype *in vivo* that corresponds to the *in vitro* data presented here. Correlating *in vitro* and *in vivo* fitness would also lend more support to the use of *in vitro* experiments as an physiologically relevant way of characterizing viral fitness. The benefits of these experiments should always be weighed against the animal sacrifice required, however, since a precondition of ethical animal experimentation is that only the most necessary experiments be conducted.

Ferrets can also be used to study antigenic properties. Ferret serum antibodies are commonly used as an alternative to the human serum used in neutralization assays, but challenge trials with vaccinated ferrets are also an effective way of measuring vaccine efficacy and antigenic drift (336, 337). One important caveat, however, is that

the antibodies raised by ferrets are not identical to those raised by humans. B cell ferret immunity more heavily targets certain antigenic sites (214), and naïve ferret vaccination and/or challenge may not accurately reflect the immunological landscape of humans who receive repeated exposure to diverse influenza antigens (338). Nevertheless, correlating the human serum neutralization assay data presented in this dissertation with differences in vaccine-conferred protection in ferrets would strengthen the antigenic drift data for these H1N1 clades.

Continued Surveillance of H1N1 into 2024

The research presented here outlines the changes to circulating H1N1 from 2019-2023. While the 2019-20 NH influenza season was a majority-H1N1 season, the United States did not see H1N1 predominate again until the most recent season, 2023-24. This season saw the continued circulation of the 5a.2a and 5a.2a.1 clades, with 5a.2a.1 comprising 75% of all H1N1 characterized (290). Trends in Baltimore, Maryland were similar; while 5a.2a continued to be reported, 5a.2a.1 gained a sizable increase in prevalence compared with past seasons. These data suggest that the 5a.2a.1 clade, while slower to take off compared with the 5a.2a clade, has a sufficiently robust phenotype to allow it to succeed.

While no new clades have yet been classified, several mutations have appeared on 5a.2a and 5a.2a.1 since the 2022-23 NH season. Most significantly are the concurrent appearances of previously seen mutations. For example, the T216A mutation has emerged at least three separate times on different evolutionary branches

of H1N1. This dissertation presented the characterization a clinical virus containing this mutation (A/Baltimore/JH-22377/2022), and it did not appear to significantly affect viral phenotype as this virus performed similarly to the other 5a.2a clade, A/Baltimore/JH-22363/2022. It is therefore still unclear what advantages are conferred by this mutation that could have led to its repeated appearance. It is certainly possible that it produces antigenic or receptor binding differences not captured by the assays employed here; T216A has shown to increase receptor affinity and improve transmission in ferrets, and is proposed to stabilize the RBS similarly to the S183P, A186T, and E224A combination discussed above (279). That study noted that the 216 and 224 residue pairs tend to either both be hydrophilic or hydrophobic, but rarely a combination of hydrophilic and hydrophobic. It is interesting that this T216A mutation only began appearing now that the predominant residue at 224 is also hydrophobic, and seems to confirms the authors' hypothesis.

Of note, the P137S and K142R mutations have both appeared independently of 5a.2a.1, and are present on their own in certain 5a.2a clusters. Again, this speaks to the putative evolutionary advantage of these mutations, and from the data presented above it is likely that the advantages conferred are largely antigenic. Considering that the vaccine clade for the 2023-24 and 2024-25 NH influenza seasons is a 5a.2a.1 virus containing these mutations, and that the 2023-24 NH influenza season was comprised largely of the 5a.2a.1 virus, it will be interesting to see whether these mutations are still advantageous to the virus now that population-level exposure to them is higher. The T120A mutation has also emerged multiple times on both 5a.2a and 5a.2a.1, though the possible phenotypic contributions of this mutation is unclear since it does not occupy an

antigenic site or the RBS. It is clear, however, that stalk and non-RBS mutations still have the potential to influence viral receptor binding (193, 194), and this mutation still merits characterization given its epidemiologic importance.

Finally, the end of the 2023-24 season saw the resurgence of H3N2. Considering the historically cyclical nature of influenza subtype predominance, there is a large possibility that H3N2 will predominate over H1N1 in the coming seasons. The tools used in this dissertation, and those proposed in this discussion, are broadly applicable to many influenza subtypes and should be employed to study the predominating virus regardless of its identity. This extends to IBV as well, which remains a public health burden despite often being overshadowed by IAV. As has been demonstrated here, influenza subtypes continue to evolve even under limited circulation, and the changes they acquire in low-circulation years can contribute to improvements in phenotype that allow it to resurge. Virological surveillance is therefore a necessary and perennial task in disease preparedness and response.

REFERENCES

1. Caini S, Kroneman M, Wiegers T, El Guerche-Séblain C, Paget J. 2018. Clinical characteristics and severity of influenza infections by virus type, subtype, and lineage: a systematic literature review. Influenza and other respiratory viruses 12:780-792.
2. WHO. 4 January 2023 2018. Influenza (seasonal), *on* World Health Organization. https://www.who.int/en/news-room/fact-sheets/detail/influenza-(seasonal). Accessed
3. Cox NJ, Subbarao K. 2000. Global epidemiology of influenza: past and present. Annual review of medicine 51:407-421.
4. Lagace-Wiens PR, Rubinstein E, Gumel A. 2010. Influenza epidemiology—past, present, and future. Critical care medicine 38:e1-e9.
5. Martínez A, Soldevila N, Romero-Tamarit A, Torner N, Godoy P, Rius C, Jané M, Domínguez À, Group SoHCoSIiCW. 2019. Risk factors associated with severe outcomes in adult hospitalized patients according to influenza type and subtype. PloS one 14:e0210353.
6. Wu P, Presanis AM, Bond HS, Lau EH, Fang VJ, Cowling BJ. 2017. A joint analysis of influenza-associated hospitalizations and mortality in Hong Kong, 1998–2013. Scientific reports 7:929.
7. Caini S, Huang QS, Ciblak MA, Kusznierz G, Owen R, Wangchuk S, Henriques CM, Njouom R, Fasce RA, Yu H. 2015. Epidemiological and virological characteristics of influenza B: results of the Global Influenza B Study. Influenza and other respiratory viruses 9:3-12.

8. Quandelacy TM, Viboud C, Charu V, Lipsitch M, Goldstein E. 2014. Age-and sex-related risk factors for influenza-associated mortality in the United States between 1997–2007. American journal of epidemiology 179:156-167.
9. Simonsen L, Clarke MJ, Schonberger LB, Arden NH, Cox NJ, Fukuda K. 1998. Pandemic versus epidemic influenza mortality: a pattern of changing age distribution. Journal of infectious diseases 178:53-60.
10. Klein SL, Hodgson A, Robinson DP. 2012. Mechanisms of sex disparities in influenza pathogenesis. Journal of leukocyte biology 92:67-73.
11. Peretz J, Hall OJ, Klein SL. 2015. Sex differences in influenza virus infection, vaccination, and therapies. Sex and gender differences in infection and treatments for infectious diseases:183-210.
12. Brankston G, Gitterman L, Hirji Z, Lemieux C, Gardam M. 2007. Transmission of influenza A in human beings. The Lancet infectious diseases 7:257-265.
13. Killingley B, Nguyen-Van-Tam J. 2013. Routes of influenza transmission. Influenza and other respiratory viruses 7:42-51.
14. Atkinson MP, Wein LM. 2008. Quantifying the routes of transmission for pandemic influenza. Bulletin of mathematical biology 70:820-867.
15. Thompson K-A, Bennett A. 2017. Persistence of influenza on surfaces. Journal of Hospital Infection 95:194-199.
16. Zhang N, Li Y. 2018. Transmission of influenza A in a student office based on realistic person-to-person contact and surface touch behaviour. International journal of environmental research and public health 15:1699.

17. Bean B, Moore B, Sterner B, Peterson L, Gerding D, Balfour Jr H. 1982. Survival of influenza viruses on environmental surfaces. Journal of infectious diseases 146:47-51.

18. Javanian M, Barary M, Ghebrehewet S, Koppolu V, Vasigala V, Ebrahimpour S. 2021. A brief review of influenza virus infection. Journal of medical virology 93:4638-4646.

19. Taubenberger JK, Morens DM. 2008. The pathology of influenza virus infections. Annu Rev Pathol Mech Dis 3:499-522.

20. Lau LL, Cowling BJ, Fang VJ, Chan K-H, Lau EH, Lipsitch M, Cheng CK, Houck PM, Uyeki TM, Peiris JM. 2010. Viral shedding and clinical illness in naturally acquired influenza virus infections. The Journal of infectious diseases 201:1509-1516.

21. Lee N, Chan PK, Hui DS, Rainer TH, Wong E, Choi K-W, Lui GC, Wong BC, Wong RY, Lam W-Y. 2009. Viral loads and duration of viral shedding in adult patients hospitalized with influenza. The Journal of infectious diseases 200:492-500.

22. Ip DK, Lau LL, Leung NH, Fang VJ, Chan K-H, Chu DK, Leung GM, Peiris JM, Uyeki TM, Cowling BJ. 2017. Viral shedding and transmission potential of asymptomatic and paucisymptomatic influenza virus infections in the community. Clinical infectious diseases 64:736-742.

23. Leung NH, Xu C, Ip DK, Cowling BJ. 2015. The fraction of influenza virus infections that are asymptomatic: a systematic review and meta-analysis. Epidemiology 26:862-872.

24. Liu Q, Zhou Y-h, Yang Z-q. 2016. The cytokine storm of severe influenza and development of immunomodulatory therapy. Cellular & molecular immunology 13:3-10.

25. Gu Y, Zuo X, Zhang S, Ouyang Z, Jiang S, Wang F, Wang G. 2021. The mechanism behind influenza virus cytokine storm. Viruses 13:1362.

26. CDC. March 30, 2023. Types of Influenza Viruses, *on* Centers for Disease Control and Prevention. https://www.cdc.gov/flu/about/viruses/types.htm. Accessed March 21, 2024.

27. Puig-Barberà J, Tormos A, Trushakova S, Sominina A, Pisareva M, Ciblak MA, Badur S, Yu H, Cowling BJ, Burtseva E. 2015. The Global Influenza Hospital Surveillance Network (GIHSN): a new platform to describe the epidemiology of severe influenza. Influenza and other respiratory viruses 9:277-286.

28. Zanobini P, Bonaccorsi G, Lorini C, Haag M, McGovern I, Paget J, Caini S. 2022. Global patterns of seasonal influenza activity, duration of activity and virus (sub) type circulation from 2010 to 2020. Influenza and Other Respiratory Viruses 16:696-706.

29. CDC. February 1, 2024. Influenza Type A Viruses, *on* Centers for Disease Control and Prevention. https://www.cdc.gov/flu/avianflu/influenza-a-virus-subtypes.htm. Accessed March 21, 2024.

30. Karlsson EA, Mook PAN, Vandemaele K, Fitzner J, Hammond A, Cozza V, Zhang W, Moen A. 2021. Review of global influenza circulation, late 2019 to 2020, and the impact of the COVID-19 pandemic on influenza circulation. Weekly epidemiological record 25:241-264.

31. Dhanasekaran V, Sullivan S, Edwards KM, Xie R, Khvorov A, Valkenburg SA, Cowling BJ, Barr IG. 2022. Human seasonal influenza under COVID-19 and the potential consequences of influenza lineage elimination. Nature communications 13:1721.

32. WHO. January 9, 2023 2023. Influenza Update N° 436, *on* World Health Organization. https://www.who.int/publications/m/item/influenza-update-n-436. Accessed March 21, 2024.

33. Ducatez M, Webster R, Webby R. 2008. Animal influenza epidemiology. Vaccine 26:D67-D69.

34. Harder TC, Vahlenkamp TW. 2010. Influenza virus infections in dogs and cats. Veterinary immunology and immunopathology 134:54-60.

35. Voorhees IE, Glaser AL, Toohey-Kurth K, Newbury S, Dalziel BD, Dubovi EJ, Poulsen K, Leutenegger C, Willgert KJ, Brisbane-Cohen L. 2017. Spread of canine influenza A (H3N2) virus, United States. Emerging infectious diseases 23:1950.

36. Crawford P, Dubovi EJ, Castleman WL, Stephenson I, Gibbs E, Chen L, Smith C, Hill RC, Ferro P, Pompey J. 2005. Transmission of equine influenza virus to dogs. science 310:482-485.

37. de Graaf M, Fouchier RA. 2014. Role of receptor binding specificity in influenza A virus transmission and pathogenesis. The EMBO journal 33:823-841.

38. Ito T. 2000. Interspecies transmission and receptor recognition of influenza A viruses. Microbiology and immunology 44:423-430.

39. Matrosovich M, Stech J, Dieter Klenk H. 2009. Influenza receptors, polymerase and host range. Revue scientifique et technique 28:203.

40. Hsieh Y-C, Wu T-Z, Liu D-P, Shao P-L, Chang L-Y, Lu C-Y, Lee C-Y, Huang F-Y, Huang L-M. 2006. Influenza pandemics: past, present and future. Journal of the Formosan Medical Association 105:1-6.

41. Kilbourne ED. 2006. Influenza pandemics of the 20th century. Emerging infectious diseases 12:9.

42. Monto AS, Fukuda K. 2020. Lessons from influenza pandemics of the last 100 years. Clinical Infectious Diseases 70:951-957.

43. Guan Y, Vijaykrishna D, Bahl J, Zhu H, Wang J, Smith GJ. 2010. The emergence of pandemic influenza viruses. Protein & cell 1:9-13.

44. Charostad J, Rukerd MRZ, Mahmoudvand S, Bashash D, Hashemi SMA, Nakhaie M, Zandi K. 2023. A comprehensive review of highly pathogenic avian influenza (HPAI) H5N1: An imminent threat at doorstep. Travel medicine and infectious disease:102638.

45. CDC. January 12, 2024. Highlights in the History of Avian Influenza (Bird Flu) Timeline – 2020-2023, *on* Centers for Disease Control and Prevention. https://www.cdc.gov/flu/avianflu/timeline/avian-timeline-2020s.htm. Accessed March 21, 2024.

46. Farahat RA, Khan SH, Rabaan AA, Al-Tawfiq JA. 2023. The resurgence of Avian influenza and human infection: A brief outlook. New Microbes and New Infections 53.

47. Vora NM, Hannah L, Walzer C, Vale MM, Lieberman S, Emerson A, Jennings J, Alders R, Bonds MH, Evans J. 2023. Interventions to reduce risk for pathogen spillover and early disease spread to prevent outbreaks, epidemics, and pandemics. Emerging infectious diseases 29.

48. Dou D, Revol R, Östbye H, Wang H, Daniels R. 2018. Influenza A virus cell entry, replication, virion assembly and movement. Frontiers in immunology 9:383042.

49. Wang M, Veit M. 2016. Hemagglutinin-esterase-fusion (HEF) protein of influenza C virus. Protein & cell 7:28-45.

50. Vasin A, Temkina O, Egorov V, Klotchenko S, Plotnikova M, Kiselev O. 2014. Molecular mechanisms enhancing the proteome of influenza A viruses: an overview of recently discovered proteins. Virus research 185:53-63.

51. Carter T, Iqbal M. 2024. The Influenza A Virus Replication Cycle: A Comprehensive Review. Viruses 16:316.

52. Moeller A, Kirchdoerfer RN, Potter CS, Carragher B, Wilson IA. 2012. Organization of the influenza virus replication machinery. Science 338:1631-1634.

53. Ruigrok RW, Barge A, Durrer P, Brunner J, Ma K, Whittaker GR. 2000. Membrane interaction of influenza virus M1 protein. Virology 267:289-298.

54. Lamb RA, Zebedee SL, Richardson CD. 1985. Influenza virus M2 protein is an integral membrane protein expressed on the infected-cell surface. Cell 40:627-633.

55. Wiley DC, Skehel JJ. 1987. The structure and function of the hemagglutinin membrane glycoprotein of influenza virus. Annual review of biochemistry 56:365-394.

56. Kilbourne ED, Murphy JS. 1960. Genetic studies of influenza viruses: I. Viral morphology and growth capacity as exchangeable genetic traits. Rapid in ovo adaptation of early passage Asian strain isolates by combination with PR8. The Journal of experimental medicine 111:387.

57. Badham MD, Rossman JS. 2016. Filamentous influenza viruses. Current clinical microbiology reports 3:155-161.

58. Skehel JJ, Wiley DC. 2000. Receptor binding and membrane fusion in virus entry: the influenza hemagglutinin. Annual review of biochemistry 69:531-569.

59. Lakadamyali M, Rust MJ, Zhuang X. 2004. Endocytosis of influenza viruses. Microbes and infection 6:929-936.

60. Edinger TO, Pohl MO, Stertz S. 2014. Entry of influenza A virus: host factors and antiviral targets. Journal of General Virology 95:263-277.

61. Sieczkarski SB, Whittaker GR. 2002. Influenza virus can enter and infect cells in the absence of clathrin-mediated endocytosis. Journal of virology 76:10455-10464.

62. Rossman JS, Leser GP, Lamb RA. 2012. Filamentous influenza virus enters cells via macropinocytosis. Journal of virology 86:10950-10960.

63. Pinto LH, Holsinger LJ, Lamb RA. 1992. Influenza virus M2 protein has ion channel activity. cell 69:517-528.

64. Manzoor R, Igarashi M, Takada A. 2017. Influenza A virus M2 protein: roles from ingress to egress. International journal of molecular sciences 18:2649.

65. Peukes J, Xiong X, Erlendsson S, Qu K, Wan W, Calder LJ, Schraidt O, Kummer S, Freund SM, Kräusslich H-G. 2020. The native structure of the assembled matrix protein 1 of influenza A virus. Nature 587:495-498.

66. Te Velthuis AJ, Fodor E. 2016. Influenza virus RNA polymerase: insights into the mechanisms of viral RNA synthesis. Nature Reviews Microbiology 14:479-493.

67. Li H, Wu Y, Li M, Guo L, Gao Y, Wang Q, Zhang J, Lai Z, Zhang X, Zhu L. 2023. An intermediate state allows influenza polymerase to switch smoothly between transcription and replication cycles. Nature structural & molecular biology 30:1183-1192.

68. Pflug A, Lukarska M, Resa-Infante P, Reich S, Cusack S. 2017. Structural insights into RNA synthesis by the influenza virus transcription-replication machine. Virus research 234:103-117.

69. Poon LL, Pritlove DC, Fodor E, Brownlee GG. 1999. Direct evidence that the poly (A) tail of influenza A virus mRNA is synthesized by reiterative copying of a U track in the virion RNA template. Journal of virology 73:3473-3476.

70. Paterson D, Fodor E. 2012. Emerging roles for the influenza A virus nuclear export protein (NEP). PLoS pathogens 8:e1003019.

71. Boulo S, Akarsu H, Ruigrok RW, Baudin F. 2007. Nuclear traffic of influenza virus proteins and ribonucleoprotein complexes. Virus research 124:12-21.

72. Hale BG, Randall RE, Ortín J, Jackson D. 2008. The multifunctional NS1 protein of influenza A viruses. Journal of general virology 89:2359-2376.

73. Rossman JS, Lamb RA. 2011. Influenza virus assembly and budding. Virology 411:229-236.

74. Rossman JS, Jing X, Leser GP, Lamb RA. 2010. Influenza virus M2 protein mediates ESCRT-independent membrane scission. Cell 142:902-913.

75. Scheiffele P, Rietveld A, Wilk T, Simons K. 1999. Influenza viruses select ordered lipid domains during budding from the plasma membrane. Journal of Biological Chemistry 274:2038-2044.

76. Chlanda P, Schraidt O, Kummer S, Riches J, Oberwinkler H, Prinz S, Kräusslich H-G, Briggs JA. 2015. Structural analysis of the roles of influenza A virus membrane-associated proteins in assembly and morphology. Journal of virology 89:8957-8966.

77. Gaymard A, Le Briand N, Frobert E, Lina B, Escuret V. 2016. Functional balance between neuraminidase and haemagglutinin in influenza viruses. Clinical Microbiology and Infection 22:975-983.

78. Guo H, Rabouw H, Slomp A, Dai M, van der Vegt F, van Lent JW, McBride R, Paulson JC, de Groot RJ, van Kuppeveld FJ. 2018. Kinetic analysis of the influenza A virus HA/NA balance reveals contribution of NA to virus-receptor binding and NA-dependent rolling on receptor-containing surfaces. PLoS pathogens 14:e1007233.

79. de Vries RP, de Vries E, Bosch BJ, de Groot RJ, Rottier PJ, de Haan CA. 2010. The influenza A virus hemagglutinin glycosylation state affects receptor-binding specificity. Virology 403:17-25.

80. Schulze IT. 1997. Effects of glycosylation on the properties and functions of influenza virus hemagglutinin. Journal of Infectious Diseases 176:S24-S28.

81. Kim JI, Park M-S. 2012. N-linked glycosylation in the hemagglutinin of influenza A viruses. Yonsei medical journal 53:886.

82. Das SR, Hensley SE, David A, Schmidt L, Gibbs JS, Puigbò P, Ince WL, Bennink JR, Yewdell JW. 2011. Fitness costs limit influenza A virus hemagglutinin glycosylation as an immune evasion strategy. Proceedings of the National Academy of Sciences 108:E1417-E1422.

83. Garten W, Klenk H-D. 2008. Cleavage activation of the influenza virus hemagglutinin and its role in pathogenesis, p 156-167, Avian influenza, vol 27. Karger Publishers.

84. Bertram S, Glowacka I, Steffen I, Kühl A, Pöhlmann S. 2010. Novel insights into proteolytic cleavage of influenza virus hemagglutinin. Reviews in medical virology 20:298-310.

85. Böttcher-Friebertshäuser E, Garten W, Matrosovich M, Klenk HD. 2014. The hemagglutinin: a determinant of pathogenicity. Influenza Pathogenesis and Control-Volume I:3-34.

86. Wu NC, Wilson IA. 2020. Influenza hemagglutinin structures and antibody recognition. Cold Spring Harbor perspectives in medicine 10:a038778.

87. Stieneke-Gröber A, Vey M, Angliker H, Shaw E, Thomas G, Roberts C, Klenk H, Garten W. 1992. Influenza virus hemagglutinin with multibasic cleavage site is activated by furin, a subtilisin-like endoprotease. The EMBO journal 11:2407-2414.

88. Chen BJ, Leser GP, Morita E, Lamb RA. 2007. Influenza virus hemagglutinin and neuraminidase, but not the matrix protein, are required for assembly and budding of plasmid-derived virus-like particles. Journal of virology 81:7111-7123.

89. Schjoldager KT, Narimatsu Y, Joshi HJ, Clausen H. 2020. Global view of human protein glycosylation pathways and functions. Nature reviews Molecular cell biology 21:729-749.

90. Reily C, Stewart TJ, Renfrow MB, Novak J. 2019. Glycosylation in health and disease. Nature Reviews Nephrology 15:346-366.

91. Martín J, Wharton SA, Lin YP, Takemoto DK, Skehel JJ, Wiley DC, Steinhauer DA. 1998. Studies of the binding properties of influenza hemagglutinin receptor-site mutants. Virology 241:101-111.

92. Powell H, Liu H, Pekosz A. 2021. Changes in sialic acid binding associated with egg adaptation decrease live attenuated influenza virus replication in human nasal epithelial cell cultures. Vaccine 39:3225-3235.

93. Kumari K, Gulati S, Smith DF, Gulati U, Cummings RD, Air GM. 2007. Receptor binding specificity of recent human H3N2 influenza viruses. Virology journal 4:1-11.

94. Glaser L, Stevens J, Zamarin D, Wilson IA, García-Sastre A, Tumpey TM, Basler CF, Taubenberger JK, Palese P. 2005. A single amino acid substitution in 1918 influenza virus hemagglutinin changes receptor binding specificity. Journal of virology 79:11533-11536.

95. Kuchipudi SV, Nelli RK, Gontu A, Satyakumar R, Surendran Nair M, Subbiah M. 2021. Sialic acid receptors: the key to solving the enigma of zoonotic virus spillover. Viruses 13:262.

96. Rogers GN, Paulson JC. 1983. Receptor determinants of human and animal influenza virus isolates: differences in receptor specificity of the H3 hemagglutinin based on species of origin. Virology 127:361-373.

97. Connor RJ, Kawaoka Y, Webster RG, Paulson JC. 1994. Receptor specificity in human, avian, and equine H2 and H3 influenza virus isolates. Virology 205:17-23.

98. Shinya K, Ebina M, Yamada S, Ono M, Kasai N, Kawaoka Y. 2006. Influenza virus receptors in the human airway. Nature 440:435-436.

99. Peteranderl C, Herold S, Schmoldt C. Human influenza virus infections, p 487-500. *In* (ed), Thieme Medical Publishers,

100. Van Poucke SG, Nicholls JM, Nauwynck HJ, Van Reeth K. 2010. Replication of avian, human and swine influenza viruses in porcine respiratory explants and association with sialic acid distribution. Virology journal 7:1-14.

101. Ito T, Couceiro JNS, Kelm S, Baum LG, Krauss S, Castrucci MR, Donatelli I, Kida H, Paulson JC, Webster RG. 1998. Molecular basis for the generation in pigs of influenza A viruses with pandemic potential. Journal of virology 72:7367-7373.

102. Ma W, Kahn RE, Richt JA. 2009. The pig as a mixing vessel for influenza viruses: Human and veterinary implications. Journal of molecular and genetic medicine: an international journal of biomedical research 3:158.

103. Abdelwhab EM, Mettenleiter TC. 2023. Zoonotic animal influenza virus and potential mixing vessel hosts. Viruses 15:980.

104. Nelson MI, Worobey M. 2018. Origins of the 1918 pandemic: revisiting the swine “mixing vessel” hypothesis. American journal of epidemiology 187:2498-2502.

105. Nelli RK, Kuchipudi SV, White GA, Perez BB, Dunham SP, Chang K-C. 2010. Comparative distribution of human and avian type sialic acid influenza receptors in the pig. BMC veterinary research 6:1-9.

106. Walther T, Karamanska R, Chan RW, Chan MC, Jia N, Air G, Hopton C, Wong MP, Dell A, Malik Peiris J. 2013. Glycomic analysis of human respiratory tract tissues and correlation with influenza virus infection. PLoS pathogens 9:e1003223.

107. Nicholls JM, Bourne AJ, Chen H, Guan Y, Peiris JM. 2007. Sialic acid receptor detection in the human respiratory tract: evidence for widespread distribution of potential binding sites for human and avian influenza viruses. Respiratory research 8:1-10.

108. Liu M, Huang LZ, Smits AA, Büll C, Narimatsu Y, van Kuppeveld FJ, Clausen H, de Haan CA, de Vries E. 2022. Human-type sialic acid receptors contribute to avian influenza A virus binding and entry by hetero-multivalent interactions. Nature Communications 13:1-12.

109. Glaser L, Conenello G, Paulson J, Palese P. 2007. Effective replication of human influenza viruses in mice lacking a major α2, 6 sialyltransferase. Virus research 126:9-18.

110. Nicholls JM, Chan RW, Russell RJ, Air GM, Peiris JM. 2008. Evolving complexities of influenza virus and its receptors. Trends in microbiology 16:149-157.

111. Lin YP, Xiong X, Wharton SA, Martin SR, Coombs PJ, Vachieri SG, Christodoulou E, Walker PA, Liu J, Skehel JJ. 2012. Evolution of the receptor binding properties of the influenza A (H3N2) hemagglutinin. Proceedings of the National Academy of Sciences 109:21474-21479.

112. Byrd-Leotis L, Gao C, Jia N, Mehta AY, Trost J, Cummings SF, Heimburg-Molinaro J, Cummings RD, Steinhauer DA. 2019. Antigenic pressure on H3N2 influenza virus drift strains imposes constraints on binding to sialylated receptors but not phosphorylated glycans. Journal of virology 93:e01178-19.

113. Nobusawa E, Ishihara H, Morishita T, Sato K, Nakajima K. 2000. Change in receptor-binding specificity of recent human influenza A viruses (H3N2): a single amino acid change in hemagglutinin altered its recognition of sialyloligosaccharides. Virology 278:587-596.

114. Bolton MJ, Ort JT, McBride R, Swanson NJ, Wilson J, Awofolaju M, Furey C, Greenplate AR, Drapeau EM, Pekosz A. 2022. Antigenic and virological properties of an H3N2 variant that continues to dominate the 2021–22 Northern Hemisphere influenza season. Cell Reports 39.

115. Gulati S, Smith DF, Air GM. 2009. Deletions of neuraminidase and resistance to oseltamivir may be a consequence of restricted receptor specificity in recent H3N2 influenza viruses. Virology journal 6:1-15.

116. Childs RA, Palma AS, Wharton S, Matrosovich T, Liu Y, Chai W, Campanero-Rhodes MA, Zhang Y, Eickmann M, Kiso M. 2009. Receptor-binding specificity of pandemic influenza A (H1N1) 2009 virus determined by carbohydrate microarray. Nature biotechnology 27:797-799.

117. Maines TR, Jayaraman A, Belser JA, Wadford DA, Pappas C, Zeng H, Gustin KM, Pearce MB, Viswanathan K, Shriver ZH. 2009. Transmission and pathogenesis of swine-origin 2009 A (H1N1) influenza viruses in ferrets and mice. Science 325:484-487.

118. Liao H-Y, Hsu C-H, Wang S-C, Liang C-H, Yen H-Y, Su C-Y, Chen C-H, Jan J-T, Ren C-T, Chen C-H. 2010. Differential receptor binding affinities of influenza hemagglutinins on glycan arrays. Journal of the American Chemical Society 132:14849-14856.

119. Gulati S, Lasanajak Y, Smith DF, Cummings RD, Air GM. 2014. Glycan array analysis of influenza H1N1 binding and release. Cancer Biomarkers 14:43-53.

120. Nunthaboot N, Rungrotmongkol T, Malaisree M, Kaiyawet N, Decha P, Sompornpisut P, Poovorawan Y, Hannongbua S. 2010. Evolution of human receptor binding affinity of H1N1 hemagglutinins from 1918 to 2009 pandemic influenza A virus. Journal of chemical information and modeling 50:1410-1417.

121. Behera AK, Basu S, Cherian SS. 2015. Molecular mechanism of the enhanced viral fitness contributed by secondary mutations in the hemagglutinin protein of oseltamivir resistant H1N1 influenza viruses: modeling studies of antibody and receptor binding. Gene 557:19-27.

122. Al Khatib HA, Al Thani AA, Yassine HM. 2018. Evolution and dynamics of the pandemic H1N1 influenza hemagglutinin protein from 2009 to 2017. Archives of virology 163:3035-3049.

123. Gambaryan A, Robertson J, Matrosovich M. 1999. Effects of egg-adaptation on the receptor-binding properties of human influenza A and B viruses. Virology 258:232-239.

124. Parker L, Wharton SA, Martin SR, Cross K, Lin Y, Liu Y, Feizi T, Daniels RS, McCauley JW. 2016. Effects of egg-adaptation on receptor-binding and antigenic properties of recent influenza A (H3N2) vaccine viruses. Journal of General Virology 97:1333-1344.

125. Gambaryan A, Tuzikov A, Piskarev V, Yamnikova S, Lvov D, Robertson J, Bovin N, Matrosovich M. 1997. Specification of receptor-binding phenotypes of influenza virus isolates from different hosts using synthetic sialylglycopolymers: non-egg-adapted human H1 and H3 influenza A and influenza B viruses share a common high binding affinity for 6′-sialyl (N-acetyllactosamine). Virology 232:345-350.

126. Wagner R, Matrosovich M, Klenk HD. 2002. Functional balance between haemagglutinin and neuraminidase in influenza virus infections. Reviews in medical virology 12:159-166.

127. Palese P, Tobita K, Ueda M, Compans RW. 1974. Characterization of temperature sensitive influenza virus mutants defective in neuraminidase. Virology 61:397-410.

128. Hinshaw V, Webster R, Naeve C, Murphy B. 1983. Altered tissue tropism of human-avian reassortant influenza viruses. Virology 128:260-263.

129. de Vries E, Du W, Guo H, de Haan CA. 2020. Influenza A virus hemagglutinin–neuraminidase–receptor balance: preserving virus motility. Trends in microbiology 28:57-67.

130. Sakai T, Nishimura SI, Naito T, Saito M. 2017. Influenza A virus hemagglutinin and neuraminidase act as novel motile machinery. Scientific reports 7:45043.

131. Powell H, Pekosz A. 2020. Neuraminidase antigenic drift of H3N2 clade 3c. 2a viruses alters virus replication, enzymatic activity and inhibitory antibody binding. PLoS pathogens 16:e1008411.

132. Xu R, Zhu X, McBride R, Nycholat CM, Yu W, Paulson JC, Wilson IA. 2012. Functional balance of the hemagglutinin and neuraminidase activities accompanies the emergence of the 2009 H1N1 influenza pandemic. Journal of virology 86:9221-9232.

133. Casalegno J-S, Ferraris O, Escuret V, Bouscambert M, Bergeron C, Linès L, Excoffier T, Valette M, Frobert E, Pillet S. 2014. Functional balance between the hemagglutinin and neuraminidase of influenza A (H1N1) pdm09 HA D222 variants. PLoS One 9:e104009.

134. McAuley JL, Gilbertson BP, Trifkovic S, Brown LE, McKimm-Breschkin JL. 2019. Influenza virus neuraminidase structure and functions. Frontiers in microbiology 10:432609.

135. Han AX, de Jong SP, Russell CA. 2023. Co-evolution of immunity and seasonal influenza viruses. Nature Reviews Microbiology 21:805-817.

136. Cohen FS, Melikyan GB. 2001. Implications of a fusion peptide structure. nature structural biology 8:653-655.

137. Krammer F. 2019. The human antibody response to influenza A virus infection and vaccination. Nature Reviews Immunology 19:383-397.

138. Zost SJ, Parkhouse K, Gumina ME, Kim K, Diaz Perez S, Wilson PC, Treanor JJ, Sant AJ, Cobey S, Hensley SE. 2017. Contemporary H3N2 influenza viruses have a glycosylation site that alters binding of antibodies elicited by egg-adapted vaccine strains. Proceedings of the National Academy of Sciences 114:12578-12583.

139. Fink AL, Liu H, Shaw-Saliba K, Mehoke T, Evans J, Liu Z-Y, Lewis M, Sauer L, Thielen P, Chen K-F. 2021. Regional differences in vaccine uptake and serological responses to vaccine and circulating strains of H1N1 viruses among patients with confirmed influenza. Journal of Clinical Virology Plus 1:100034.

140. Bedford T, Riley S, Barr IG, Broor S, Chadha M, Cox NJ, Daniels RS, Gunasekaran CP, Hurt AC, Kelso A. 2015. Global circulation patterns of seasonal influenza viruses vary with antigenic drift. Nature 523:217-220.

141. Carrat F, Flahault A. 2007. Influenza vaccine: the challenge of antigenic drift. Vaccine 25:6852-6862.

142. Medina RA, Stertz S, Manicassamy B, Zimmermann P, Sun X, Albrecht RA, Uusi-Kerttula H, Zagordi O, Belshe RB, Frey SE. 2013. Glycosylations in the globular head of the hemagglutinin protein modulate the virulence and antigenic properties of the H1N1 influenza viruses. Science translational medicine 5:187ra70-187ra70.

143. Krammer F, Palese P. 2013. Influenza virus hemagglutinin stalk-based antibodies and vaccines. Current opinion in virology 3:521-530.

144. Margine I, Hai R, Albrecht RA, Obermoser G, Harrod AC, Banchereau J, Palucka K, García-Sastre A, Palese P, Treanor JJ. 2013. H3N2 influenza virus infection induces broadly reactive hemagglutinin stalk antibodies in humans and mice. Journal of virology 87:4728-4737.

145. Nguyen-Van-Tam JS, Hampson AW. 2003. The epidemiology and clinical impact of pandemic influenza. Vaccine 21:1762-1768.

146. Shapshak P, Chiappelli F, Somboonwit C, Sinnott J. 2011. The influenza pandemic of 2009: lessons and implications. Molecular diagnosis & therapy 15:63-81.

147. Girard MP, Tam JS, Assossou OM, Kieny MP. 2010. The 2009 A (H1N1) influenza virus pandemic: A review. Vaccine 28:4895-4902.

148. Xu R, Ekiert DC, Krause JC, Hai R, Crowe Jr JE, Wilson IA. 2010. Structural basis of preexisting immunity to the 2009 H1N1 pandemic influenza virus. Science 328:357-360.

149. Guo F, Yang J, Pan J, Liang X, Shen X, Irwin DM, Chen R-A, Shen Y. 2020. Origin and evolution of H1N1/pdm2009: a codon usage perspective. Frontiers in Microbiology 11:1615.

150. Hadfield J, Megill C, Bell SM, Huddleston J, Potter B, Callender C, Sagulenko P, Bedford T, Neher RA. 2018. Nextstrain: real-time tracking of pathogen evolution. Bioinformatics 34:4121-4123.

151. Hay AJ, McCauley JW. 2018. The WHO global influenza surveillance and response system (GISRS)—a future perspective. Influenza and other respiratory viruses 12:551-557.

152. WHO. 2019. Recommended composition of influenza virus vaccines for use in the 2019-2020 northern hemisphere influenza season.

153. WHO. 2020. Recommended composition of influenza virus vaccines for use in the 2020-2021 northern hemisphere influenza season.

154. WHO. 24 September 2021 2021. Recommended composition of influenza virus vaccines for use in the 2022 southern hemisphere influenza season. https://www.who.int/publications/m/item/recommended-composition-of-influenza-virus-vaccines-for-use-in-the-2022-southern-hemisphere-influenza-season. Accessed

155. WHO. 23 September 2022 2022. Recommended composition of influenza virus vaccines for use in the 2023 southern hemisphere influenza season. https://www.who.int/publications/m/item/recommended-composition-of-influenza-virus-vaccines-for-use-in-the-2023-southern-hemisphere-influenza-season. Accessed

156. Milián E, Kamen AA. 2015. Current and emerging cell culture manufacturing technologies for influenza vaccines. BioMed research international 2015.

157. Gouma S, Weirick M, Hensley SE. 2020. Potential antigenic mismatch of the H3N2 component of the 2019 southern hemisphere influenza vaccine. Clinical Infectious Diseases 70:2432-2434.

158. Tricco AC, Chit A, Soobiah C, Hallett D, Meier G, Chen MH, Tashkandi M, Bauch CT, Loeb M. 2013. Comparing influenza vaccine efficacy against mismatched and matched strains: a systematic review and meta-analysis. BMC medicine 11:1-19.

159. Tenforde MW, Kondor RJG, Chung JR, Zimmerman RK, Nowalk MP, Jackson ML, Jackson LA, Monto AS, Martin ET, Belongia EA. 2021. Effect of antigenic drift on influenza vaccine effectiveness in the United States—2019–2020. Clinical Infectious Diseases 73:e4244-e4250.

160. Tenforde MW, Talbot HK, Trabue CH, Gaglani M, McNeal TM, Monto AS, Martin ET, Zimmerman RK, Silveira FP, Middleton DB. 2021. Influenza vaccine effectiveness against hospitalization in the United States, 2019–2020. The Journal of infectious diseases 224:813-820.

161. CDC. February 8, 2023. Vaccine Effectiveness: How Well Do Flu Vaccines Work?, *on* Centers for Disease Control and Prevention. https://www.cdc.gov/flu/vaccines-work/vaccineeffect.htm. Accessed March 24, 2024.

162. Zimmerman RK, Nowalk MP, Chung J, Jackson ML, Jackson LA, Petrie JG, Monto AS, McLean HQ, Belongia EA, Gaglani M. 2016. 2014–2015 influenza vaccine effectiveness in the United States by vaccine type. Clinical Infectious Diseases:ciw635.

163. Rajaram S, Boikos C, Gelone DK, Gandhi A. 2020. Influenza vaccines: the potential benefits of cell-culture isolation and manufacturing. Therapeutic advances in vaccines and immunotherapy 8:2515135520908121.

164. Jin H, Subbarao K. 2014. Live attenuated influenza vaccine. Influenza Pathogenesis and Control-Volume II:181-204.

165. CDC. November 3, 2022. How Influenza (Flu) Vaccines Are Made, *on* CDC. https://www.cdc.gov/flu/prevent/how-fluvaccine-made.htm. Accessed

166. Hegde NR. 2015. Cell culture-based influenza vaccines: A necessary and indispensable investment for the future. Human vaccines & immunotherapeutics 11:1223-1234.

167. Ortiz de Lejarazu-Leonardo R, Montomoli E, Wojcik R, Christopher S, Mosnier A, Pariani E, Trilla Garcia A, Fickenscher H, Gärtner BC, Jandhyala R. 2021. Estimation of reduction in influenza vaccine effectiveness due to egg-adaptation changes—Systematic literature review and expert consensus. Vaccines 9:1255.

168. Filip R, Leluk J. 2017. Phylogenetic and variability study on all known hemagglutinin subtypes of influenza A virus. Bio-Algorithms and Med-Systems 13:153-159.

169. Igarashi M, Ito K, Yoshida R, Tomabechi D, Kida H, Takada A. 2010. Predicting the antigenic structure of the pandemic (H1N1) 2009 influenza virus hemagglutinin. PloS one 5:e8553.

170. Ha Y, Stevens DJ, Skehel JJ, Wiley DC. 2002. H5 avian and H9 swine influenza virus haemagglutinin structures: possible origin of influenza subtypes. The EMBO journal 21:865-875.

171. Koel BF, Burke DF, Bestebroer TM, Van Der Vliet S, Zondag GC, Vervaet G, Skepner E, Lewis NS, Spronken MI, Russell CA. 2013. Substitutions near the

receptor binding site determine major antigenic change during influenza virus evolution. Science 342:976-979.

172. Wang X, Ilyushina NA, Lugovtsev VY, Bovin NV, Couzens LK, Gao J, Donnelly RP, Eichelberger MC, Wan H. 2017. Amino acids in hemagglutinin antigenic site B determine antigenic and receptor binding differences between A (H3N2) v and ancestral seasonal H3N2 influenza viruses. Journal of Virology 91:10.1128/jvi. 01512-16.

173. Hensley SE, Das SR, Bailey AL, Schmidt LM, Hickman HD, Jayaraman A, Viswanathan K, Raman R, Sasisekharan R, Bennink JR. 2009. Hemagglutinin receptor binding avidity drives influenza A virus antigenic drift. Science 326:734-736.

174. Peacock TP, Benton DJ, James J, Sadeyen J-R, Chang P, Sealy JE, Bryant JE, Martin SR, Shelton H, Barclay WS. 2017. Immune escape variants of H9N2 influenza viruses containing deletions at the hemagglutinin receptor binding site retain fitness in vivo and display enhanced zoonotic characteristics. Journal of virology 91:10.1128/jvi. 00218-17.

175. Myers JL, Wetzel KS, Linderman SL, Li Y, Sullivan CB, Hensley SE. 2013. Compensatory hemagglutinin mutations alter antigenic properties of influenza viruses. Journal of virology 87:11168-11172.

176. Underwood PA, Skehel J, Wiley D. 1987. Receptor-binding characteristics of monoclonal antibody-selected antigenic variants of influenza virus. Journal of virology 61:206-208.

177. Li Y, Bostick DL, Sullivan CB, Myers JL, Griesemer SB, StGeorge K, Plotkin JB, Hensley SE. 2013. Single hemagglutinin mutations that alter both antigenicity and receptor binding avidity influence influenza virus antigenic clustering. Journal of virology 87:9904-9910.

178. O'Donnell CD, Vogel L, Wright A, Das SR, Wrammert J, Li G-M, McCausland M, Zheng N-Y, Yewdell JW, Ahmed R. 2012. Antibody pressure by a human monoclonal antibody targeting the 2009 pandemic H1N1 virus hemagglutinin drives the emergence of a virus with increased virulence in mice. MBio 3:10.1128/mbio. 00120-12.

179. Gong LI, Suchard MA, Bloom JD. 2013. Stability-mediated epistasis constrains the evolution of an influenza protein. Elife 2:e00631.

180. Koel BF, Burke DF, van der Vliet S, Bestebroer TM, Rimmelzwaan GF, Osterhaus AD, Smith DJ, Fouchier RA. 2019. Epistatic interactions can moderate the antigenic effect of substitutions in haemagglutinin of influenza H3N2 virus. Journal of General Virology 100:773-777.

181. Liu T, Wang Y, Tan TJ, Wu NC, Brooke CB. 2022. The evolutionary potential of influenza A virus hemagglutinin is highly constrained by epistatic interactions with neuraminidase. Cell host & microbe 30:1363-1369. e4.

182. Visher E, Whitefield SE, McCrone JT, Fitzsimmons W, Lauring AS. 2016. The mutational robustness of influenza A virus. PLoS pathogens 12:e1005856.

183. Das SR, Hensley SE, Ince WL, Brooke CB, Subba A, Delboy MG, Russ G, Gibbs JS, Bennink JR, Yewdell JW. 2013. Defining influenza A virus hemagglutinin

antigenic drift by sequential monoclonal antibody selection. Cell host & microbe 13:314-323.

184. Hensley SE, Das SR, Gibbs JS, Bailey AL, Schmidt LM, Bennink JR, Yewdell JW. 2011. Influenza A virus hemagglutinin antibody escape promotes neuraminidase antigenic variation and drug resistance. PLoS One 6:e15190.

185. Neverov AD, Kryazhimskiy S, Plotkin JB, Bazykin GA. 2015. Coordinated evolution of influenza A surface proteins. PLoS Genetics 11:e1005404.

186. Wolf YI, Viboud C, Holmes EC, Koonin EV, Lipman DJ. 2006. Long intervals of stasis punctuated by bursts of positive selection in the seasonal evolution of influenza A virus. Biology direct 1:1-19.

187. Kryazhimskiy S, Dushoff J, Bazykin GA, Plotkin JB. 2011. Prevalence of epistasis in the evolution of influenza A surface proteins. PLoS genetics 7:e1001301.

188. Shih AC-C, Hsiao T-C, Ho M-S, Li W-H. 2007. Simultaneous amino acid substitutions at antigenic sites drive influenza A hemagglutinin evolution. Proceedings of the National Academy of Sciences 104:6283-6288.

189. Tria F, Pompei S, Loreto V. 2013. Dynamically correlated mutations drive human Influenza A evolution. Scientific reports 3:2705.

190. Wu NC, Xie J, Zheng T, Nycholat CM, Grande G, Paulson JC, Lerner RA, Wilson IA. 2017. Diversity of functionally permissive sequences in the receptor-binding site of influenza hemagglutinin. Cell host & microbe 21:742-753. e8.

191. Stevens J, Blixt O, Tumpey TM, Taubenberger JK, Paulson JC, Wilson IA. 2006. Structure and receptor specificity of the hemagglutinin from an H5N1 influenza virus. science 312:404-410.

192. Martínez-Romero C, de Vries E, Belicha-Villanueva A, Mena I, Tscherne DM, Gillespie VL, Albrecht RA, de Haan CA, García-Sastre A. 2013. Substitutions T200A and E227A in the hemagglutinin of pandemic 2009 influenza A virus increase lethality but decrease transmission. Journal of virology 87:6507-6511.

193. de Vries RP, de Vries E, Moore KS, Rigter A, Rottier PJ, de Haan CA. 2011. Only two residues are responsible for the dramatic difference in receptor binding between swine and new pandemic H1 hemagglutinin. Journal of Biological Chemistry 286:5868-5875.

194. Lee C-Y, Raghunathan V, Caceres CJ, Geiger G, Seibert B, Cargnin Faccin F, Gay LC, Ferreri LM, Kaul D, Wrammert J. 2023. Epistasis reduces fitness costs of influenza A virus escape from stem-binding antibodies. Proceedings of the National Academy of Sciences 120:e2208718120.

195. Akand EH, Downard KM. 2018. Identification of epistatic mutations and insights into the evolution of the influenza virus using a mass-based protein phylogenetic approach. Molecular phylogenetics and evolution 121:132-138.

196. Kosik I, Ince WL, Gentles LE, Oler AJ, Kosikova M, Angel M, Magadán JG, Xie H, Brooke CB, Yewdell JW. 2018. Influenza A virus hemagglutinin glycosylation compensates for antibody escape fitness costs. PLoS pathogens 14:e1006796.

197. Koelle K, Cobey S, Grenfell B, Pascual M. 2006. Epochal evolution shapes the phylodynamics of interpandemic influenza A (H3N2) in humans. Science 314:1898-1903.

198. Koelle K, Rasmussen DA. 2015. The effects of a deleterious mutation load on patterns of influenza A/H3N2's antigenic evolution in humans. Elife 4:e07361.

199. Wu NC, Thompson AJ, Xie J, Lin C-W, Nycholat CM, Zhu X, Lerner RA, Paulson JC, Wilson IA. 2018. A complex epistatic network limits the mutational reversibility in the influenza hemagglutinin receptor-binding site. Nature communications 9:1264.

200. Nakajima K, Nobusawa E, Nagy A, Nakajima S. 2005. Accumulation of amino acid substitutions promotes irreversible structural changes in the hemagglutinin of human influenza AH3 virus during evolution. Journal of virology 79:6472-6477.

201. WHO. March 24, 2024. Virus detections by subtype reported to FluNet, *on* Worl Health Organization. https://app.powerbi.com/view?r=eyJrIjoiZTkyODcyOTEtZjA5YS00ZmI0LWFkZGUtODIxNGI5OTE3YjM0IiwidCI6ImY2MTBjMGI3LWJkMjQtNGIzOS04MTBiLTNkYzI4MGFmYjU5MCIsImMiOjh9. Accessed March 24, 2024.

202. CDC. February 28, 2024. Disease Burden of Flu. https://www.cdc.gov/flu/about/burden/index.html. Accessed

203. Swanson NJ, Marinho P, Dziedzic A, Jedlicka A, Liu H, Fenstermacher K, Rothman R, Pekosz A. 2023. 2019–2020 H1N1 clade A5a. 1 viruses have better in vitro fitness compared with the co-circulating A5a. 2 clade. Scientific Reports 13:10223.

204. Iuliano AD, Roguski KM, Chang HH, Muscatello DJ, Palekar R, Tempia S, Cohen C, Gran JM, Schanzer D, Cowling BJ. 2018. Estimates of global seasonal influenza-associated respiratory mortality: a modelling study. The Lancet 391:1285-1300.

205. Shao W, Li X, Goraya MU, Wang S, Chen J-L. 2017. Evolution of influenza a virus by mutation and re-assortment. International journal of molecular sciences 18:1650.

206. Wohlgemuth N, Ye Y, Fenstermacher KJ, Liu H, Lane AP, Pekosz A. 2017. The M2 protein of live, attenuated influenza vaccine encodes a mutation that reduces replication in human nasal epithelial cells. Vaccine 35:6691-6699.

207. Liu H, Grantham ML, Pekosz A. 2018. Mutations in the influenza A virus M1 protein enhance virus budding to complement lethal mutations in the M2 cytoplasmic tail. Journal of virology 92:e00858-17.

208. WHO. 2020. Influenza Update N° 365.

209. Strengell M, Ikonen N, Ziegler T, Julkunen I. 2011. Minor changes in the hemagglutinin of influenza A (H1N1) 2009 virus alter its antigenic properties. PLoS One 6:e25848.

210. Ramos AP, Herrera BA, Ramírez OV, García AA, Jiménez MM, Valdés CS, Fernández AG, González G, Fernández SIO, Báez GG. 2013. Molecular and phylogenetic analysis of influenza A H1N1 pandemic viruses in Cuba, May 2009 to August 2010. International Journal of Infectious Diseases 17:e565-e567.

211. Guarnaccia T, Carolan LA, Maurer-Stroh S, Lee RT, Job E, Reading PC, Petrie S, McCaw JM, McVernon J, Hurt AC. 2013. Antigenic drift of the pandemic 2009 A (H1N1) influenza virus in A ferret model. PLoS pathogens 9:e1003354.

212. Wedde M, Oh D-Y, Buda S, Thürmer A, Kaiser S, Biere B, Duwe S, Reiche J, Wolff T, Dürrwald R. 2021. Characterization of influenza A (H1N1) pdm09 viruses

in Germany in season 2019-2020–co-circulation of an antigenic drift variant. Authorea Preprints.

213. Mohan T, Nguyen HT, Kniss K, Mishin VP, Merced-Morales AA, Laplante J, George KS, Blevins P, Chesnokov A, Juan A. 2021. Cluster of Oseltamivir-Resistant and Hemagglutinin Antigenically Drifted Influenza A (H1N1) pdm09 Viruses, Texas, USA, January 2020. Emerging Infectious Diseases 27:1953.

214. Liu ST, Behzadi MA, Sun W, Freyn AW, Liu W-C, Broecker F, Albrecht RA, Bouvier NM, Simon V, Nachbagauer R. 2018. Antigenic sites in influenza H1 hemagglutinin display species-specific immunodominance. The Journal of clinical investigation 128:4992-4996.

215. Skowronski DM, Zou M, Sabaiduc S, Murti M, Olsha R, Dickinson JA, Gubbay JB, Croxen MA, Charest H, Jassem A. 2020. Interim estimates of 2019/20 vaccine effectiveness during early-season co-circulation of influenza A and B viruses, Canada, February 2020. Eurosurveillance 25:2000103.

216. Matrosovich M, Matrosovich T, Carr J, Roberts NA, Klenk H-D. 2003. Overexpression of the α-2, 6-sialyltransferase in MDCK cells increases influenza virus sensitivity to neuraminidase inhibitors. Journal of virology 77:8418-8425.

217. Ibricevic A, Pekosz A, Walter MJ, Newby C, Battaile JT, Brown EG, Holtzman MJ, Brody SL. 2006. Influenza virus receptor specificity and cell tropism in mouse and human airway epithelial cells. Journal of virology 80:7469-7480.

218. Fischer II WA, King LS, Lane AP, Pekosz A. 2015. Restricted replication of the live attenuated influenza A virus vaccine during infection of primary differentiated human nasal epithelial cells. Vaccine 33:4495-4504.

219. Blumenkrantz DR, Mehoke T, Shaw-Saliba K, Powell H, Wohlgemuth N, Liu H, Macias E, Evans J, Lewis M, Medina R. 2021. Identification of H3N2 NA and PB1-F2 genetic variants and their association with disease symptoms during the 2014–15 influenza season. Virus Evolution 7:veab047.

220. Liu H, Gong Y-N, Shaw-Saliba K, Mehoke T, Evans J, Liu Z-Y, Lewis M, Sauer L, Thielen P, Rothman R. 2021. Differential disease severity and whole-genome sequence analysis for human influenza A/H1N1pdm virus in 2015–2016 influenza season. Virus evolution 7:veab044.

221. Forero A, Fenstermacher K, Wohlgemuth N, Nishida A, Carter V, Smith EA, Peng X, Hayes M, Francis D, Treanor J. 2017. Evaluation of the innate immune responses to influenza and live-attenuated influenza vaccine infection in primary differentiated human nasal epithelial cells. Vaccine 35:6112-6121.

222. Canaday LM, Resnick JD, Liu H, Powell H, McCoy AM, Nguyen D, Pekosz A. 2022. HA and M2 sequences alter the replication of 2013–16 H1 live attenuated influenza vaccine infection in human nasal epithelial cell cultures. Vaccine 40:4544-4553.

223. Bradley KC, Jones CA, Tompkins SM, Tripp RA, Russell RJ, Gramer MR, Heimburg-Molinaro J, Smith DF, Cummings RD, Steinhauer DA. 2011. Comparison of the receptor binding properties of contemporary swine isolates and early human pandemic H1N1 isolates (Novel 2009 H1N1). Virology 413:169-182.

224. Matrosovich M, Gambaryan A, Teneberg S, Piskarev V, Yamnikova S, Lvov D, Robertson J, Karlsson K-A. 1997. Avian influenza A viruses differ from human

viruses by recognition of sialyloligosaccharides and gangliosides and by a higher conservation of the HA receptor-binding site. Virology 233:224-234.

225. Gulati S, Smith DF, Cummings RD, Couch RB, Griesemer SB, St. George K, Webster RG, Air GM. 2013. Human H3N2 influenza viruses isolated from 1968 to 2012 show varying preference for receptor substructures with no apparent consequences for disease or spread. PloS one 8:e66325.

226. Amonsen M, Smith DF, Cummings RD, Air GM. 2007. Human parainfluenza viruses hPIV1 and hPIV3 bind oligosaccharides with α2-3-linked sialic acids that are distinct from those bound by H5 avian influenza virus hemagglutinin. Journal of virology 81:8341-8345.

227. Alymova IV, York IA, Air GM, Cipollo JF, Gulati S, Baranovich T, Kumar A, Zeng H, Gansebom S, McCullers JA. 2016. Glycosylation changes in the globular head of H3N2 influenza hemagglutinin modulate receptor binding without affecting virus virulence. Scientific Reports 6:1-15.

228. Song X, Yu H, Chen X, Lasanajak Y, Tappert MM, Air GM, Tiwari VK, Cao H, Chokhawala HA, Zheng H. 2011. A sialylated glycan microarray reveals novel interactions of modified sialic acids with proteins and viruses. Journal of Biological Chemistry 286:31610-31622.

229. Heimburg-Molinaro J, Tappert M, Song X, Lasanajak Y, Air G, Smith DF, Cummings RD. 2012. Probing virus–glycan interactions using glycan microarrays. Carbohydrate Microarrays: Methods and Protocols:251-267.

230. Pekosz A, Newby C, Bose PS, Lutz A. 2009. Sialic acid recognition is a key determinant of influenza A virus tropism in murine trachea epithelial cell cultures. Virology 386:61-67.

231. Doyle TM, Li C, Bucher DJ, Hashem AM, Van Domselaar G, Wang J, Farnsworth A, She Y-M, Cyr T, He R. 2013. A monoclonal antibody targeting a highly conserved epitope in influenza B neuraminidase provides protection against drug resistant strains. Biochemical and biophysical research communications 441:226-229.

232. Aytay S, Schulze IT. 1991. Single amino acid substitutions in the hemagglutinin can alter the host range and receptor binding properties of H1 strains of influenza A virus. Journal of virology 65:3022-3028.

233. Daniels R, Douglas A, Skehel J, Wiley D, Naeve C, Webster R, Rogers G, Paulson J. 1984. Antigenic analyses of influenza virus haemagglutinins with different receptor-binding specificities. Virology 138:174-177.

234. Bolton MJ, Ort JT, McBride R, Swanson NJ, Wilson J, Awofolaju M, Furey C, Greenplate AR, Drapeau EM, Pekosz A. 2022. Antigenic and virological properties of an H3N2 variant that continues to dominate the 2021–22 Northern Hemisphere influenza season. Cell Reports 39:110897.

235. Skowronski DM, Janjua NZ, De Serres G, Sabaiduc S, Eshaghi A, Dickinson JA, Fonseca K, Winter A-L, Gubbay JB, Krajden M. 2014. Low 2012–13 influenza vaccine effectiveness associated with mutation in the egg-adapted H3N2 vaccine strain not antigenic drift in circulating viruses. PloS one 9:e92153.

236. Wan H, Gao J, Yang H, Yang S, Harvey R, Chen Y-Q, Zheng N-Y, Chang J, Carney PJ, Li X. 2019. The neuraminidase of A (H3N2) influenza viruses circulating since 2016 is antigenically distinct from the A/Hong Kong/4801/2014 vaccine strain. Nature microbiology 4:2216-2225.

237. Wang TL, Jing L, Bocchini JA. 2017. Mandatory influenza vaccination for all healthcare personnel: a review on justification, implementation and effectiveness. Current Opinion in Pediatrics 29:606-615.

238. Huang K-YA, Chang S-C, Huang Y-C, Chiu C-H, Lin T-Y. 2017. Antibody responses to trivalent inactivated influenza vaccine in health care personnel previously vaccinated and vaccinated for the first time. Scientific Reports 7:1-10.

239. Kuo H, Shapiro JR, Dhakal S, Morgan R, Fink AL, Liu H, Westerbeck JW, Sylvia KE, Park H-S, Ursin RL. 2022. Sex-specific effects of age and body mass index on antibody responses to seasonal influenza vaccines in healthcare workers. Vaccine 40:1634-1642.

240. Leung VK, Carolan LA, Worth LJ, Harper SA, Peck H, Tilmanis D, Laurie KL, Slavin MA, Sullivan SG. 2017. Influenza vaccination responses: Evaluating impact of repeat vaccination among health care workers. Vaccine 35:2558-2568.

241. Goh KCM, Tang CK, Norton DC, Gan ES, Tan HC, Sun B, Syenina A, Yousuf A, Ong XM, Kamaraj US. 2016. Molecular determinants of plaque size as an indicator of dengue virus attenuation. Scientific reports 6:1-11.

242. Rijsbergen LC, Van Dijk LL, Engel MF, De Vries RD, De Swart RL. 2021. In Vitro Modelling of Respiratory Virus Infections in Human Airway Epithelial Cells–A Systematic Review. Frontiers in Immunology:3301.

243. Bradley KC, Galloway SE, Lasanajak Y, Song X, Heimburg-Molinaro J, Yu H, Chen X, Talekar GR, Smith DF, Cummings RD. 2011. Analysis of influenza virus hemagglutinin receptor binding mutants with limited receptor recognition properties and conditional replication characteristics. Journal of virology 85:12387-12398.

244. Reed LJ, Muench H. 1938. A simple method of estimating fifty per cent endpoints. American journal of epidemiology 27:493-497.

245. McCown MF, Pekosz A. 2005. The influenza A virus M2 cytoplasmic tail is required for infectious virus production and efficient genome packaging. Journal of virology 79:3595-3605.

246. Mehta AY, Cummings RD. 2019. GLAD: GLycan Array Dashboard, a visual analytics tool for glycan microarrays. Bioinformatics 35:3536-3537.

247. Paul Glezen W, Schmier JK, Kuehn CM, Ryan KJ, Oxford J. 2013. The burden of influenza B: a structured literature review. American journal of public health 103:e43-e51.

248. Tokars JI, Olsen SJ, Reed C. 2018. Seasonal incidence of symptomatic influenza in the United States. Clinical Infectious Diseases 66:1511-1518.

249. Petrova VN, Russell CA. 2018. The evolution of seasonal influenza viruses. Nature Reviews Microbiology 16:47-60.

250. Conenello GM, Tisoncik JR, Rosenzweig E, Varga ZT, Palese P, Katze MG. 2011. A single N66S mutation in the PB1-F2 protein of influenza A virus increases virulence by inhibiting the early interferon response in vivo. Journal of virology 85:652-662.

251. Wei K, Sun H, Sun Z, Sun Y, Kong W, Pu J, Ma G, Yin Y, Yang H, Guo X. 2014. Influenza A virus acquires enhanced pathogenicity and transmissibility after serial passages in swine. Journal of virology 88:11981-11994.

252. Meng F, Yang H, Qu Z, Chen Y, Zhang Y, Zhang Y, Liu L, Zeng X, Li C, Kawaoka Y. 2022. A Eurasian avian-like H1N1 swine influenza reassortant virus became pathogenic and highly transmissible due to mutations in its PA gene. Proceedings of the National Academy of Sciences 119:e2203919119.

253. Hu J, Hu Z, Wei Y, Zhang M, Wang S, Tong Q, Sun H, Pu J, Liu J, Sun Y. 2022. Mutations in PB2 and HA are crucial for the increased virulence and transmissibility of H1N1 swine influenza virus in mammalian models. Veterinary Microbiology 265:109314.

254. Griffin EF, Tompkins SM. 2023. Fitness Determinants of Influenza A Viruses. Viruses 15:1959.

255. Merced-Morales A, Daly P, Abd Elal AI, Ajayi N, Annan E, Budd A, Barnes J, Colon A, Cummings CN, Iuliano AD. 2022. Influenza activity and composition of the 2022–23 influenza vaccine—United States, 2021–22 season. Morbidity and Mortality Weekly Report 71:913.

256. Zolotarova O, Budzanivska I, Leibenko L, Radchenko L, Mironenko A. 2019. Antigenic site variation in the hemagglutinin of pandemic influenza A (H1N1) pdm09 viruses between 2009–2017 in Ukraine. Pathogens 8:194.

257. El Moussi A, Ben Hadj Kacem MA, Pozo F, Ledesma J, Cuevas MT, Casas I, Slim A. 2013. Genetic diversity of HA1 domain of heammaglutinin gene of influenza A (H1N1) pdm09 in Tunisia. Virology journal 10:1-9.

258. Guldemir D, Kalaycioglu AT, Altas AB, Korukluoglu G, Durmaz R. 2013. Monitoring genetic diversity of influenza A (H1N1) pdm09 virus circulating during the post-pandemic period in Turkey. Japanese journal of infectious diseases 66:299-305.

259. Ledesma J, Pozo F, Reina G, Blasco M, Rodríguez G, Montes M, López-Miragaya I, Salvador C, Reina J, de Lejarazu RO. 2012. Genetic diversity of influenza A (H1N1) 2009 virus circulating during the season 2010–2011 in Spain. Journal of clinical virology 53:16-21.

260. Wedde M, Biere B, Wolff T, Schweiger B. 2015. Evolution of the hemagglutinin expressed by human influenza A (H1N1) pdm09 and A (H3N2) viruses circulating between 2008–2009 and 2013–2014 in Germany. International Journal of Medical Microbiology 305:762-775.

261. van Doremalen N, Shelton H, Roberts KL, Jones IM, Pickles RJ, Thompson CI, Barclay WS. 2011. A single amino acid in the HA of pH1N1 2009 influenza virus affects cell tropism in human airway epithelium, but not transmission in ferrets. PLoS One 6:e25755.

262. Shang C, Whittleston CS, Sutherland-Cash KH, Wales DJ. 2015. Analysis of the Contrasting Pathogenicities Induced by the D222G Mutation in 1918 and 2009 Pandemic Influenza A Viruses. Journal of Chemical Theory and Computation 11:2307-2314.

263. de Vries RP, de Vries E, Martínez-Romero C, McBride R, van Kuppeveld FJ, Rottier PJ, García-Sastre A, Paulson JC, de Haan CA. 2013. Evolution of the hemagglutinin protein of the new pandemic H1N1 influenza virus: maintaining

optimal receptor binding by compensatory substitutions. Journal of virology 87:13868-13877.

264. Martínez-Baz I, Fernández-Huerta M, Navascués A, Pozo F, Trobajo-Sanmartín C, Casado I, Echeverria A, Ezpeleta C, Castilla J. 2023. Influenza vaccine effectiveness in preventing laboratory-confirmed influenza cases and hospitalizations in Navarre, Spain, 2022–2023. Vaccines 11:1478.

265. Opanda S, Bulimo W, Gachara G, Ekuttan C, Amukoye E. 2020. Assessing antigenic drift and phylogeny of influenza A (H1N1) pdm09 virus in Kenya using HA1 sub-unit of the hemagglutinin gene. PLoS One 15:e0228029.

266. Monamele CG, Munshili Njifon HL, Vernet M-A, Njankouo MR, Kenmoe S, Yahaya AA, Deweerdt L, Nono R, Mbacham W, Anong DN. 2019. Molecular characterization of influenza A (H1N1) pdm09 in Cameroon during the 2014-2016 influenza seasons. PloS one 14:e0210119.

267. Kao C-L, Chan T-C, Tsai C-H, Chu K-Y, Chuang S-F, Lee C-C, Li Z-RT, Wu K-W, Chang L-Y, Shen Y-H. 2012. Emerged HA and NA mutants of the pandemic influenza H1N1 viruses with increasing epidemiological significance in Taipei and Kaohsiung, Taiwan, 2009–10. PLoS One 7:e31162.

268. Tapia R, Torremorell M, Culhane M, Medina RA, Neira V. 2020. Antigenic characterization of novel H1 influenza A viruses in swine. Scientific reports 10:4510.

269. Ortiz L, Geiger G, Ferreri L, Moran D, Alvarez D, Gonzalez-Reiche AS, Mendez D, Rajao D, Cordon-Rosales C, Perez DR. 2023. Evolution and Introductions of

Influenza A Virus H1N1 in a Farrow-to-Finish Farm in Guatemala. Microbiology Spectrum 11:e02878-22.

270. Kowalczyk A, Urbaniak K, Markowska-Daniel I, Pejsak Z. 2015. Substitution in position 222 of haemagglutinin of pandemic influenza A (H1N1) viruses isolated from pigs in Poland. Journal of Veterinary Research 59:451-456.

271. Markin A, Ciacci Zanella G, Arendsee ZW, Zhang J, Krueger KM, Gauger PC, Vincent Baker AL, Anderson TK. 2023. Reverse-zoonoses of 2009 H1N1 pandemic influenza A viruses and evolution in United States swine results in viruses with zoonotic potential. PLoS Pathogens 19:e1011476.

272. Fischer W. 2015. II; King, LS; Lane, AP; Pekosz, A. Restricted replication of the live attenuated influenza a virus vaccine during infection of primary differentiated human nasal epithelial cells. Vaccine 33:4495-4504.

273. van Riel D, den Bakker MA, Leijten LM, Chutinimitkul S, Munster VJ, de Wit E, Rimmelzwaan GF, Fouchier RA, Osterhaus AD, Kuiken T. 2010. Seasonal and pandemic human influenza viruses attach better to human upper respiratory tract epithelium than avian influenza viruses. The American journal of pathology 176:1614-1618.

274. Lee C-W. 2014. Reverse genetics of influenza virus. Animal influenza virus:37-50.

275. Neumann G, Watanabe T, Ito H, Watanabe S, Goto H, Gao P, Hughes M, Perez DR, Donis R, Hoffmann E. 1999. Generation of influenza A viruses entirely from cloned cDNAs. Proceedings of the National Academy of Sciences 96:9345-9350.

276. Fodor E, Devenish L, Engelhardt OG, Palese P, Brownlee GG, García-Sastre A. 1999. Rescue of influenza A virus from recombinant DNA. J Virol 73:9679-82.

277. WHO. 28 February 2020 2020. Recommended composition of influenza virus vaccines for use in the 2020-2021 northern hemisphere influenza season. https://www.who.int/publications/m/item/recommended-composition-of-influenza-virus-vaccines-for-use-in-the-2020-2021-northern-hemisphere-influenza-season. Accessed

278. McQuillan AM, Byrd-Leotis L, Heimburg-Molinaro J, Cummings RD. 2019. Natural and synthetic sialylated glycan microarrays and their applications. Frontiers in Molecular Biosciences 6:88.

279. Jayaraman A, Pappas C, Raman R, Belser JA, Viswanathan K, Shriver Z, Tumpey TM, Sasisekharan R. 2011. A single base-pair change in 2009 H1N1 hemagglutinin increases human receptor affinity and leads to efficient airborne viral transmission in ferrets. PloS one 6:e17616.

280. Lyons DM, Lauring AS. 2018. Mutation and epistasis in influenza virus evolution. Viruses 10:407.

281. Fall A, Han L, Yunker M, Gong Y-N, Li T-J, Norton JM, Abdullah O, Rothman RE, Fenstermacher KZ, Morris CP. Evolution of influenza A (H3N2) viruses in 2 consecutive seasons of genomic surveillance, 2021–2023, p ofad577. *In* (ed), Oxford University Press US,

282. Zhou B, Donnelly ME, Scholes DT, St. George K, Hatta M, Kawaoka Y, Wentworth DE. 2009. Single-reaction genomic amplification accelerates

sequencing and vaccine production for classical and Swine origin human influenza a viruses. Journal of virology 83:10309-10313.

283. Fall A, Gallagher N, Morris CP, Norton JM, Pekosz A, Klein E, Mostafa HH. 2022. Circulation of enterovirus D68 during period of increased influenza-like illness, Maryland, USA, 2021. Emerging infectious diseases 28:1525.

284. Klein EY, Blumenkrantz D, Serohijos A, Shakhnovich E, Choi J-M, Rodrigues JV, Smith BD, Lane AP, Feldman A, Pekosz A. 2018. Stability of the influenza virus hemagglutinin protein correlates with evolutionary dynamics. Msphere 3:10.1128/mspheredirect. 00554-17.

285. McCown MF, Pekosz A. 2006. Distinct domains of the influenza a virus M2 protein cytoplasmic tail mediate binding to the M1 protein and facilitate infectious virus production. Journal of virology 80:8178-8189.

286. Resnick JD, Beer MA, Pekosz A. 2023. Early transcriptional responses of human nasal epithelial cells to infection with Influenza A and SARS-CoV-2 virus differ and are influenced by physiological temperature. Pathogens 12:480.

287. Heimburg-Molinaro J, Tappert M, Song X, Lasanajak Y, Air G, Smith DF, Cummings RD. 2012. Probing virus–glycan interactions using glycan microarrays, p 251-267, Carbohydrate Microarrays. Springer.

288. Wilson JL, Akin E, Zhou R, Jedlicka A, Dziedzic A, Liu H, Fenstermacher KZ, Rothman RE, Pekosz A. 2023. The influenza B virus victoria and yamagata lineages display distinct cell tropism and infection-induced host gene expression in human nasal epithelial cell cultures. Viruses 15:1956.

289. WHO. Global Influenza Surveillance and Response System (GISRS), *on* WHO. https://www.who.int/initiatives/global-influenza-surveillance-and-response-system. Accessed March 29, 2024.

290. CDC. March 29, 2024. Weekly U.S. Inflluenza Surveillance Report. https://www.cdc.gov/flu/weekly/index.htm. Accessed March 29, 2024.

291. WHO. 2017. WHO public health research agenda for influenza: 2017 update. World Health Organization,

292. Suphaphiphat P, Franti M, Hekele A, Lilja A, Spencer T, Settembre E, Palmer G, Crotta S, Tuccino AB, Keiner B. 2010. Mutations at positions 186 and 194 in the HA gene of the 2009 H1N1 pandemic influenza virus improve replication in cell culture and eggs. Virology journal 7:1-5.

293. Ye J, Sorrell EM, Cai Y, Shao H, Xu K, Pena L, Hickman D, Song H, Angel M, Medina RA. 2010. Variations in the hemagglutinin of the 2009 H1N1 pandemic virus: potential for strains with altered virulence phenotype? PLoS pathogens 6:e1001145.

294. Ilyushina NA, Khalenkov AM, Seiler JP, Forrest HL, Bovin NV, Marjuki H, Barman S, Webster RG, Webby RJ. 2010. Adaptation of pandemic H1N1 influenza viruses in mice. Journal of virology 84:8607-8616.

295. Thompson AJ, Wu NC, Canales A, Kikuchi C, Zhu X, de Toro BF, Worth C, Wang S, McBride R, Peng W. 2024. Evolution of human H3N2 influenza virus receptor specificity has substantially expanded the receptor-binding domain site. Cell Host & Microbe 32:261-275. e4.

296. Lewis NS, Anderson TK, Kitikoon P, Skepner E, Burke DF, Vincent AL. 2014. Substitutions near the hemagglutinin receptor-binding site determine the antigenic evolution of influenza A H3N2 viruses in US swine. Journal of virology 88:4752-4763.

297. Gambaryan A, Tuzikov A, Pazynina G, Bovin N, Balish A, Klimov A. 2006. Evolution of the receptor binding phenotype of influenza A (H5) viruses. Virology 344:432-438.

298. Pushan SS, Samantaray M, Rajagopalan M, Ramaswamy A. 2023. Evolution of Indian influenza A (H1N1) hemagglutinin strains: A comparative analysis of the pandemic Californian HA strain. Frontiers in Molecular Biosciences 10:1111869.

299. Dai M, Du W, Martínez-Romero C, Leenders T, Wennekes T, Rimmelzwaan GF, van Kuppeveld FJ, Fouchier RA, Garcia-Sastre A, de Vries E. 2021. Analysis of the evolution of pandemic influenza A (H1N1) virus neuraminidase reveals entanglement of different phenotypic characteristics. MBio 12:10.1128/mbio. 00287-21.

300. Hung S-J, Hsu Y-M, Huang S-W, Tsai H-P, Lee LYY, Hurt AC, Barr IG, Shih S-R, Wang J-R. 2020. Genetic variations on 31 and 450 residues of influenza A nucleoprotein affect viral replication and translation. Journal of biomedical science 27:1-13.

301. Xu J, Christman MC, Donis RO, Lu G. 2011. Evolutionary dynamics of influenza A nucleoprotein (NP) lineages revealed by large-scale sequence analyses. Infection, genetics and Evolution 11:2125-2132.

302. Smith DF, Song X, Cummings RD. 2010. Use of glycan microarrays to explore specificity of glycan-binding proteins. Methods in enzymology 480:417-444.

303. Air GM. 2014. Influenza virus–glycan interactions. Current opinion in virology 7:128-133.

304. Song X, Lasanajak Y, Xia B, Heimburg-Molinaro J, Rhea JM, Ju H, Zhao C, Molinaro RJ, Cummings RD, Smith DF. 2011. Shotgun glycomics: a microarray strategy for functional glycomics. Nature methods 8:85-90.

305. Byrd-Leotis L, Jia N, Dutta S, Trost JF, Gao C, Cummings SF, Braulke T, Müller-Loennies S, Heimburg-Molinaro J, Steinhauer DA. 2019. Influenza binds phosphorylated glycans from human lung. Science advances 5:eaav2554.

306. Byrd-Leotis L, Liu R, Bradley KC, Lasanajak Y, Cummings SF, Song X, Heimburg-Molinaro J, Galloway SE, Culhane MR, Smith DF. 2014. Shotgun glycomics of pig lung identifies natural endogenous receptors for influenza viruses. Proceedings of the National Academy of Sciences 111:E2241-E2250.

307. CFG. Microarrays, *on* National Center for Functional Glycomics. https://research.bidmc.org/ncfg/microarrays. Accessed April 2, 2024.

308. Matrosovich MN, Gambaryan AS. 2012. Solid-phase assays of receptor-binding specificity. Influenza Virus: Methods and Protocols:71-94.

309. Gambaryan AS, Matrosovich MN. 1992. A solid-phase enzyme-linked assay for influenza virus receptor-binding activity. Journal of virological methods 39:111-123.

310. de Vries E, Guo H, Du W, Liu M, van Kuppeveld FJ, de Haan CA. 2022. Quantification of receptor association, dissociation, and NA-dependent motility of

influenza A particles by biolayer interferometry, p 123-140, Glycovirology: Methods and Protocols. Springer.

311. Benton DJ, Wharton SA, Martin SR, McCauley JW. 2017. Role of neuraminidase in influenza A (H7N9) virus receptor binding. Journal of virology 91:10.1128/jvi. 02293-16.

312. Rapoport E, Mochalova L, Gabius H-J, Romanova J, Bovin N. 2006. Search for additional influenza virus to cell interactions. Glycoconjugate journal 23:115-125.

313. Londrigan SL, Turville SG, Tate MD, Deng Y-M, Brooks AG, Reading PC. 2011. N-linked glycosylation facilitates sialic acid-independent attachment and entry of influenza A viruses into cells expressing DC-SIGN or L-SIGN. Journal of virology 85:2990-3000.

314. Upham JP, Pickett D, Irimura T, Anders EM, Reading PC. 2010. Macrophage receptors for influenza A virus: role of the macrophage galactose-type lectin and mannose receptor in viral entry. Journal of virology 84:3730-3737.

315. Karakus U, Pohl MO, Stertz S. 2020. Breaking the convention: sialoglycan variants, coreceptors, and alternative receptors for influenza a virus entry. Journal of virology 94:10.1128/jvi. 01357-19.

316. Karakus U, Thamamongood T, Ciminski K, Ran W, Günther SC, Pohl MO, Eletto D, Jeney C, Hoffmann D, Reiche S. 2019. MHC class II proteins mediate cross-species entry of bat influenza viruses. Nature 567:109-112.

317. Stray SJ, Cummings RD, Air GM. 2000. Influenza virus infection of desialylated cells. Glycobiology 10:649-658.

318. Liu Y, Childs RA, Matrosovich T, Wharton S, Palma AS, Chai W, Daniels R, Gregory V, Uhlendorff J, Kiso M. 2010. Altered receptor specificity and cell tropism of D222G hemagglutinin mutants isolated from fatal cases of pandemic A (H1N1) 2009 influenza virus. Journal of virology 84:12069-12074.

319. Wan H, Perez DR. 2007. Amino acid 226 in the hemagglutinin of H9N2 influenza viruses determines cell tropism and replication in human airway epithelial cells. Journal of virology 81:5181-5191.

320. Sieben C, Kappel C, Zhu R, Wozniak A, Rankl C, Hinterdorfer P, Grubmüller H, Herrmann A. 2012. Influenza virus binds its host cell using multiple dynamic interactions. Proceedings of the National Academy of Sciences 109:13626-13631.

321. Kim I, Dubrow A, Zuniga B, Zhao B, Sherer N, Bastiray A, Li P, Cho J-H. 2022. Energy landscape reshaped by strain-specific mutations underlies epistasis in NS1 evolution of influenza A virus. Nature Communications 13:5775.

322. Pauly MD, Lyons DM, Fitzsimmons WJ, Lauring AS. 2017. Epistatic interactions within the influenza A virus polymerase complex mediate mutagen resistance and replication fidelity. MSphere 2:10.1128/msphere. 00323-17.

323. Müller NF, Stolz U, Dudas G, Stadler T, Vaughan TG. 2020. Bayesian inference of reassortment networks reveals fitness benefits of reassortment in human influenza viruses. Proceedings of the National Academy of Sciences 117:17104-17111.

324. Potter BI, Kondor R, Hadfield J, Huddleston J, Barnes J, Rowe T, Guo L, Xu X, Neher RA, Bedford T. 2019. Evolution and rapid spread of a reassortant A

(H3N2) virus that predominated the 2017–2018 influenza season. Virus evolution 5:vez046.

325. Bouvier NM, Lowen AC. 2010. Animal models for influenza virus pathogenesis and transmission. Viruses 2:1530-1565.

326. Herfst S, Schrauwen EJ, Linster M, Chutinimitkul S, De Wit E, Munster VJ, Sorrell EM, Bestebroer TM, Burke DF, Smith DJ. 2012. Airborne transmission of influenza A/H5N1 virus between ferrets. science 336:1534-1541.

327. Zhu H, Wang D, Kelvin D, Li L, Zheng Z, Yoon S-W, Wong S-S, Farooqui A, Wang J, Banner D. 2013. Infectivity, transmission, and pathology of human-isolated H7N9 influenza virus in ferrets and pigs. Science 341:183-186.

328. Belser JA, Gustin KM, Pearce MB, Maines TR, Zeng H, Pappas C, Sun X, Carney PJ, Villanueva JM, Stevens J. 2013. Pathogenesis and transmission of avian influenza A (H7N9) virus in ferrets and mice. Nature 501:556-559.

329. Pearce MB, Jayaraman A, Pappas C, Belser JA, Zeng H, Gustin KM, Maines TR, Sun X, Raman R, Cox NJ. 2012. Pathogenesis and transmission of swine origin A (H3N2) v influenza viruses in ferrets. Proceedings of the National Academy of Sciences 109:3944-3949.

330. Wan H, Sorrell EM, Song H, Hossain MJ, Ramirez-Nieto G, Monne I, Stevens J, Cattoli G, Capua I, Chen L-M. 2008. Replication and transmission of H9N2 influenza viruses in ferrets: evaluation of pandemic potential. PloS one 3:e2923.

331. Otte A, Marriott AC, Dreier C, Dove B, Mooren K, Klingen TR, Sauter M, Thompson K-A, Bennett A, Klingel K. 2016. Evolution of 2009 H1N1 influenza

viruses during the pandemic correlates with increased viral pathogenicity and transmissibility in the ferret model. Scientific reports 6:28583.

332. Duan S, Boltz DA, Seiler P, Li J, Bragstad K, Nielsen LP, Webby RJ, Webster RG, Govorkova EA. 2010. Oseltamivir–resistant pandemic H1N1/2009 influenza virus possesses lower transmissibility and fitness in ferrets. PLoS pathogens 6:e1001022.

333. van der Vries E, Veldhuis Kroeze EJ, Stittelaar KJ, Linster M, Van der Linden A, Schrauwen EJ, Leijten LM, van Amerongen G, Schutten M, Kuiken T. 2011. Multidrug resistant 2009 A/H1N1 influenza clinical isolate with a neuraminidase I223R mutation retains its virulence and transmissibility in ferrets. PLoS pathogens 7:e1002276.

334. Belser JA, Jayaraman A, Raman R, Pappas C, Zeng H, Cox NJ, Katz JM, Sasisekharan R, Tumpey TM. 2011. Effect of D222G mutation in the hemagglutinin protein on receptor binding, pathogenesis and transmissibility of the 2009 pandemic H1N1 influenza virus. PloS one 6:e25091.

335. Abed Y, Pizzorno A, Hamelin M-E, Leung A, Joubert P, Couture C, Kobasa D, Boivin G. 2011. The 2009 pandemic H1N1 D222G hemagglutinin mutation alters receptor specificity and increases virulence in mice but not in ferrets. Journal of Infectious Diseases 204:1008-1016.

336. Baras B, Stittelaar KJ, Kuiken T, Jacob V, Bernhard R, Giannini S, de Waal L, van Amerongen G, Simon JH, Osterhaus AD. 2011. Longevity of the protective immune response induced after vaccination with one or two doses of AS03A-adjuvanted split H5N1 vaccine in ferrets. Vaccine 29:2092-2099.

337. Isakova-Sivak I, Matyushenko V, Kotomina T, Kiseleva I, Krutikova E, Donina S, Rekstin A, Larionova N, Mezhenskaya D, Sivak K. 2019. Sequential immunization with universal live attenuated influenza vaccine candidates protects ferrets against a high-dose heterologous virus challenge. Vaccines 7:61.

338. Hensley SE. 2014. Challenges of selecting seasonal influenza vaccine strains for humans with diverse pre-exposure histories. Current opinion in virology 8:85-89.

CURRICULUM VITAE

Nico Swanson

EDUCATION

PhD, Molecular Microbiology and Immunology **May 2024 (anticipated)**

Johns Hopkins Bloomberg School of Public Health

MSPH, Global Disease Epidemiology and Control **May 2024 (anticipated)**

Johns Hopkins Bloomberg School of Public Health

BS, Biochemistry **May 2018**

Florida State University

BA, French **May 2018**

Florida State University

RESEARCH EXPERIENCE

Doctoral Researcher
August 2020-Present

Johns Hopkins Bloomberg School of Public Health, Laboratory of Dr. Andrew Pekosz

Investigating Viral Evolution of Seasonal Influenza

- Performed surveillance of circulating influenza strains in Baltimore, Maryland, through the isolation and characterization of clinical samples on a primary human nasal epithelial cell culture system.
- Discovered phenotypic changes in emergent H1N1 clades, including differences in replication, receptor binding, and antigenic drift.
- Identified individual mutations responsible for differences in viral fitness and antigenic profile through the construction and site-directed mutagenesis of an H1N1 infectious clone system.

Analytical Chemist
June 2018-July 2020

Florida Department of Agriculture and Consumer Services, Division of Food Safety

Detection of Chemical Residues in Food Products

- Screened foods for chemical residues, including pesticides in produce and antibiotics in seafood, using high performance liquid chromatography (HPLC) coupled with mass spectrometry.
- Developed and implemented methods for the analysis of cannabinoids in hemp oils and plant matter using HPLC coupled with UV/Vis spectroscopy.

TEACHING AND MENTORING

Graduate Mentor
2021-2024

- Mentored and trained over ten undergraduate and graduate students in the Pekosz laboratory.
- Taught students the fundamentals of molecular biology, virology, and immunology, developed their proficiency in laboratory techniques, and aided in professional development.

Diversity Summer Internship Program
2022

- Served as a mentor for students enrolled in the Diversity Summer Internship Program, which recruits highly qualified undergraduates from traditionally underrepresented groups in STEM.
- Provided training on microbiological assays and facilitated a research project over the course of eight weeks.
- Coached students on scientific communication in preparation for their presentations at a national conference.

Teaching Assistant
2022-2024

Advanced Virology; Immunity, Infection, and Disease

- Held office hours and one-on-one tutoring sessions for graduate students.
- Led journal discussions and provided individualized guidance and feedback to student presenters.
- Organized guest lecturers, managed technical considerations, and graded student progress.

PUBLICATIONS

Swanson, N. J., Girish, J., Yunker, M., Liu, H., Norton, J., Han, C., Mostafa, H., Fenstermacher, K., Rothman, R., Pekosz, A. (2024). Evaluating the phenotypic contributions of recent clade-defining seasonal H1N1 mutations on in vitro fitness and antigenic drift. *In preparation.*

Swanson, N. J., Marinho, P., Dziedzic, A., Jedlicka, A., Liu, H., Fenstermacher, K., ... & Pekosz, A. (2023). 2019–2020 H1N1 clade A5a. 1 viruses have better in vitro fitness compared with the co-circulating A5a. 2 clade. *Scientific Reports*, 13(1), 10223.

Bolton, M. J., Ort, J. T., McBride, R., **Swanson, N. J.**, Wilson, J., Awofolaju, M., Furey, C., Greenplate, A.R., Drapeau, E.M., Pekosz, A., Paulson, J.C., Hensley, S. E. (2022). Antigenic and virological properties of an H3N2 variant that continues to dominate the 2021–22 Northern Hemisphere influenza season. *Cell Reports*, 39(9), 110897.

Mostafa H.H., Luo C.H., Morris C.P., Li M., **Swanson N.J.**, Amadi A., Gallagher N., Pekosz A.. (2022). SARS-CoV-2 infections in mRNA vaccinated individuals are biased for viruses encoding spike E484K and associated with reduced infectious virus loads that correlate with respiratory antiviral IgG levels. *Journal of Clinical Virology*, 150, 105151.

Fall, A., Eldesouki, R.E., Sachithanandham, J., Morris, C.P., Norton, J.M., Gaston, D.C., Forman, M., Abdullah, O., Gallagher, N., Li, M., **Swanson, N.J.**, Pekosz, A., Klein, E.Y., Mostafa, H.H., (2022). The displacement of the SARS-CoV-2 variant Delta with Omicron: An investigation of hospital admissions and upper respiratory viral loads. *EBioMedicine*, 79, 104008.

Luo, C.H., Morris, C.P., Sachithanandham, J., Amadi, A., Gaston, D.C., Li, M., **Swanson, N.J.**, Schwartz, M., Klein, E.Y., Pekosz, A. and Mostafa, H.H., (2021). Infection with the SARS-CoV-2 delta variant is associated with higher recovery of infectious virus compared to the alpha variant in both unvaccinated and vaccinated individuals. *Clinical Infectious Diseases: An Official Publication of the Infectious Diseases Society of America*.

PRESENTATIONS

Oral:

Swanson N.J. Applying the R3 Framework to the Evaluation of Seasonal Influenza Viral Fitness. Presented as a Katharine Welsh Scholar at the Katharine E. Welsh Symposium of the R³ Center for Innovation in Science Education in Baltimore, MD (2023).

Swanson N.J (January 20, 2023). Interviewee. Virology Series, Part 1: The Flu and You [The Sci'more Podcast].

Swanson N.J., Girish J., Liu H., Fall A., Norton J., Han C. Mostafa H., Pekosz A. Differences in viral fitness between recently circulating seasonal H1N1 strains. Presented at the American Society for Virology conference in Athens, GA (2023)

Swanson N.J., Girish J., Liu H., Fall A., Norton J., Han C. Mostafa H., Pekosz A. Differences in viral fitness between recently circulating seasonal H1N1 strains. Presented at the Centers for Excellence in Influenza Research and Response conference in Baltimore, MD (2023).

Swanson N.J., Marinho P, Liu H, Pekosz A. Differences in viral fitness between two cocirculating H1N1 clades in the 2019-20 influenza season. Presented at the American Society for Virology conference in Madison, WI (2022).

Poster:

Swanson N.J., Girish J., Liu H., Fall A., Norton J., Han C. Mostafa H., Pekosz A. Differences in viral fitness between recently circulating seasonal H1N1 strains. Presented at the American Society for Virology conference in Athens, GA (2023)

Swanson N.J., Girish J., Liu H., Fall A., Norton J., Han C. Mostafa H., Pekosz A. Differences in viral fitness between recently circulating seasonal H1N1 strains. Presented at the Centers for Excellence in Influenza Research and Response conference in Baltimore, MD (2023).

Swanson N.J., Marinho P, Liu H, Pekosz A. Differences in viral fitness between two cocirculating H1N1 clades in the 2019-20 influenza season. Presented at the American Society for Virology conference in Madison, WI (2022)

Swanson N.J., Marinho P, Liu H, Pekosz A. Differences in viral fitness between two cocirculating H1N1 clades in the 2019-20 influenza season. Presented at the Centers for Excellence in Influenza Research and Response conference in Memphis, TN (2022).

AWARDS AND ACTIVITIES

Delegate, Johns Hopkins Science Policy and Diplomacy Group 2024 Capitol Hill Day

- Chosen as a delegate to promote congressional support for the PLANT Act and broader investment in the development of alternative proteins in agriculture.
- Coauthoring a policy memo advocating alternative protein investment which will be submitted for publication in May, 2024.

Katharine E. Welsh Scholar

- Completed the R3 program in the Center for Innovation in Science Education, which involves a series of courses focused on rigor, reproducibility, and responsibility in immunology and microbiology research.
- Presented at the Katharine E. Welsh Symposium of the R³ Center for Innovation in Science Education to discuss the application of the R3 framework to my doctoral research.

T32 Ruth L. Kirschstein National Research Service Award

- Accepted onto the departmental T32 educational training grant funded by the National Institutes of Health, which covers the majority of the financial support for my position as a predoctoral fellow.

SKILLS

Molecular Biology:

- Cell and Tissue Culture
- Molecular Cloning
- Viral Titer Assays (TCID50, PFU)
- PCR/qPCR
- Western Blot Assays
- Protein Labeling and Purification

Communication:

- Scientific writing
- Scientific presentation
- Intermediate French fluency

Analytical Chemistry:

- High Performance Liquid Chromatography
- Gas Chromatography
- Mass Spectrometry
- Uv/Vis Spectroscopy
- Chemical Extractions

General Laboratory:

- Sample Custody
- Statistical Analysis
- Biosafety levels 2 and 3